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EVALUATION OF GENE REGULATORY NETWORK INFERENCE METHODS FOR SPATIAL TRANSCRIPTOMICS

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Preface

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List of Abbreviations and Symbols

API	Application programming interface
ARACNe	Algorithm for the Reconstruction of Accurate Cellular Networks
ATAC-seq	Assay for transposase-accessible chromatin using sequencing
AUC	Area under the curve
AUCell	Area under the recovery curve for cell-level gene set enrichment
AUPRC	Area under the precision–recall curve
AUROC	Area under the receiver operating characteristic curve
BANJO	Bayesian Network Inference with Java Objects
BANKSY	Building Aggregates with a Neighborhood Kernel and Spatial Yardstick
BEELINE	Benchmarking framework for gene regulatory network inference algorithms
Bin2cell	Bin-to-cell reconstruction tool for Visium HD data
BP	Biological process (Gene Ontology domain)
CB	Cerebellum
ChIP-Atlas	Integrated database of public ChIP-seq and related binding data
ChIP-seq	Chromatin immunoprecipitation sequencing
CNU	Cerebral nuclei
CollecTRI	Collection of transcriptional regulatory interactions
CTX	Cerebral cortex
CTXsp	Cortical subplate
CUT&Tag	Cleavage under targets and tagmentation
CV	Coefficient of variation
DAG	Directed acyclic graph
DBiT-seq	Deterministic Barcoding in Tissue sequencing
DNA	Deoxyribonucleic acid
DoRothEA	Discriminant regulon expression analysis
DPI	Data processing inequality
eGRN	Enhancer-driven gene regulatory network
ENACT	End-to-end analysis of Visium HD data
EP	Early precision
EPR	Early precision ratio

ER	Early recall
FDR	False discovery rate
FISH	Fluorescence in situ hybridization
GENIE3	Gene Network Inference with Ensemble of Trees
GO	Gene Ontology
g:Profiler	Web server for functional enrichment analysis
GRN	Gene regulatory network
GRNBoost2	Gradient boosting-based gene regulatory network inference
HD	High definition (Visium HD)
H&E	Haematoxylin and eosin
HPF	Hippocampal formation
HVG	Highly variable gene
HY	Hypothalamus
IB	Interbrain
ISH	In situ hybridization
ISS	In situ sequencing
KNN	K-nearest neighbors
KnockTF	Database of transcription factor knockdown and knockout expression profiles
MapMyCells	Allen Institute cell type mapping service
MB	Midbrain
MERFISH	Multiplexed error-robust fluorescence in situ hybridization
MI	Mutual information
MRF	Markov random field
mRNA	Messenger RNA
MY	Medulla
NGS	Next-generation sequencing
nnSVG	Nearest-neighbor Gaussian process method for spatially variable genes
ODE	Ordinary differential equation
OLF	Olfactory areas
OmniPath	Integrated database of molecular interactions and regulatory knowledge
PAL	Pallidum
pySCENIC	Python implementation of SCENIC
QC	Quality control
RCTD	Robust cell type decomposition
RE	Regulatory element
RNA	Ribonucleic acid
RRID	Research resource identifier
scATAC-seq	Single-cell assay for transposase-accessible chromatin using sequencing

SCENIC	Single-Cell Regulatory Network Inference and Clustering
SCODE	Single-cell Ordinary Differential Equation-based inference
SCRIPro	Single Cell Regulatory Network Inference Profiling
scRNA-seq	Single-cell RNA sequencing
seqFISH	Sequential fluorescence in situ hybridization
SFINN	Shared Factor neighborhood and Integrated Neural Network
smFISH	Single-molecule fluorescence in situ hybridization
SpaGRN	Spatially informed gene regulatory network inference
SpaGTL	Spatially Aligned Graph Transfer Learning
SPARK-X	Non-parametric method for detecting spatially variable genes
SpatialDE	Method for identifying spatially variable genes
ST	Spatial transcriptomics
STalign	Diffeomorphic alignment method for spatial transcriptomics data
STR	Striatum
SVG	Spatially variable gene
TF	Transcription factor
TH	Thalamus
TRN	Transcriptional regulatory network
TSS	Transcription start site
UMAP	Uniform manifold approximation and projection
UMI	Unique molecular identifier
VAE	Variational autoencoder
VIPER	Virtual inference of protein activity by enriched regulon analysis
WGCNA	Weighted Gene Co-expression Network Analysis

Abstract

Gene regulatory networks (GRNs) link transcription factors (TFs), target genes, and regulatory elements controlling gene expression. This thesis focuses on transcriptional regulatory networks (TRNs), defined as directed TF–target relationships. Spatial transcriptomics measures gene expression together with tissue location, enabling GRN inference methods to incorporate regional and neighborhood information rather than relying on expression alone. Direct comparison among spatially informed GRN inference methods is difficult because real datasets rarely have a complete reference network for the tissue and condition being studied. Methods are also evaluated in separate studies using different datasets, references, and metrics, making their reported performance difficult to compare directly.

This thesis therefore compared networks inferred by three GRN strategies that differ in their use of spatial information, using the same real dataset and a common evaluation framework. Without a complete ground-truth network, complementary forms of evidence were used to assess the performance and examine what each evaluation revealed and where its interpretation was limited.

After data-driven quality control and single-cell reference annotation, 30,000 bins were sampled from a 10x Genomics Visium HD mouse brain section. GRNBoost2 used expression alone, SpaGRN added spatial statistics to gene selection and edge scoring, and SpaGTL combined pretraining and transfer learning with spatially constrained representation learning. pySCENIC then constructed regulons and scored their activity per bin with AUCell. Evaluation covered reproducibility across runs, support from curated, perturbation, and binding resources, and the spatial organization of regulon activity, including agreement with in situ hybridization (ISH).

GRNBoost2 produced more consistent network sizes and retained more repeated edges across runs, whereas SpaGTL most consistently retained its highest-ranked edges. Assigned edges of TFs were not identical across different runs, implying no single run was definitive.

All three recovered reference-supported TF-target relationships above random expectation, although absolute precision remained low. GRNBoost2 agreed most with curated and binding evidence, SpaGRN with perturbation evidence.

SpaGRN produced the strongest spatial coherence, but activity maps for target gene sets of different TFs were often similar, so coherence did not always reflect a TF-specific pattern. Some GRNBoost2 regulons were spatially organized despite receiving no coordinates, since coexpressed targets concentrated in the same regions produce localized activity. SpaGTL showed

the most consistent regional agreement with ISH TF expression.

Overall, No method performed best across all evaluation. Three different validation captured different network properties, and they should be interpreted as complementary evidence rather than interchangeable measures of accuracy. Method performance therefore cannot be summarized by a single ranking, and the most relevant criterion depends on the analytical aim.

Chapter 1

Literature Review

1.1 Gene Regulatory Network (GRN)

The human genome contains approximately 19,000 to 20,000 protein-coding genes (Viggiano, 2022). However, these genes are not uniformly expressed, but their expression is dynamically regulated across cell types, developmental stages, and environmental conditions. Gene regulation is controlled at multiple levels of genome organization. At the genomic level, regulatory elements such as promoters, enhancers, and the three-dimensional organization of chromatin shape the regulatory landscape. At the epigenomic level, chromatin accessibility and epigenetic modifications influence the accessibility of regulatory regions and transcription factor binding. At the transcriptional level, transcription factors (TFs) serve as the primary regulatory drivers because they directly bind to specific DNA sequences within promoters and enhancers to modulate transcriptional activity (I. R. Wolf, Simões, and Valente, 2021). Upon binding, TFs influence RNA polymerase recruitment and activity, often through interactions with cofactors, thereby enhancing or repressing gene expression (T.-M. Kim and Park, 2011). Importantly, gene expression is rarely governed by a single TF. Instead, regulatory control follows a many-to-many relationship. Multiple TFs can act on the same gene, while individual TFs regulate sets of target genes. Through this combinatorial regulation, TFs influence not only genes but also one another in structured ways, giving rise to gene regulatory networks (GRNs).

Broadly defined, a GRN is a graph over the molecular players that set transcript level. Nodes stand for regulatory elements (RE), transcription factors, and target genes. Edges run in several directions at once, linking a regulatory element to the TF that occupies it, a TF to the gene it modulates, and a target gene back to a regulatory element when its own product acts as a regulator. A graph of this breadth accommodates chromatin state, cofactor binding, and post-transcriptional effects alongside direct transcriptional control. When regulatory elements are modeled explicitly, together with the transcription factors that occupy them and the genes they act on, such a network is usually called an enhancer GRN, or eGRN (Loers and Vermeirssen, 2024). Transcriptional regulatory network (TRN) is a narrower sense of GRN, a network restricted to relationships of TFs and their target genes (T.-M. Kim and Park, 2011). In this con-

text, a GRN represents these transcriptional regulatory interactions as a network structure in which nodes correspond to genes, including TFs and their target genes, and edges represent directed regulatory relationships between them (I. R. Wolf, Simões, and Valente, 2021). Figure 1.1 illustrates these two views of the same regulatory system.

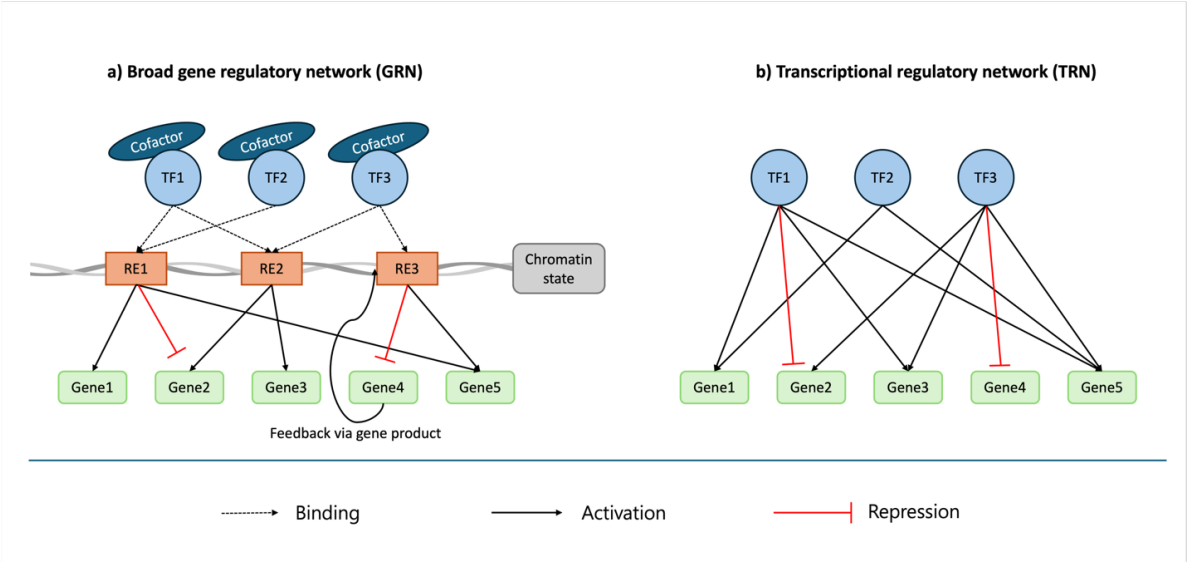

Figure 1.1: Broad and narrow views of a gene regulatory network. *(a) In the broad view, nodes include regulatory elements (RE), transcription factors (TF) together with their cofactors, and target genes. Edges run from a TF to the element it occupies, from an element to the gene it controls, and back from a gene product to a regulatory element as feedback, while chromatin state modulates how accessible the elements are. (b) In the transcriptional regulatory network, the same system is reduced to direct relationships between TFs and their target genes. Dashed arrows denote binding, solid arrows activation, and red bars repression.*

GRNs have several properties that shape their interpretation. Unlike simple co-expression networks, they represent directed regulatory influence from a TF to its targets. They are also sparse, since direct regulation connects only a small fraction of all possible gene pairs. Finally, GRNs are context dependent. The targets and effects of a TF can change across cell types, developmental stages, and environmental conditions. A network inferred from one dataset should therefore be treated as a condition specific model rather than a universal regulatory map (Marbach et al., 2012)

GRNs provide important insights into how gene expression is coordinated to control cellular functions and biological processes, including the regulation of cell identity, cell fate decisions, and disease progression (D. Kim et al., 2023). Understanding these regulatory networks can therefore contribute to identifying key regulatory genes involved in disease mechanisms and may support applications such as therapeutic target discovery and drug development (Nguyen et al., 2021; Triantafyllidis et al., 2023). For example, analysis of the TP53-centered regulatory network in cancer has revealed that mutant p53 and its downstream targets form a critical node driving tumor progression, which has led to the development of drugs that specifically target mutant p53 for cancer therapy (Triantafyllidis et al., 2023). However, the regulatory interactions that form GRNs are not directly observable and therefore need to be reconstructed

computationally. This motivates the development of computational approaches for GRN inference.

1.2 GRN inference

GRN inference generally aims to identify regulatory relationships between genes using molecular datasets (De Smet and Marchal, 2010). These datasets can originate from different molecular modalities, including transcriptomic and epigenomic measurements. The increasing availability of large-scale high-throughput datasets has enabled the inference of genome-scale GRNs and stimulated the development of diverse inference strategies based on different statistical models and underlying assumptions (Marbach et al., 2012).

1.2.1 Modeling frameworks

GRN inference methods can be grouped by how they represent regulation and infer edges from expression data. One group contains mechanistic or dynamical models. Boolean models, for example, represent gene activity as discrete states, usually on or off. This simplification makes large regulatory systems easier to analyze, but it cannot capture the continuous range of gene expression (Huynh-Thu et al., 2010; Kauffman, 2011).

Dynamical models employ systems of ordinary or stochastic differential equations to describe temporal changes in gene expression. In these models, the expression of a target gene is modeled as a function of the activity or concentration of upstream regulators. While these approaches can explicitly capture regulatory mechanisms and dynamic behavior, they require estimating many parameters, such as interaction strengths and kinetic rates. As the number of genes increases, parameter estimation becomes computationally demanding and difficult to perform reliably given limited data, which limits their applicability to large-scale transcriptomic datasets. A representative example is SCODE which infers regulatory networks from single-cell RNA-seq data using an ordinary differential equation framework (Matsumoto et al., 2017).

A second category relies on statistical analysis of gene expression data. Early approaches often inferred candidate regulatory relationships indirectly by first constructing gene co-expression networks based on expression correlations. In these approaches, strongly correlated gene pairs were treated as potential regulatory links rather than confirmed regulatory interactions. For example, WGCNA (Weighted Gene Co-expression Network Analysis) infers networks by connecting genes with highly correlated expression profiles (Langfelder and Horvath, 2008). Although computationally efficient and widely used, correlation-based methods cannot distinguish direct from indirect interactions and are limited to detecting linear relationships (Huynh-Thu et al., 2010).

Mutual information (MI) based approaches were later introduced as an alternative to correlation-based methods. MI measures how strongly the expression levels of two genes are statistically related. They extend correlation-based approaches by capturing nonlinear statistical depen-

dencies between gene expression profiles. A representative example is ARACNe (Algorithm for the Reconstruction of Accurate Cellular Networks), which infers regulatory interactions by identifying gene pairs with high MI and attempts to reduce indirect associations using the data processing inequality (DPI), which inspects every gene triplet and discards the weakest of the three edges on the assumption that it reflects an indirect path through the other two (Margolin et al., 2006). However, MI-based methods still primarily rely on pairwise comparisons, and do not directly encode regulatory directions.

A third group comprises probabilistic graphical models. These extend beyond pairwise analysis by adopting a multivariate framework. Rather than evaluating gene pairs independently, these models estimate conditional dependencies among multiple genes simultaneously (Friedman et al., 2000). A representative example is Bayesian networks, which represent genes as nodes in a directed acyclic graph (DAG) where edges encode probabilistic dependencies between regulators and targets. However, because the graph must be acyclic, feedback loops are not present (Auliac et al., 2008; Perrin et al., 2003). An example of such probabilistic frameworks is BANJO (Bayesian Network Inference with Java Objects), which applies Bayesian network structure learning to infer regulatory interactions (Smith et al., 2006). Gaussian graphical models provide an alternative probabilistic framework. These models use undirected graphs to represent conditional dependencies under the assumption that gene expression follows a multivariate Gaussian distribution. While statistically robust and suitable for large gene sets, Gaussian graphical models rely on linearity and normality assumptions and do not directly encode regulatory directionality (Ambroise, Chiquet, and Matias, 2009; Opgen-Rhein and Strimmer, 2007).

A fourth group is prediction-based methods. These approaches formulate GRN inference as a prediction problem, in which a target gene's expression level is predicted from the expression levels of candidate transcription factors. An edge is inferred when a TF significantly contributes to explaining the target's expression. Methods such as GENIE3 (Huynh-Thu et al., 2010) and GRNBoost2 (Moerman et al., 2019) employ tree-based ensemble techniques to assess the importance of TFs in predicting target gene expression. As TFs are treated as predictors and target genes as responses, directionality is introduced by design.

More recent deep learning-based methods extend this idea by modeling complex nonlinear relationships between regulators and targets. Neural network models belong to this same group rather than forming a separate one. For example, DeepDRIM applies convolutional neural networks to learn regulatory patterns from gene expression data by representing gene pair relationships in a structured input format (Jiaxing Chen et al., 2021). These methods offer greater flexibility, enabling the capture of higher-order dependencies that may not be detected by tree-based approaches. However, this increased modeling capacity often comes at the cost of reduced interpretability and a higher risk of overfitting, particularly in settings with limited sample sizes.

As each modeling framework relies on distinct assumptions regarding dependency structure, noise, and network topology, applying different methods to the same dataset can yield net-

works with systematically different structural properties. For expression-based methods in particular, the immediate output is often ranked list of candidate TF-target association.

1.2.2 From ranked associations to regulons

Many gene regulatory network (GRN) inference methods return a scored or weighted list of candidate associations between transcription factors (TFs) and target genes. These lists help prioritize possible regulatory links, but the meaning of an association score depends on the method. It may reflect coexpression, predictive importance, spatial association, or another relationship learned from the data. Regardless of how it is calculated, the score provides evidence for a candidate relationship rather than proof of direct regulation. A high score does not establish that a TF binds to the predicted target or directly controls its expression (Pratapa et al., 2020). Moreover, a further decision is needed when the output is a ranked list. A threshold must be chosen to determine which edges are retained, and no single threshold is suitable for every method or dataset (Xue, Y. Wu, and Y. Lin, 2023). A ranked list therefore remains a collection of candidate relationships. Regulon analysis provides a way to organize these candidates by regulator and interpret them as a regulatory program rather than as isolated edges.

A regulon consists of a regulator and a set of genes that are known or predicted to respond to it. A regulon is a widely used unit for interpreting the combined responses of multiple target genes. The Algorithm for the Reconstruction of Accurate Cellular Networks (ARACNe) infers context specific networks between TFs and target genes from expression data (Margolin et al., 2006). The targets linked to each regulator can then be treated as a regulon. Virtual Inference of Protein Activity by Enriched Regulon Analysis (VIPER) uses the combined expression patterns of activated and repressed targets within these regulons to estimate regulator activity (Alvarez et al., 2016). More recently, Epiregulon has combined single cell RNA expression, chromatin accessibility, and TF binding information to construct weighted regulons and estimate regulator activity in individual cells (Włodarczyk et al., 2025).

Single Cell Regulatory Network Inference and Clustering (SCENIC) is a regulon-based workflow originally developed for single-cell RNA sequencing data. It combines three main tasks in one workflow. These are the inference of regulons tailored to the dataset, their refinement using TF binding motifs, and the measurement of their activity in individual cells.

SCENIC first applies a network inference algorithm to generate weighted candidate associations between TFs and target genes. Candidate targets linked to the same TF are then organized into coexpression modules. cisTarget examines whether the regulatory regions of genes in each module are enriched for a binding motif associated with the TF. This process is known as motif pruning. Candidate targets without sufficient motif support are removed, while genes that contribute to motif enrichment are retained. The TF and its remaining candidate targets form a regulon. Motif pruning adds sequence-based support to relationships initially inferred from expression data and gives priority to possible direct targets. It does not, however, provide experimental evidence of TF binding or direct regulation (Aibar et al., 2017).

SCENIC then uses AUCell to estimate the activity of each regulon in each cell. AUCell is a method that uses the area under the curve (AUC) to measure gene set activity. Genes are first ranked by their expression within each cell. AUCell then examines whether the genes in a regulon are concentrated near the top of that ranking by calculating the area under a recovery curve. A regulon receives a high score when many of its target genes are highly ranked in the same cell. Repeating this analysis across all cells produces a matrix containing an activity score for every regulon in every cell.

At the single cell level, regulon based scoring helps counteract sparsity by combining expression signals from multiple target genes. A single target may not be detected in a given cell, but the broader target set can still capture a coordinated regulatory signal. Regulon activity therefore provides a more robust readout than the expression of any single gene (Aibar et al., 2017). This approach reduces, but does not completely remove, the effect of sparsity. The resulting activity scores still depend on the quality of the expression data and the accuracy of the regulon.

The Python implementation, pySCENIC, follows the same general workflow and supports its use with large single cell datasets (Van de Sande et al., 2020). Its modular design allows different algorithms to be used during the initial network inference step while retaining motif pruning and AUCell. Nevertheless, the regulons produced by SCENIC remain computational predictions. True targets may be excluded when relevant motifs are missing or poorly annotated in the reference database. TFs from the same family may also be difficult to distinguish because they often recognize similar motifs. Confirmation therefore requires additional experimental evidence, such as TF perturbation (Chevalley et al., 2025). Figure 1.2 summarizes the full workflow, beginning with scored candidate associations, continuing through motif-based regulon refinement, and ending with a matrix of regulon activity across cells.

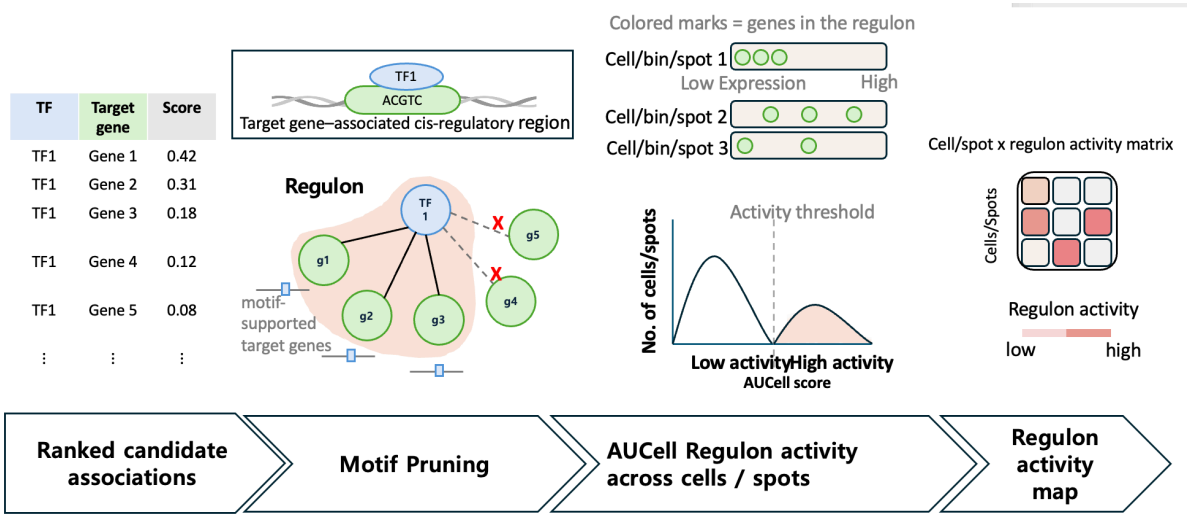

Figure 1.2: From ranked candidate associations to a regulon activity map. *A ranked table of TF to target scores is first checked for binding motif support in the cis regulatory regions of the candidate targets. Targets without support are removed, and the TF together with its retained targets forms a regulon. AUCell then ranks all genes within each cell, bin, or spot and scores how strongly the members of a regulon concentrate at the top of that ranking, which yields a cell by regulon activity matrix that can be projected back onto the tissue as a regulon activity map.*

1.2.3 Data contexts for GRN inference

Early GRN studies were primarily based on bulk transcriptomic data, in which each sample represents the averaged gene expression across heterogeneous cell populations. While bulk measurements provide relatively stable expression estimates, averaging across a heterogeneous population cancels out the very differences that define a cell-type-specific regulatory program. A TF active in one cell type and silent in a neighboring one registers as a moderate signal in the mixture, and its regulon cannot be separated from the background (Nguyen et al., 2021).

Single-cell RNA sequencing (scRNA-seq) enables gene expression profiling at the resolution of individual cells, allowing the reconstruction of cell-type-specific or state-specific regulatory programs. However, scRNA-seq datasets are characterized by high levels of sparsity. In particular, dropout events occur when genes that are biologically expressed are not detected due to technical limitations, such as inefficient RNA capture or variability in reverse transcription and amplification efficiency (Nguyen et al., 2021). Thus, while single-cell resolution enhances biological specificity, it introduces additional statistical challenges that remain only partially resolved.

Multiomics based GRN inference integrates transcriptomic measurements with additional molecular layers that capture different aspects of gene regulation, such as epigenetic features (e.g., ATAC-seq), TF binding (e.g., ChIP-seq), or proteomics. Combining multiple layers of molecular regulation can provide complementary evidence for regulatory interactions and improve confidence in network reconstruction (D. Kim et al., 2023). In single cell multiome data, paired measurements of gene expression and chromatin accessibility have become a standard basis for GRN inference (Loers and Vermeirssen, 2024). These data connect target gene expression with accessible regulatory regions in the same cellular context. Representative examples include STARNet, which integrates spatial transcriptomic (RNA-seq) and chromatin accessibility (ATAC-seq) data to infer spatially resolved GRNs, and CeSpGRN, which infers cell-specific GRNs from paired scRNA-seq/scATAC-seq or spatial transcriptomic data (Hu et al., 2025; Ziqi Zhang et al., 2026).

A common limitation of bulk and single cell protocols is the need for tissue dissociation. Cells are separated from their original neighbors, and their coordinates are lost (Lähnemann et al., 2020). Regulatory signals that depend on position, tissue gradients, or local neighbors can therefore be missed when expression is measured from dissociated cells alone. Spatial transcriptomics was developed to retain this spatial axis.

1.3 Spatial Transcriptomics

Spatial transcriptomics (ST) recovers a record of where in the tissue each expression profile came from by measuring gene expression within an intact tissue section (Ståhl et al., 2016). Unlike conventional transcriptomic approaches that provide only a gene expression matrix, ST data include both gene expression matrix and the spatial coordinates of each location. This

data structure allows gene expression to be interpreted within the anatomical context of the tissue and enables intuitive identification of spatial patterns and location-dependent expression changes. Furthermore, by avoiding tissue dissociation, ST data reduces experimental artifacts and enables the observation of gene expression patterns that more closely reflect the in vivo state (Brink et al., 2017).

Spatial transcriptomics technologies are broadly classified into two main approaches, which are barcoding-based and imaging-based methods (Sang-aram et al., 2024).

1.3.1 Barcoding based spatial transcriptomics

Barcoding-based spatial transcriptomics is a next-generation sequencing (NGS)-based approach in which tissue sections are placed onto surfaces containing spatially arranged barcoded oligonucleotides. Messenger RNA (mRNA) molecules captured at each location are tagged with positional barcodes and subsequently sequenced to reconstruct spatially resolved gene expression profiles (Moses and Pachter, 2022; Moffitt, Lundberg, and Heyn, 2022). In this context, each barcoded location serves as a spatial capture unit, meaning that transcripts assigned to the same barcode are interpreted as originating from the same position or local region within the tissue. Depending on how spatial barcodes are assigned, these approaches can be broadly categorized into array-based, bead-based, and deterministic barcoding strategies.

In array-based platforms, barcodes are pre-positioned at fixed locations on a surface. A representative example is the 10x Genomics Visium platform, which uses spots of approximately 55 µm in diameter as a spatial capture unit (Polański et al., 2024). As each spot is larger than an individual cell containing multiple cells, the measured expression values represent a mixture of signals from several cells.

In contrast, bead-based platforms such as Slide-seq (Rodriques et al., 2019) use small, barcoded beads that are randomly distributed across a surface. The spatial positions of these beads are determined beforehand, allowing gene expression to be reconstructed at each location after sequencing. This approach provides higher spatial resolution (approximately 10–15 µm) and reduces the degree of signal mixing, although the observation unit still does not perfectly correspond to individual cell boundaries (Rodriques et al., 2019; Stickels et al., 2021).

Deterministic barcoding approaches, such as DBiT-seq (Deterministic Barcoding in Tissue sequencing), use microfluidic channels to deliver barcodes sequentially across the tissue in two perpendicular directions, thereby creating a grid of spatially indexed locations (Y. Liu et al., 2020). Each position is defined by a unique combination of two barcodes, enabling precise spatial localization without requiring prefabricated arrays.

More recently, high-resolution platforms such as 10x Genomics Visium HD, which is also based on an array-based design, offer bin sizes down to approximately 2 µm, approaching subcellular-scale resolution and substantially reducing cell mixing (Polański et al., 2024).

Overall, barcoding-based approaches enable transcriptome-wide profiling through capture oligonucleotides, but this comes with an inherent trade-off: as spatial resolution increases, data spar-

sity and variability also increase, introducing additional statistical challenges (Moses and Pachter, 2022; Sang-aram et al., 2024)

1.3.2 Imaging based spatial transcriptomics

Imaging-based technologies, in contrast, directly visualize RNA molecules within tissue sections using fluorescently labeled probes that hybridize to specific target transcripts (Moses and Pachter, 2022). Through repeated cycles of hybridization and imaging, these methods can detect and localize individual RNA molecules within cells at high spatial resolution (Moses and Pachter, 2022).

These approaches can be broadly categorized into two groups. First is in situ hybridization (ISH) based imaging methods and in situ sequencing (ISS) – based method. Early ISH approaches such as single-molecule FISH (smFISH) enabled the direct visualization of individual RNA transcripts (Femino et al., 1998). SeqFISH extended this principle by using sequential rounds of probe hybridization for highly multiplexed transcript detection (Lubeck et al., 2014). MERFISH similarly relies on successive rounds of hybridization and imaging but uses combinatorial, error-robust barcodes to identify RNA molecules (K. H. Chen et al., 2015). In contrast, ISS-based methods, like ISH-based methods, preserve transcript locations through microscopy but differ in identifying transcripts by in situ sequencing rather than target-specific probe hybridization (Du et al., 2023).

Gene-specific probes give imaging-based approaches their subcellular resolution and high sensitivity. However, a key limitation is that the number of detectable genes is typically restricted to a predefined panel, usually ranging from hundreds to a few thousand targets(Sang-aram et al., 2024). In addition, probe design requires prior knowledge of target sequences and careful optimization, which can limit applicability in less-characterized systems (Du et al., 2023). As a result, imaging-based methods provide high spatial resolution, but more limited transcriptome coverage compared to sequencing-based approaches, reflecting a trade-off between resolution and gene coverage (Moses and Pachter, 2022).

1.4 Spatial transcriptomics in GRN inference

Incorporating spatial information into GRN inference may allow tissue context to be considered when inferring regulatory programs. This spatial perspective is particularly relevant in heterogeneous tissues, where expression patterns can reflect local cell-type composition, microenvironmental cues, or region-specific signaling rather than a uniform regulatory program. For example, a TF-target association observed only within a tumor boundary, vascular niche, or developmental gradient may represent context-specific regulation that would be obscured if all locations were analyzed together (Y. Li et al., 2025).

1.4.1 Strategies for incorporating spatial information

Recent advances in spatial GRN inference have led to a growing number of methods that incorporate spatial information. However, these methods differ mainly in where spatial information are used in the analysis. For example, SCRIPro and SpaGRN use statistical strategies, although they introduce spatial context at different stages (Chang et al., 2022; Y. Li et al., 2025). SFINN and SpaGTL instead learn spatial representations through neural network models (W. Huang et al., 2024; Wang, F. Zhou, and Guan, 2024).

SCRIPro uses spatial information primarily to improve the stability of sparse single-cell and spatial data rather than to score regulatory relationships directly. Its main inputs are a gene expression matrix and spatial coordinates. Cells or spots with similar expression profiles and nearby coordinates are first grouped into locally informed SuperCells, and GRN inference is then performed at this aggregated level. For transcriptomic data without paired chromatin measurements, SCRIPro uses a LISA-based approach to reconstruct the likely chromatin landscape from public reference data (Qin et al., 2020). It then estimates how much the predicted regulatory potential changes when the binding signal of a transcriptional regulator is removed. This regulatory importance is combined with regulator and target expression to calculate transcriptional regulator activity and identify likely target genes. The main outputs are SuperCell-level regulator activity scores and regulator-centered regulatory relationships (Chang et al., 2022)

SpaGRN is also a statistical framework, but it brings spatial information more directly into GRN inference. It takes spatial gene expression profiles and coordinates as input, together with TF and motif reference databases. The aim is to identify regulatory relationships supported by both expression patterns and spatial organization. Spatial information is used in two stages. First, statistics such as Moran's I and Geary's C measure whether gene expression is spatially autocorrelated, meaning that nearby locations show related expression values. Getis-Ord G* detects local clusters of high or low expression, while Hotspot identifies genes with local spatial dependence. These analyses select genes with non-random spatial patterns. Next, bivariate Moran's I or bivariate Geary's C examines whether a TF and a potential target also vary together across neighboring locations (Y. Li et al., 2025)

Beyond these statistical strategies, another group of methods incorporates spatial context through neural representation learning. These methods consider not only the expression pattern of a gene pair but also the neighborhood structure formed by cells or spots in the tissue.

SFINN is a supervised model that uses gene expression profiles, spatial coordinates, and labeled gene pairs for training. Spatial information enters the model through a cell-cell adjacency graph. SFINN first constructs an expression-based graph from shared factor neighborhoods and a second graph from the Euclidean distances between spatial coordinates. The two graphs are then combined. This integrated graph and the expression profiles of each gene pair are processed by two parallel neural networks. A graph convolutional network learns how gene-pair expression patterns are distributed across cellular neighborhoods, while a fully connected network learns directly from the expression values of the pair. Their outputs are combined to classify gene interactions and predict regulatory direction. When the evaluated regulators

are TFs, the resulting directional relationships can be interpreted as candidate TF-target edges (Wang, F. Zhou, and Guan, 2024).

Unlike SFINN, which trains a supervised model using labeled gene pairs for the dataset being analyzed, SpaGTL transfers representations learned from a large external collection of spatial transcriptomic datasets to a target dataset through fine tuning. For high-resolution spatial data, local coordinate information can first be incorporated by reconstructing expression from spatial neighborhoods. SpaGTL then connects a gene-level relation model with a cell or spot level variational autoencoder. The gene level component represents potential gene relationships, whereas the autoencoder learns a lower dimensional representation of expression variation across locations. Alignment of these components allows the pretrained relation structure to be adapted to the target tissue. Fine tuning produces a context adapted gene relation matrix and reconstructed expression profiles that can subsequently be used for regulon inference and activity analysis (W. Huang et al., 2024).

1.5 Validation

The most reliable way to validate an inferred GRN is to compare it directly against an experimentally derived ground truth network constructed from data matched to the same cell type, tissue, and condition. However, constructing such a tissue-matched ground truth network is rarely feasible in practice. This practical limitation has led to validation frameworks that approximate ground truth comparison using partial reference data, curated databases, or benchmark datasets with predefined synthetic networks.

1.5.1 Partial reference data as a substitute for ground truth

In the absence of a complete, tissue-matched ground truth network, several types of partial reference data are commonly used to evaluate individual TF–target interactions.

Prior-knowledge resources such as CollecTRI, TRRUST, and DoRothEA provide reference TF–target interactions collected from the published literature and other regulatory data sources. TRRUST contains manually curated TF–target relationships for human and mouse, based on sentence-level text mining followed by expert review (H. Han et al., 2018). DoRothEA integrates several forms of evidence, including curated databases, TF-binding data, motif predictions, and reverse-engineered regulons, and assigns confidence levels to each TF–target interaction (Garcia-Alonso et al., 2019). CollecTRI is a human-centered meta-resource that combines interactions from multiple databases and text-mined literature. Evidence reported for mouse or rat genes is mapped to human orthologs during construction, and the resulting network can subsequently be translated to mouse or rat through orthologous gene pairs Müller-Dott (Müller-Dott et al., 2023). A predicted TF–target interaction can therefore be compared with these resources to assess its agreement with existing prior knowledge. However, agreement provides supporting evidence rather than definitive validation, since these resources are

incomplete and may not reflect the tissue or condition under study.

Binding data provide a different type of evidence. ChIP-seq experiments identify genomic regions bound by a TF (Furey, 2012). These binding regions are commonly linked to nearby genes to generate a TF–gene table, often by assigning each binding peak to the nearest transcription start site (TSS) within a predefined genomic window, such as 10 kb (Bravo González-Blas et al., 2023). A binding peak near a gene promoter or another regulatory region indicates that the TF is physically associated with a region that could influence the expression of that gene. ChIP-seq can therefore provide indirect support for a candidate TF–target relationship, although it does not establish that TF binding changes gene expression.

Perturbation experiments identify genes whose expression changes significantly after the specific TF is removed or reduced compared to a control condition. Knock TF 2.0 collects such perturbation data from TF and transcriptional cofactor knockdown or knockout experiments across multiple species, including human and mouse (Feng et al., 2024). By showing that alteration of a TF is followed by a measurable change in gene expression, perturbation data provides stronger evidence for regulation than ChIP-seq alone. However, this data does not confirm a direct regulatory relationship, as the observed response could be due to downstream regulators or other indirect effects.

Each of these data types captures a different aspect of regulation, namely curated prior knowledge, physical binding, and functional perturbation response. They are therefore used in combination rather than individually, providing complementary, though still partial, evidence for the reliability of inferred TF–target interactions.

1.5.2 Benchmark frameworks and validation metrics

Partial reference data cover only a subset of known regulatory interactions. Benchmark frameworks start with a regulatory network whose edges are already known. They then generate gene expression data from this network and compare the inferred edges with the original network. This reference network may be curated from well-characterized biological systems, or fully synthetic networks. A representative example is BEELINE (Pratapa et al., 2020), which includes curated Boolean models such as mCAD, a network of mammalian cortical area development, as well as fully synthetic networks.

GRN inference methods usually return a ranked list of candidate TF to target gene relationships. Predicted networks are evaluated using early precision, the early precision ratio, the area under the precision recall curve (AUPRC), and the area under the receiver operating characteristic curve (AUROC) (Pratapa et al., 2020).

Early precision focuses on the leading part of the ranking. It measures the proportion of reference interactions among the top k predicted pairs, where BEELINE sets k to the number of edges in the reference network. The early precision ratio shows how much higher this precision is than would be expected if the same number of TF to target gene pairs were selected at random.

The AUPRC and the AUROC both evaluate the full ranked edge list using the importance scores assigned by the inference method. The area under the precision recall curve summarizes how precision changes as additional predicted edges are included and recall increases. It therefore reflects whether the candidate list remains enriched for reference interactions as more edges are retrieved. The area under the receiver operating characteristic curve evaluates whether reference interactions generally receive higher importance scores than pairs that are absent from the reference network.

This difference between AUPRC and AUROC is particularly important for GRNs, where regulatory interactions represent only a small fraction of all possible gene pairs. The AUROC scales false predictions against the large number of edges that are not in the reference network. It can therefore remain high even when many incorrect edges appear early in the ranking. The AUPRC and early precision instead reflect the proportion of correct interactions among the edges selected for follow up. They are often more informative when evaluating GRN inference performance under strong class imbalance (Y. Li et al., 2025; Saito and Rehmsmeier, 2015).

1.5.3 Spatial evaluation

The approaches described above generally assess individual TF to target relationships by comparing predictions with external regulatory reference. This type of evaluation is useful for assessing predicted regulatory relationships, but it does not examine how regulatory activity is organized within a tissue.

Spatial evaluation provides a different axis of evidence. Rather than testing whether each predicted edge is correct, it examines whether regulon activity occurs in the parts of a tissue where the corresponding regulatory program would be biologically expected.

Evaluating spatial organization presents a different set of challenges. A gold standard regulon activity map is rarely available for the exact tissue section being analyzed. Independent spatial evidence often comes from a different section, animal, developmental stage, or experimental platform. Spatial technologies also differ in gene coverage, sensitivity, and resolution, ranging from multicellular spots to individual cells or subcellular locations (Walker et al., 2022; You et al., 2024). Spatial evaluation should therefore be understood as a complementary and largely indirect form of evidence.

The form of the comparison depends on the available spatial reference. When measurements originate from the same tissue section or can be registered to a common coordinate system, regulon activity and reference maps can be compared at corresponding spots, cells, pixels, or atlas voxels. Spatial registration supports comparisons across sections, platforms, and three dimensional atlases, but its accuracy depends on tissue similarity, section orientation, spatial resolution, and local deformation (Clifton et al., 2023). Exact coordinate matching is therefore often unreliable when datasets originate from different sections, animals, or platforms. Although registration can reduce positional disagreement, it cannot fully remove biological variation, sectioning differences, tissue deformation, or platform specific measurement effects.

When exact spatial alignment is not reliable, regulon activity can instead be summarized within corresponding anatomical regions and compared through its relative regional distribution. This approach is less sensitive to small registration errors and differences in resolution, but it loses finer spatial detail. Cheng et al. (2022), for example, mapped regulon activity across anatomically annotated regions of the postnatal mouse brain and identified regulatory programs associated with particular brain structures. This represents an assessment of anatomical consistency rather than an independent validation of the underlying regulatory relationships.

For mouse brain studies, the Allen Mouse Brain Atlas provides a useful source of external spatial evidence. The atlas contains genome wide in situ hybridization measurements registered to an anatomical reference framework, allowing gene expression to be examined through tissue images, spatial grids, or summaries of annotated brain regions (Lein et al., 2007). Regulon activity maps may be examined alongside the spatial expression pattern of the corresponding TF. These measurements are not quantitatively equivalent. TF in situ hybridization measures TF transcript abundance, whereas regulon activity summarizes the coordinated expression of the inferred target-gene set. However, they describe biologically connected components of a proposed regulatory process. TF transcript expression indicates where the regulator is transcribed and may be available to influence downstream gene expression, while target-gene expression can provide an indirect readout of the regulatory program associated with that TF (Aibar et al., 2017; Garcia-Alonso et al., 2019). Based on this relationship, similarity between the spatial distribution of TF expression and that of its inferred target-gene program can be interpreted as indirect evidence that the regulon is anatomically plausible. Studies have used this type of comparison. Luo et al. (2021), for example, identified cell state specific regulons during postnatal cerebellar development and examined the in situ hybridization patterns of selected TFs using the Allen Developing Mouse Brain Atlas. SCENIC studies have accordingly examined TF expression alongside target-derived regulon activity, while emphasizing that regulon activity may provide information not captured by TF expression alone (Aibar et al., 2017).

Evaluating a target-gene program reduces dependence on the potentially nonspecific pattern of a single target gene. However, such concordance does not demonstrate TF activation or confirm individual TF–target relationships. TF transcript abundance may differ from TF protein abundance or activity because of translation, protein stability, post-translational regulation, cofactors, and chromatin accessibility. Target-gene expression may also be influenced by other regulators, while shared cell-type composition or broad anatomical structure can produce similar spatial patterns without direct regulation.

1.6 Methods compared in this thesis

Three methods were selected to represent different uses of spatial information in GRN inference. GRNBoost2 uses expression alone, SpaGRN directly includes spatial statistics in edge scoring, and SpaGTL learns spatially informed gene relationships through transfer learning. Figure 1.3 summarizes the inputs, use of spatial information, and output of each method.

1.6.1 GRNBoost2: Expression-only inference

GRNBoost2 is a regression-based method that infers associations between candidate TFs and target genes from gene expression data (Moerman et al., 2019). Its main inputs are a gene expression matrix and a list of candidate TFs. Spatial coordinates are not used. Each cell or spot is therefore analyzed as an independent expression profile without considering its tissue location or neighboring observations.

GRNBoost2 does not learn the entire network in a single model. Instead, it builds a separate regression model for each target gene. The expression of the selected target is used as the response variable, while the expression values of candidate TFs measured in the same cells or spots are used as predictors. For example, when Gene A is selected as the target, its expression in each cell is predicted from the expression of TF1, TF2, and the remaining candidate TFs measured in that cell.

Each regression tree divides the observations according to TF expression. A TF can be selected as a splitting variable when separating observations with higher and lower expression of that TF improves the prediction of the target. Further divisions are then made using other TFs. This process creates groups with different TF expression patterns, and each final group is assigned an expected target expression value. Regression trees can therefore capture nonlinear patterns as well as simpler relationships.

A single tree is unlikely to explain all variation in target expression. GRNBoost2 adds trees in sequence. The first tree produces an initial prediction, and the next tree learns the remaining errors. Later trees continue to correct the parts that were overestimated or underestimated by the earlier model. This boosting process combines many simple trees into one gradient boosted regression model.

After training, GRNBoost2 calculates how much each candidate TF contributed to predicting the target. A TF receives a high importance score when it is repeatedly selected by the trees and substantially reduces prediction error. The procedure is repeated for every target gene. The final output is a ranked table of candidate TF to target associations and their importance scores. An importance score indicates how useful a TF was for predicting the expression of a target gene.

1.6.2 SpaGRN: Spatial-statistics-based inference

SpaGRN is a statistical method that combines gene expression with spatial coordinates to infer TF to target associations from spatial transcriptomic data (Y. Li et al., 2025). Its inputs are a gene expression matrix, the spatial coordinates of each cell, spot, or bin, and a list of candidate TFs. SpaGRN first connects nearby observations in a spatial neighborhood graph. This graph is represented by a spatial weight matrix W which defines neighboring locations and their relative influence.

Spatial information enters both main stages of SpaGRN inference directly. The first stage uses

W to identify genes with structured spatial expression patterns. Spatial autocorrelation is evaluated using statistics such as Hotspot, Moran's I, Geary's C, and Getis–Ord G. Moran's I measures similarity between nearby locations, while Geary's C instead emphasizes local differences between neighboring bins. Results from several methods can be combined to obtain a more conservative set of spatially variable genes.

The second stage calculates spatial co-expression between selected candidate TFs and potential target genes. The spatial weight matrix W is also used directly in this calculation. Bivariate Moran's I measures the association between TF expression at one location and weighted target expression at nearby locations. The score therefore incorporates both gene expression and information about which locations are spatial neighbors. Bivariate Geary's C evaluates a similar association but gives greater weight to local expression differences between connected locations.

This approach differs from ordinary co-expression analysis. A conventional correlation compares the expression profiles of two genes across the same observations. SpaGRN also considers where those observations are located. Two genes may show a strong overall correlation but receive weak spatial support when they are expressed in different tissue regions. In contrast, a TF and target gene that occur within the same region or across neighboring locations may receive a high spatial co-expression score.

The final output contains the evaluated combinations of selected TFs and target genes together with their spatial co-expression scores. Each score describes how strongly the expression patterns of the two genes are associated across the spatial neighborhood graph. SpaGRN therefore incorporates spatial information directly into both gene selection and TF to target scoring.

1.6.3 SpaGTL: Spatially constrained transfer learning

Spatially Aligned Graph Transfer Learning (SpaGTL) transfers gene relationships learned from a large collection of spatial transcriptomic data to a new dataset and adapts them to its expression and spatial structure (W. Huang et al., 2024). SpaGTL has approximately 100 million parameters and was pretrained using about 96 million cells and spots collected across different species, tissues, and experimental platforms. The pretrained weights provide the starting point for fine tuning on a new dataset.

The main inputs are a gene expression matrix, spatial coordinates, and pretrained weights. A candidate TF list is later used to extract potential TF to target relationships. The core architecture contains three parts. These are a gene level graph transformer, a manifold dominated variational autoencoder at the cell or spot level, and deep spatial distribution alignment.

The gene-level graph transformer treats genes as nodes and learns their relationships through attention weights. This produces a gene attention matrix S, in which each value indicates how information from one gene contributes to reconstructing the expression of another. The transformer applies S to the original expression data to generate an augmented expression matrix. This augmentation is based on learned gene–gene relationships rather than spatial neighbor-

hoods.

Spatial information is incorporated separately through a graph B constructed at the cell or spot level. SpaGTL calculates distances between locations and selects the nearest neighbors of each cell or spot. Physical distance determines which locations are connected, while similarity between their expression profiles contributes to the weight of each connection.

The original and augmented expression matrices are both passed to the manifold-dominated variational autoencoder. This module learns a shared cell or spot relation matrix, $Z \in R^{N \times N}$, where N is the number of cells or spots. Each value, Z_{ij}, represents the learned relationship between two locations based on the original and augmented expression views. Unlike B, which is constructed directly from spatial proximity, Z is learned during training and reflects both global expression relationships and local spatial constraints. The spatial graph B encourages nearby locations to remain strongly related in Z, allowing the learned cell or spot representation to preserve local spatial organization.

Deep spatial distribution alignment connects the representations learned from the two expression views. It first encourages their overall distributions to agree. It then compares the representations within the spatial neighborhoods defined by graph B. This step helps the original and augmented expression views retain the same local spatial structure. These spatial constraints are optimized together with the rest of the model and therefore also influence the gene level graph transformer. Spatial coordinates do not enter S directly, but they shape the final gene relationships through graph B, the variational autoencoder, and local spatial alignment.

After fine tuning, SpaGTL produces a gene attention matrix S adapted to the expression and spatial structure of the target dataset. The candidate TF list is used to extract combinations of TFs and potential target genes, and each combination is assigned an attention score. The final output is a ranked table of candidate TF to target associations.

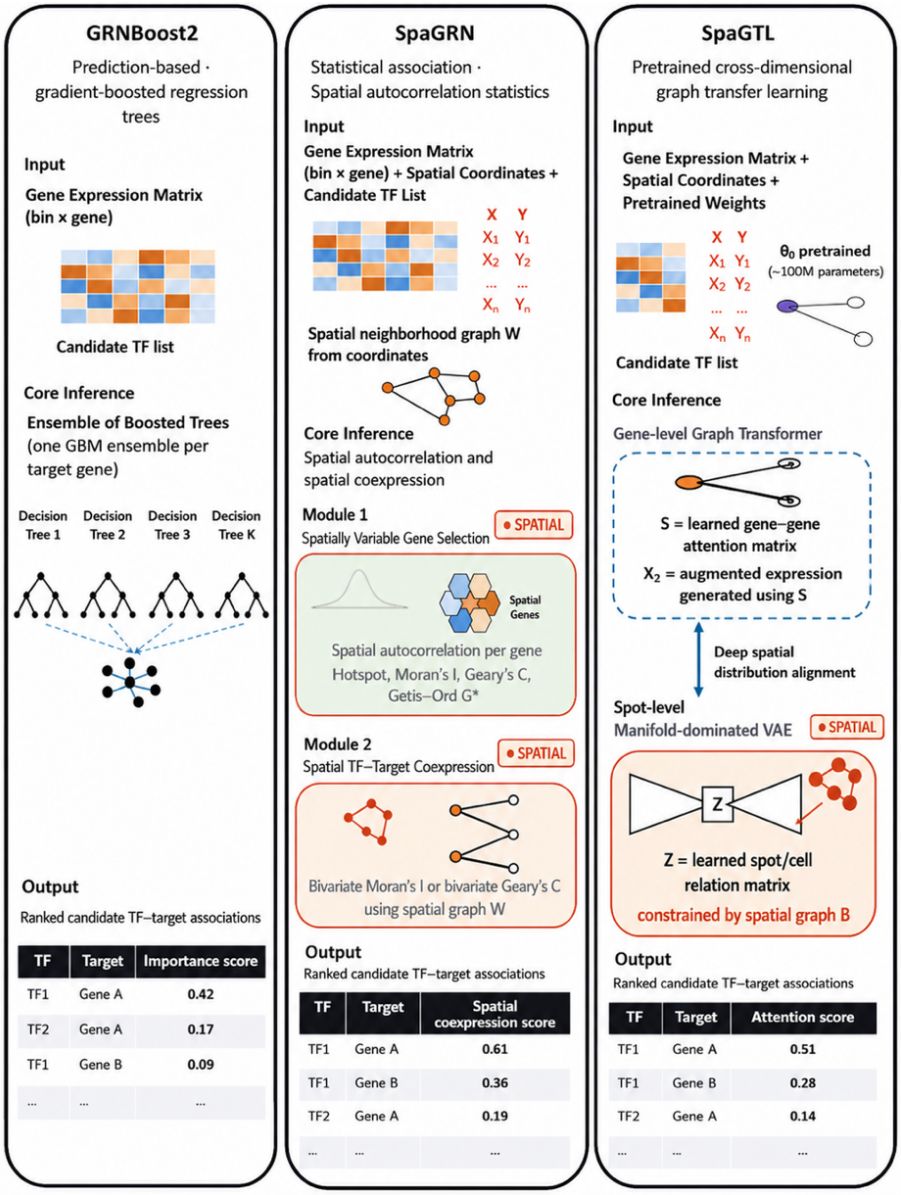

Figure 1.3: Overview of the inputs, use of spatial information, and outputs of the three GRN inference methods. *GRNBoost2 uses expression alone, SpaGRN applies spatial statistics to gene selection and edge scoring, and SpaGTL adapts pretrained gene relationships through spatially constrained fine tuning. All candidate edge tables undergo the same motif pruning and AUCell steps.*

Chapter 2

Aims

Gene regulatory networks, or GRNs, describe regulatory relationships among transcription factors, regulatory elements, and target genes. This thesis focuses on directed relationships between transcription factors and their target genes. These networks are also known as transcriptional regulatory networks, or TRNs. In this thesis, GRN inference specifically refers to the identification of candidate relationships between transcription factors and target genes. Traditional expression-based GRN inference methods rely on gene expression patterns observed across cells or measurement locations but do not directly consider their spatial positions within a tissue. Spatial transcriptomics combines gene expression measurements with tissue coordinates. Using spatial transcriptomic data for GRN inference provides additional spatial information and allows inferred networks to be interpreted in their spatial context.

Spatial GRN inference methods utilize spatial information in various ways. Some methods identify separate networks for different cell types or tissue regions. Others apply spatial statistics or local neighborhood structures to select or weight candidate regulatory relationships. Some approaches learn gene expression in conjunction with the tissue's spatial arrangement. These methods have been tested using both simulated and real spatial transcriptomic data. However, real data often lacks a complete reference network that corresponds to the tissue and biological condition being studied. The datasets, reference resources, and evaluation measures used in previous studies also differ, making comparisons of their reported performance challenging. A direct comparison of different GRN inference strategies using the same real spatial transcriptomic data is therefore necessary, along with an evaluation of their results across multiple criteria.

This thesis had two related aims. The first was to compare GRNs inferred by strategies that use spatial information in different ways, using the same real spatial transcriptomic data. The second was to examine how complementary evaluation criteria could be used together to assess differences among the inferred GRNs, and the limitations of their interpretation, in the absence of a complete ground-truth network.

Three different GRN inference methods, GRNBoost2, SpaGRN, and SpaGTL were compared using the same Visium HD mouse brain dataset to assess differences in their inferred networks.

The three methods were chosen to represent different strategies for utilizing spatial information. GRNBoost2 represented expression-based inference without spatial coordinates. SpaGRN involved the direct use of spatial statistics in network inference. SpaGTL combined pretraining, transfer learning, and spatially constrained representation learning.

The inferred networks were evaluated from three perspectives. The reproducibility across repeated runs, external support for individual transcription factor–target gene relationships, and the spatial properties of regulon activity.

Chapter 3

Materials and Methods

The analysis followed the five stages shown in fig. 3.1. The input dataset was a publicly available 10x Visium HD mouse brain section at 8 µm bin resolution. Quality control, annotation, and sampling formed one shared data-preparation stage that filtered bins and genes, assigned each retained bin a cell type label and a broad anatomical region, and reduced the section to a stratified 30,000-bin subset, yielding the shared, quality-controlled AnnData object supplied to every method (Section 3.1). GRN inference was then run separately with GRNBoost2, SpaGRN, and SpaGTL on that object, each returning a TF-target score table, with ten repeated runs for the two stochastic methods. All result tables from each three methods were passed through shared motif pruning and AUCell scoring, giving motif-pruned edges and a regulon activity matrix (Section 3.2). Validation last compared the resulting regulons in three categories, namely reproducibility, agreement with external TF-target databases, and spatial organization of regulon activity (Section 3.3). The subsections below describe these stages in the same order.

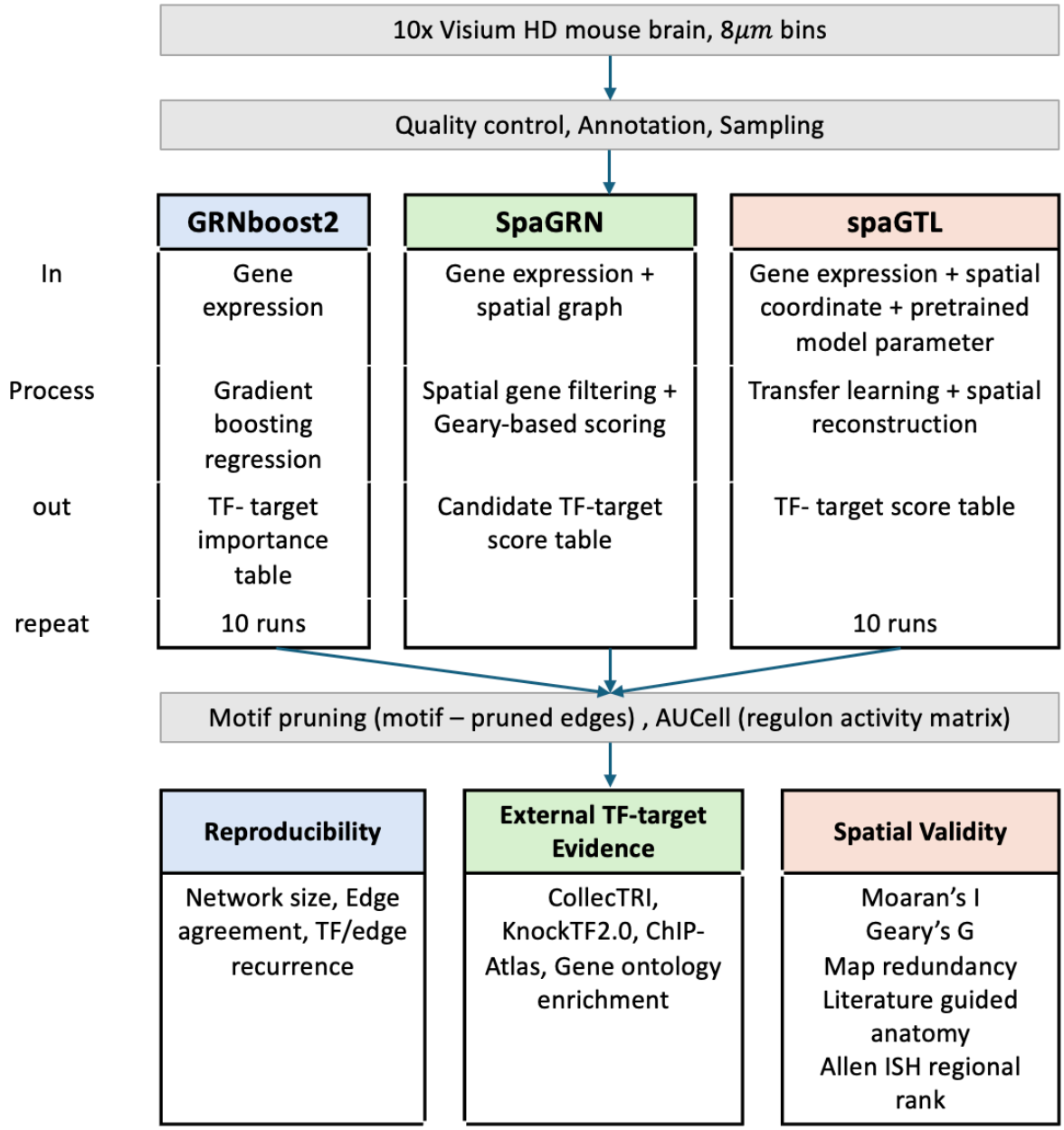

Figure 3.1: Study workflow from Visium HD preprocessing to GRN inference and validation. *Blue, green, and orange denote GRNBoost2, SpaGRN, and SpaGTL. All branches were converted to the same TF-target-importance format before cisTarget motif pruning, and AUCell scores were used for spatial validation.*

3.1 Data acquisition, preprocessing, annotation, and sampling

These steps produced the shared, quality-controlled AnnData object used for GRN inference. The following subsections describe how the tissue section was obtained, filtered, annotated, and reduced to a computationally tractable sample.

Data acquisition

The publicly available Visium HD Spatial Gene Expression mouse brain fresh frozen dataset was downloaded from 10x Genomics (10x Genomics, 2024). It was generated from a hematoxylin and eosin (H&E) imaged 10 µm coronal section from an 8 week old male C57BL/6 mouse and processed with Space Ranger version 3.1.1 (10x Genomics, 2024). The 8 µm binned output was used. Each bin aggregates sixteen native 2 µm squares, increasing transcript recovery while retaining finer spatial detail than the 16 µm output. This resolution provided a practical balance between transcript coverage, spatial detail, and computational cost.

Preprocessing

Visium HD bins are smaller and sparser than conventional Visium spots, and no single quality threshold applies across tissues. Filtering was therefore conservative and guided by spatial maps and distributions of total unique molecular identifier (UMI) counts, detected genes, and mitochondrial read fraction. Preprocessing was performed in Python with Scanpy (F. A. Wolf, Angerer, and Theis, 2018). The final criteria required at least 300 counts, at least 250 detected genes, and no more than 10% mitochondrial counts per bin.

After filtering, 378,185 of 453,797 bins were retained, corresponding to 83.3% of bins, and 16,051 of 19,059 genes were retained, corresponding to 84.2% of genes. Thus, the final filtered count matrix contained 378,185 bins and 16,051 genes.

Annotation

Labels for individual 8 µm bins were assigned using the Allen Institute's MapMyCells, a reference-mapping tool for annotation based on single-cell RNA-seq data. The Allen Institute's Whole Mouse Brain taxonomy, CCN20230722, was used as the reference (Daniel et al., 2026; Yao et al., 2023). Each Visium HD bin was treated as an individual query for cell-type annotation based on its gene expression profile. Assignments with a confidence score below 0.50 were marked as unresolved. Predicted cell classes were linked to anatomical region labels using a predefined mapping based on their known anatomical associations. The resulting labels were then placed back at the original Visium HD coordinates. The Allen Mouse Brain Atlas was used only as an anatomical reference for visual comparison, and no direct atlas registration or segmentation was performed. For spatial validation, the region labels were consolidated into three anatomical divisions.

Sampling

Repeated genome-scale inference was computationally prohibitive on all retained bins. A 30,000-bin subset was therefore sampled proportionally from strata defined by gene-expression-based clusters and broad anatomical regions. This sampling strategy was intended to preserve both transcriptional and anatomical heterogeneity more effectively than unstratified sampling. The

same sampled AnnData object was supplied to all methods.

3.2 GRN inference and regulon construction

All three methods started from the shared AnnData object and the same Aerts Lab mouse TF list (Aerts Lab, 2022c). No common highly variable gene (HVG) or spatially variable gene (SVG) prefilter was applied, so that all methods were compared on the same post-QC gene universe. An HVG prefilter can remove transcription factors and low-variance targets that may still be regulatory, and an SVG prefilter would have restricted every method, including the non-spatial GRNBoost2 baseline, to genes already selected for spatial structure.

GRNBoost2 was run as an expression-only baseline and SpaGRN and SpaGTL as spatially informed methods. The subsections below describe how each method was configured, from input representation through edge scoring, and how the resulting edge lists were pruned into motif-supported regulons.

3.2.1 GRNBoost2: Expression-only inference

GRNBoost2 was run through the Arboreto Python API (Moerman et al., 2019). The model principle is described in section 1.6.1. The sampled count-like expression matrix was supplied without normalization, log transformation, scaling, or highly variable gene selection. Candidate TFs were restricted to genes from the Aerts Lab list that were present in the matrix, while every retained gene was evaluated as a possible target. Spatial coordinates were not supplied. GRNBoost2 therefore served as the expression-only baseline and could not use bin proximity or tissue position during inference. For each target, stochastic gradient boosted regression predicted its expression from candidate TF expression and returned a candidate TF-target importance score. Default stochastic gradient boosting settings and an early stopping window of 25 iterations were used.

Target genes were divided into ten non-overlapping chunks and processed with Arboreto. All chunks within a run used the same random seed and were merged after checking that target assignments were complete and did not overlap. Ten independent runs used seeds 777 to 786. Self-loops were removed after merging, leaving a ranked table of candidate TF-target pairs, with TF, target, and importance columns, for each run.

3.2.2 SpaGRN: Spatial-statistics-based inference

SpaGRN was run through its Python API following the official high-resolution tutorial (Y. Li et al., 2025), with raw counts and x–y bin coordinates supplied as the spatial representation. Detailed model architecture is described in section 1.6.2. The spatial weight matrix W was built by connecting each bin to its ten nearest neighbors. In the first stage, selection of genes with structured spatial expression, SpaGRN applied W to test every gene for spatial autocorrelation

with Moran's I, Geary's C, and Getis-Ord G. A gene was kept only if it passed a false discovery rate below 0.05 under all three statistics, taken as an intersection rather than a union, so that only genes with spatial structure supported by all three tests entered the next stage.

Candidate TFs were taken from the Aerts Lab mouse TF list described above, and the genes retained in the first stage formed the target pool. Each TF-target pair then entered the second stage, spatial co-expression scoring, where the pair was scored with bivariate Geary's C using the same weight matrix W, which rewards a TF and a target that covary across neighboring bins rather than across bins in general. Three settings were fixed at this stage. The Bernoulli expression model treats each bin measurement as a detection event and serves as the null model against which spatial covariation is assessed. Local spatial calculation computes the statistic within each bin's own neighborhood rather than only across the section. Geary mode selects that statistic rather than bivariate Moran's I. The output contained every evaluated combination between a candidate TF and a spatially selected target gene, reported as TF, target, and importance columns, where the importance value is the spatial association score of the pair. It ranks candidate relationships but does not by itself establish direct regulation.

The optional ligand-receptor component was disabled by leaving the niche input empty. SpaGRN was deterministic under these settings and produced one candidate network.

3.2.3 SpaGTL: Spatially constrained transfer learning

SpaGTL was implemented with its high-resolution spatial reconstruction workflow (W. Huang et al., 2024). The model principle is described in section 1.6.3. During input preparation, genes with zero counts in every bin were removed. The remaining values were transformed with log1p, without library size normalization or highly variable gene selection. Spatial reconstruction then used the bin coordinates to identify ten nearest neighbors for each bin, while similarity among these neighbors was measured in a space of 15 principal components. A reconstruction weight of 8 controlled the contribution of expression aggregated from the spatial neighbors relative to the original expression of each bin. This produced the spatially reconstructed expression matrix supplied to the model. Genes were subsequently restricted to the pretrained gene set and ordered as expected by the model.

The pretrained model was then fine-tuned on this tissue section for 1,000 epochs. During this process, the gene relation module learned the gene attention matrix S from expression, while the internal spatial graph B constrained the shared bin representation Z. Spatial coordinates therefore did not enter S directly; instead, they influenced the learned gene relationships through the joint optimization of the gene and spatial components. The model used 128 hidden units and a 10-dimensional latent representation for Z. Batch correction and additional covariates were not used.

Each run produced a ranked table of candidate TF-target pairs. The positive, attention-derived QK value for each pair was reported as its importance score, and self-loops were removed. Ten independent runs used seeds 777 to 786 so that run-to-run variability could be quantified.

3.2.4 Motif pruning and AUCell scoring

Each final edge list was processed with the pySCENIC cisTarget workflow to identify motif-supported regulons (Van de Sande et al., 2020). Candidate TF-target relationships were grouped into TF-centered modules and evaluated for motif enrichment using mouse mm10/refseq_r80 gene-ranking databases and the corresponding mouse motif-to-TF annotation (Aerts Lab, 2022a; Aerts Lab, 2022b). Associations without motif support were removed, yielding final TF-centered regulons.

3.3 Validation and comparison of inferred networks

Every step described so far produces a network. Everything in this section is validation and each of the three parts asks a different question about the same set of inferred regulons. Reproducibility comes first. Database-based validation then compares the edges against external experimental evidence. Spatial validation examines whether regulon activity is organized across tissue in a way that matches known anatomy.

3.3.1 Reproducibility of stochastic inference pipelines

GRNBoost2 and SpaGTL both can return different networks from repeated runs. Reproducibility was therefore examined. Each method was run ten times with different random seeds while input data and parameters were held constant, giving 45 unique pairwise comparisons per method. As SpaGRN inference is a deterministic method, it was excluded from this assessment. All comparisons used motif-pruned edges.

Network size stability across runs

Network size was described by the number of TF-target edges, unique TFs, and unique target genes in each run. For each quantity, the mean, sample standard deviation, and coefficient of variation (CV) were calculated across ten runs. CV allows variability to be compared between methods whose network sizes differ.

$$\text{Coefficient of Variation (CV)} = \frac{\text{standard deviation}}{\text{mean}} * 100 \tag{3.1}$$

CV expresses variability relative to the mean and is reported as a percentage. Values start at 0, which would indicate identical network sizes in every run. No field-wide cutoff separates acceptable from unacceptable CV for inferred networks, so the values were read comparatively between the two stochastic methods rather than against a fixed standard.

Edge membership and rank stability across runs

Similar network size does not guarantee that the same edges were selected. Whether the ten runs of each method recovered the same edges, and ordered them in the same way, was assessed separately. Within each run, edges were ranked by reported importance score. Pairwise stability was evaluated across matched top-K edge sets, and direct comparison between methods was restricted to K values available in every run of both. The Jaccard index measures overlap in edge identity. For two equal-sized top-K sets A and B it was calculated as follows.

$$\text{Jaccard } (\text{run}_n\,(K)\,, \text{run}_{n+1}(\text{K})) = \frac{|\ \text{run}_n(\text{K})\ \cap \text{run}_{n+1}(\text{K})|}{|\ \text{run}_n(\text{K}) \cup\ \text{run}_{n+1}\,(K)\ |} \tag{3.2}$$

Jaccard ranges from 0 to 1. A value of 0 means two runs shared no edges at all, while 1 means they returned exactly the same edge set.

Spearman rank correlation was then calculated only for edges present in both top-K sets. It measures whether the common edges keep a similar order. Jaccard/shared fraction and conditional Spearman were reported together to distinguish membership stability from within-overlap ranking stability.

$$\rho_{\text{rank}}\,(K) = \text{Spearman(importance ranking of run}_n\text{(K)} \cap\ \text{run}_{\text{n+1}}\text{(K))} \tag{3.3}$$

Spearman ρ ranges from −1 to +1. Values near +1 indicate that shared edges were ranked almost identically in the two runs, values near 0 indicate that the shared edges carried unrelated importance orderings, and negative values indicate reversed ordering.

Target set consistency of recurrently detected transcription factors

A method may repeatedly recover the same TF but assign different target genes to it. Target set consistency was therefore evaluated at every TF occurrence threshold from 1 to 10 runs. For each threshold K from 1 to 10, the proportion of TFs detected in at least K runs was calculated. This indicated how many TFs remained detectable as the required number of repeated detections increased.

For each occurrence count K, Jaccard indices were calculated between the target sets from every pair of runs in which each eligible TF appeared and then averaged across run pairs. The resulting values were subsequently averaged across eligible TFs. Values near 1 indicate that a TF retained similar targets across runs, whereas values near 0 indicate substantial changes in its assigned targets.

3.3.2 Validation against external regulatory evidence

Reproducibility shows whether a method gives the same answer repeatedly, but not whether that answer is biologically correct. The inferred edges were therefore compared with three

independent types of experimental evidence. Curated interactions show whether an edge has been reported in the literature, perturbation data show whether removing a TF changes a target's expression, and binding data show whether a TF occupies DNA near a target gene.

CollecTRI holds literature-curated TF–target relationships for human, mouse, and rat (Müller-Dott et al., 2023). The mouse network was retrieved through OmniPath, where it is generated by orthology translation of the human reference network. This resource indicates whether an inferred edge has already been reported in the published literature.

KnockTF 2.0 collects genes whose expression changes significantly after a specific TF is knocked down or knocked out (Feng et al., 2024). Only pairs with $p < 0.05$ were retained, and independent experiments for the same TF were combined by union. Evidence of this kind is causal rather than correlational, since it records what happens to a target once its candidate regulator is removed.

ChIP-Atlas supplies physical DNA-binding evidence near target genes from integrated public ChIP-seq data (Oki et al., 2018). Target genes within ±5 kb of the transcription start site were collected for the inferred TF list. Absolute binding-score scales differ substantially between TFs, so targets were ranked within each TF and retained above a TF-specific knee identified by the Kneedle algorithm (Satopaa et al., 2011). Binding evidence shows that a TF occupies DNA near a gene, which is weaker than showing that it regulates that gene.

Eligible edge universe and prediction ranking

Precision and recall depend on what counts as a possible answer, so the comparison space was fixed before any metric was calculated. For each of the three resources, the possible edge space was defined as every pairwise combination of eligible TFs and genes, excluding self-loops, after restriction to genes present in the gene universe used for inference. True edges were defined as reference interactions falling inside that space.

Early precision, early precision ratio, recall, and area under the precision–recall curve

Predicted network size differed across methods by up to roughly twentyfold, so evaluation at one fixed number of edges could not represent each method fairly. Following the BEELINE evaluation framework, the number of predicted edges evaluated was cut equal to the number of edges in the reference network. Supplementary top-K thresholds of 500, 1,000, 2,000, 5,000, 10,000, 20,000, 50,000, and 100,000 (Pratapa et al., 2020), to see if there is any changes while low ranked edges are gradually added. Predicted edges were ranked in descending order of absolute importance, and all edges tied with the K-th ranked edge were retained. The actual number of selected edges could therefore exceed K, and early precision used the actual number selected as its denominator. Where a network held fewer than K eligible edges, all available eligible edges were evaluated. Early precision (EP) and early recall (ER) were calculated at each cutoff as follows.

$$\text{Early Precision } (\text{EP}) = \frac{\text{True positive edges among the selected predictions}}{\text{Actual number of selected predictions}} \tag{3.4}$$

$$\text{Early Recall } (\text{ER}) = \frac{\text{True positive edges among the selected predictions}}{\text{Total true edges in the filtered reference network}} \tag{3.5}$$

Both quantities range from 0 to 1. Early precision reports the proportion of selected predictions that are supported by the reference, so a value of 0.05 means one supported edge in twenty. Early recall reports the proportion of all reference edges that were recovered. Reading the two across increasing K shows how reference support changes as evaluation expands from the highest-ranked predictions toward broader and lower-ranked parts of the network.

As summary metrics, early precision and the early precision ratio (EPR) were also computed at a fixed cutoff, set to the size of each reference network. EP and EPR were calculated as follows.

$$\text{random baseline precision} = \frac{K}{\text{possible edges in reference network}} \tag{3.6}$$

$$\text{Early Precision Ratio } (\text{EPR}) = \frac{\text{EP}}{\text{random base line precision}} \tag{3.7}$$

EPR divides observed early precision by the precision expected from random edge selection in the same reference. A value of 1 indicates chance-level performance, values above 1 indicate enrichment for supported edges, and values below 1 indicate performance below random expectation.

AUPRC summarizes precision and recall across the full networks obtained by three different GRN inference methods. Higher values indicate better recovery. AUPRC ratio is then normalized to the random baseline.

$$N^{\text{AUPRC}}_{\text{possible edges}} = G * (G - 1) \tag{3.8}$$

Here G is the combined set of TF and target genes in the reference network. The candidate universe contains all directed gene pairs in G except self-loops, following BEELINE's own AUPRC implementation, which does not restrict the source of a candidate edge to known TFs (unlike EP/EPR above). A random predictor has expected precision equal to the prevalence of true edges in this universe, so its expected AUPRC is:

$$\text{random baseline AUPRC} = \frac{N^{\text{ref}}_{\text{true}}}{N^{\text{AUPRC}}_{\text{possible}}} \tag{3.9}$$

where $N^{\text{ref}}_{\text{true}}$ is the number of true reference edges. The normalized value was calculated as:

$$\text{AUPRC ratio} = \frac{\text{AUPRC}}{\text{random baseline AUPRC}} \tag{3.10}$$

An AUPRC ratio near 1 is indistinguishable from random prediction on that reference network. Larger values indicate stronger recovery relative to the reference-specific random baseline.

For GRNBoost2 and SpaGTL, all metrics were calculated separately for each of the ten runs and reported as the mean across runs. SpaGRN was evaluated on its single available network.

Gene Ontology biological process face validity

Before turning to the Gene Ontology enrichment, a single consensus network was built for each stochastic method. Building on the recurrence counts above, an edge was kept in the consensus network only if it appeared in at least five of the ten runs. The single deterministic SpaGRN network needed no such step and was carried forward directly. This five-run threshold was a pragmatic choice. The pySCENIC multi-run guidance suggests treating edges recurring in more than 80% of runs as high confidence (Van de Sande et al., 2020), but applying same threshold directly retained only a small subset of the inferred edges. These consensus networks, not any single run, are what the Gene Ontology enrichment part and the spatial evaluation part that follows it are evaluated on.

Exact edge matching can be too strict because a correct TF-target relationship may simply be missing from the reference. Gene Ontology (GO) biological process enrichment was therefore added as a qualitative complement. The top 20 regulons by this ranking were retained for each method. For each retained regulon, its top 100 targets by edge importance were queried against g:Profiler for GO:BP enrichment, using the full measured gene panel as the background (custom domain scope) and an FDR threshold of 0.05. The single most significant term for each regulon was kept (Raudvere et al., 2019). When multiple regulons within the same method matched the same top term, only the occurrence with the smallest p-value was retained and the rest were dropped, so that one recurring biological process would not be counted multiple times toward the same method's enrichment picture.

3.3.3 Spatial validation of regulon activity

Reproducibility and database support do not use the spatial layout of the tissue. Spatial validation therefore addressed three separate properties. It was carried out using the consensus networks for GRNBoost2 and SpaGTL and the single deterministic network for SpaGRN. Regulon activity per bin was computed using AUCell, as implemented in pySCENIC (Van de Sande et al., 2020).

These maps were subsequently evaluated in terms of internal spatial organization, distinctness between regulon activity patterns, and concordance with anatomical and external spatial evidence.

Internal spatial organization of regulon activity

The internal spatial analysis was intended to test whether differences in method design were reflected in the spatial organization of the final regulon activity maps. A binary K nearest neighbor graph was built from the spatial coordinates and made undirected. The weight was 1 when either bin belonged to the K nearest neighbors of the other, and 0 otherwise. Values of K equal to 5, 10, and 15 were tested. K equal to 10 was used for the main comparison.

Two complementary statistics were calculated for every regulon. Moran's I measures global spatial autocorrelation. Positive values indicate that bins with similar activity tend to occur near one another, while values near zero indicate little global structure. Geary's C places more weight on differences between neighboring bins. It has an expected value of 1 under spatial randomness and moves towards 0 as local similarity becomes stronger.

Permutation standardization placed the two statistics on a comparable scale. For each regulon, AUCell values were held fixed and randomly reassigned to bin locations 999 times, and the observed statistic was standardized against the resulting null distribution. The Geary's C z-score was sign-reversed so that larger values carry the same meaning for both measures.

$$z_{\text{Moran's I}} = \frac{I_{\text{observed}} - \text{mean}(I_{\text{perm}})}{\text{standard deviation}(I_{\text{perm}})} \tag{3.11}$$

$$z_{\text{Geary's C}} = \frac{\text{mean}\,(C_{\text{perm}}) - C_{\text{observed}}}{\text{standard deviation}(C_{\text{perm}})} \tag{3.12}$$

The standardized scores were treated as continuous measures of spatial organization. The conventional absolute z-score threshold of 1.96 marks the central 95% of a standard normal distribution and was reported as a descriptive reference only. With 30,000 bins, even very small spatial effects were enough to pass this threshold, so almost every regulon exceeded it. Method comparisons therefore relied on score magnitude rather than the number of nominally significant regulons.

Influence of regulon size on spatial coherence

Regulon size was examined as a potential confounder because differences in target count may affect AUCell activity profiles and the resulting spatial-coherence scores. First, within each method, the Spearman correlation between log-transformed target count and Moran's I z-score was calculated. Second, for the 22 TFs shared across all three methods, regulons were reduced to the smallest target count observed for the corresponding TF, retaining the highest-ranked targets. AUCell activity and Moran's I were then recalculated to determine whether differences between methods remained after controlling for regulon size.

Similarity among regulon activity maps

Map similarity was assessed to determine whether regulons from the same method retained distinct spatial activity patterns or repeatedly followed a shared tissue-wide pattern. For each method, pairwise correlations were calculated among the AUCell profiles of the 22 shared TFs across all bins. For each regulon i, R_i was defined as its median correlation with the other 21 regulon maps. Higher R_i values indicate greater similarity to other maps from the same method, whereas values near zero indicate greater spatial distinction. These scores were interpreted alongside spatial coherence and anatomical concordance because similar maps may reflect either shared biological programs or limited TF-specific spatial distinction.

Literature-guided anatomical assessment of individual regulons

To examine whether differences in spatial coherence translated into more anatomically interpretable activity patterns, maps for the 22 shared TFs were reviewed together with their spatial-coherence scores and relevant mouse brain literature. Nine post hoc examples were selected to illustrate three recurring patterns: increased spatial coherence accompanied by clearer anatomical compatibility, increased coherence without clear TF-specific anatomical support, and anatomically interpretable spatial patterns already recovered by expression-only-based GRNBoost2, with no consistent improvement from the spatially informed methods. These categories were descriptive rather than exhaustive and were not used for quantitative method ranking. Agreement with published evidence was interpreted as anatomical compatibility, not validation of a TF-specific regulatory domain.

External regional concordance using Allen in situ hybridization data

The preceding spatial analyses characterized the inferred regulon activity maps internally but did not assess their concordance with an external reference. Allen Mouse Brain Atlas in situ hybridization (ISH) data were therefore used to obtain broad regional expression profiles for the corresponding TFs (Lein et al., 2007). Allen ISH measures TF transcript expression, whereas AUCell summarizes the enrichment of the inferred target genes in each bin. The comparison assessed whether the TF transcript and its inferred downstream target program showed similar relative distributions across broad anatomical compatibility rather than validation of individual TF-target edges (Garcia-Alonso et al., 2019; Luo et al., 2021).

Comparison was performed at the level of three broad anatomical divisions, which are the cerebral cortex (CTX), the cerebral nuclei (CNU), and the interbrain (IB). Broad region labels assigned during annotation placed each bin into its division. Mean AUCell activity was calculated separately within CTX, CNU, and IB for each regulon, producing a three-region activity profile. TFs lacking eligible Allen data or missing a value for any region were excluded.

Spearman's rank correlation was calculated between the three-region AUCell profile and the corresponding Allen expression energy profile. A rank-based measure suits this comparison since AUCell and ISH expression sit on different absolute scales. With only three regions, Spear-

man ρ can take just four values, which are −1, −0.5, +0.5, and +1, so ties between methods are common. Therefore, a Pearson score reflecting the relative spacing between regions rather than their order alone was used as a secondary tie-break. This measure rests on the same three points, so it is also limited and was treated as supplementary evidence for separating otherwise tied methods rather than as an absolute measure of agreement.

Chapter 4

Results

The results below follow the order in which the analysis was carried out. A single Visium HD coronal mouse brain section was filtered, annotated, and reduced to a 30,000 bin sample. That sample was then given to three inference methods, GRNBoost2, SpaGRN, and SpaGTL, and each returned a set of motif pruned regulons. Section 4.1 reports what the input data looked like once quality filtering and annotation were complete. Section 4.2 reports the comparison itself, organized as three validation categories. Reproducibility comes first (section 4.2.1), followed by agreement with external evidence databases (section 4.2.2), and then spatial organization of regulon activity across the tissue (section 4.2.3).

4.1 Data quality and input characteristics

Quality filtering outcome

Quality control thresholds for spatial transcriptomics are not universal across tissues and datasets (Crowell et al., 2025). Library size per bin may also reflect biological differences in addition to technical quality (Bhuva et al., 2024). Published cutoff values were therefore not adopted directly. Instead, total UMI counts, detected genes, and mitochondrial percentage were used as the main quality control metrics, while the exact thresholds were selected from their observed distributions and the proportion of bins removed at each candidate value. The aim was not to achieve a fixed retention rate. Both the filtering effect and the resulting data loss were considered.

Total UMI count cutoffs of 200, 300, and 500 removed 4.4%, 13.0%, and 30.4% of bins, respectively. The effect was limited at 200, whereas the proportion removed increased markedly at 500. A cutoff of 300 was therefore selected as an intermediate value. Minimum detected-gene thresholds of 200, 250, and 300 genes per bin removed 7.0%, 12.1%, and 17.3% of bins, respectively. A minimum threshold of 250 detected genes per bin was selected to balance data retention with the exclusion of bins with low gene detection. Mitochondrial percentage cutoffs of 5%, 10%, and 15% removed 35.9%, 7.5%, and 1.4% of bins, respectively. The 5% cutoff

resulted in extensive removal, while the effect of the 15% cutoff was limited. The intermediate value of 10% was selected.

Applying all three criteria retained 378,185 of the original 453,797 bins, corresponding to 83.3% of the dataset. Genes detected in fewer than ten retained bins were then removed using a minimal prevalence criterion. This left 16,051 of the original 19,059 genes, or 84.2%. For context, another Visium HD study retained 50.7% of its 8 µm bins after applying criteria suited to its own dataset (Tong et al., 2026). This value was not used to validate the thresholds in the present analysis. It was included only as an example showing that retention rates can vary across tissues and datasets.

Anatomical annotation and grouping

Visium HD provided the gene expression measurements and spatial coordinates for each 8 µm bin analyzed in this study. Using MapMyCells, the expression profile of each bin were compared with the Allen Institute 10X Whole Mouse Brain reference taxonomy and assigned a transcriptomic cell class (Yao et al., 2023; Daniel et al., 2026). These cell classes were then converted into proxy region labels based on their known anatomical associations. Each label was displayed at the original spatial coordinates provided by Visium HD. It served as an independent anatomical reference for assessing whether the predicted locations and shapes of the regions agreed with known mouse brain anatomy.

Visual comparison with the Allen Mouse Brain Atlas indicated that the broad organization of the forebrain was recognizable. However, some differences remained. The striatum (STR) covered a larger area than shown in the reference atlas, extending into portions of the pallidum (PAL) and olfactory areas (OLF) territories. In contrast, the thalamus (TH) and hypothalamus (HY) were represented by smaller and more fragmented areas than expected.

These differences may partly arise from the mismatch between the MapMyCells reference and the spatial units measured by Visium HD. The MapMyCells reference taxonomy was constructed from expression profiles of individual cells. An 8 µm Visium HD bin, however, is a fixed spatial capture unit rather than a cell defined by segmentation. A bin may capture transcripts from only part of one cell or from several neighboring cells when it crosses a cellular boundary. Such partial or mixed expression profiles can make annotation less clear, especially near anatomical boundaries. Further uncertainty may arise when transcriptomic cell classes are converted into region labels, since a cell class is not always restricted to a single anatomical region.

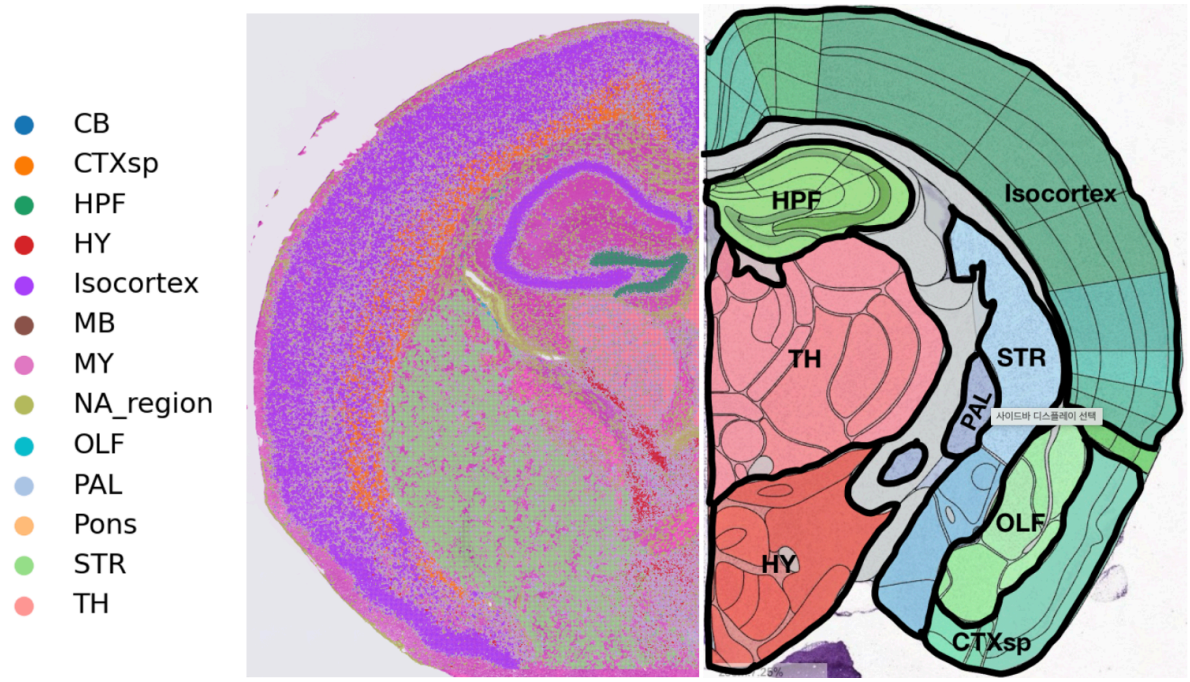

Figure 4.1: Comparison between MapMyCells-derived anatomical annotations and the Allen Mouse Brain Atlas. *Left: MapMyCells-derived broad anatomical region labels assigned to individual 8-µm bins.* ***Right:*** *Corresponding Allen Mouse Brain Atlas reference (Yao et al., 2023).*

To improve the robustness of downstream spatial validation, anatomically related regions were consolidated into three broad anatomical divisions according to the Allen Mouse Brain Atlas hierarchy: Cerebral Cortex (CTX), comprising Isocortex, CTXsp, HPF, and OLF; Cerebral Nuclei (CNU), comprising STR and PAL; and Interbrain (IB), comprising TH and HY (fig. 4.2). These broad anatomical groups preserved the major anatomical organization of the tissue while reducing the influence of bin-level annotation uncertainty during region-level comparisons.

The analyzed Visium HD tissue section contained predominantly forebrain structures. Thus, the midbrain (MB), cerebellum (CB), medulla (MY), and pons were represented by only a negligible number of annotated bins and were therefore excluded from the broad anatomical grouping used for downstream spatial validation.

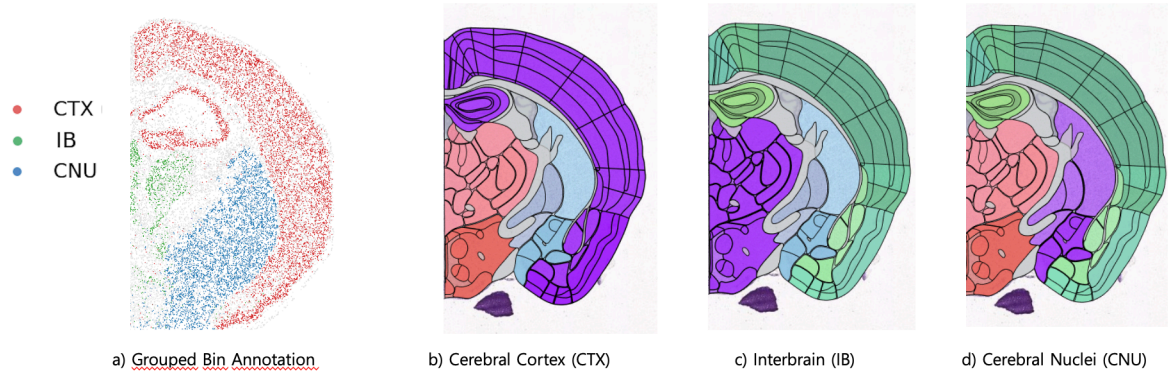

Figure 4.2: Broad anatomical grouping used for spatial validation. *(A) Broad anatomical groups assigned to individual 8-µm spatial bins after merging the original MapMyCells-derived annotations. (B–D) Correspondence between each broad anatomical division and the Allen Mouse Brain Atlas hierarchy: (B) Cerebral Cortex (CTX), (C) Interbrain (IB), and (D) Cerebral Nuclei (CNU).*

Sampling outcome

The filtered matrix remained too large for repeated genome scale network inference. The downloaded 10x Genomics dataset included graph-based cluster labels generated by Space Ranger v3.1.1 as part of the original vendor processing (10x Genomics, 2024) (Space Ranger, RRID: SCR_025848). Stratified sampling was used to select 30,000 bins in proportion to strata jointly defined by the vendor supplied Space Ranger cluster and the broad anatomical division. This approach retained representation from both transcriptomic clusters and anatomical regions. The same sampled matrix was used as input for all network inference methods.

4.2 Comparative evaluation of gene regulatory network inference methods

Using the sampled input generated in Section 4.1, transcription factor target edges were inferred with each of the three GRN methods and then processed through the same motif pruning and AUCell scoring workflow. Performance was evaluated in three categories. Reproducibility measured consistency across repeated runs. Data validation assessed the agreement between inferred edges and external reference data. Spatial evaluation examined the anatomical coherence of regulon activity.

4.2.1 Reproducibility of stochastic inference pipelines

Reproducibility was assessed for the two stochastic methods, GRNBoost2 and SpaGTL, using motif pruned outputs from ten seeded runs. The analysis examined whether repeated runs produced networks of similar size, recovered the same highly ranked regulatory relationships and repeatedly identified the same TFs and exact TF target edges. As SpaGRN is deterministic, it took no part in the reproducibility comparison and enters the analysis from the database

validation onward.

Network size across runs

This first comparison asks whether a stochastic method returns a network of comparable size each time it is run. Across ten runs GRNBoost2 produced considerably larger motif pruned networks than SpaGTL, holding roughly 3.7 times as many edges, 1.6 times as many transcription factors (TFs), and 3.1 times as many target genes (fig. 4.3; table 4.1). Size also held steadier from run to run. Coefficients of variation for GRNBoost2 were 7.7% for edge count, 2.0% for TF count, and 1.9% for target gene count. The corresponding values for SpaGTL were 20.3%, 18.6%, and 12.9%. SpaGTL therefore varied more in the size of the network it returned.

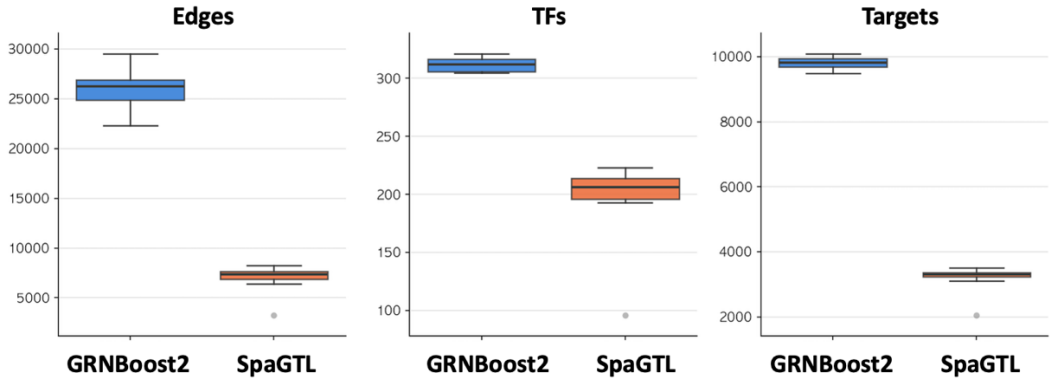

Figure 4.3: Run to run variability in network size after motif pruning. *(A) Edge, TF, and target gene counts obtained from ten runs of GRNBoost2 and SpaGTL. Box plots show the median and interquartile range across runs.*

Table 4.1: Summary statistics for network size across ten runs. *Mean, standard deviation, and coefficient of variation are reported for edge, transcription factor, and target gene counts for GRNBoost2 and SpaGTL. Lower coefficients of variation indicate greater consistency across runs.*

Method	Edges, mean ± Sd (CV)	TFs, mean ± Sd (CV)	Targets, mean ± Sd (CV)
GRNBoost2	25,940.4 ± 2,002.4 (7.7%)	312 ± 6.2 (2.0%)	9,814.5 ± 189.7 (1.9%)
SpaGTL	6,961.1 ± 1,409.7 (20.3%)	196.3 ± 36.6 (18.6%)	3,182 ± 409.9 (12.9%)

Consistency of edge selection and ranking across runs

This analysis asked whether each method selected similar edges and assigned them similar ranks across repeated runs. At matched top K thresholds, similarity in edge selection was measured using pairwise Jaccard indices, while rank consistency among shared edges was measured using Spearman correlations.

The two methods behaved in opposite directions (fig. 4.4). GRNBoost2 started low, with a pairwise Jaccard index near 0.12 at the smallest K, rose to about 0.31 at intermediate K, then

eased back to roughly 0.27 at the largest K. SpaGTL began higher at approximately 0.20 but fell quickly to a plateau near 0.08.

Rank stability showed a different pattern. The mean Spearman correlation for SpaGTL was close to 1.0 at the smallest K and remained above approximately 0.84 across the evaluated range, although it gradually decreased as K increased. GRNBoost2 started much lower, at approximately 0.20, but increased steadily to about 0.72 at the largest K. Thus, SpaGTL showed stronger rank stability among its highest ranked edges, but the overlap in edge membership decreased as the ranking expanded. GRNBoost2 showed limited agreement at the top of the ranking, while both edge overlap and rank consistency improved as more edges were included.

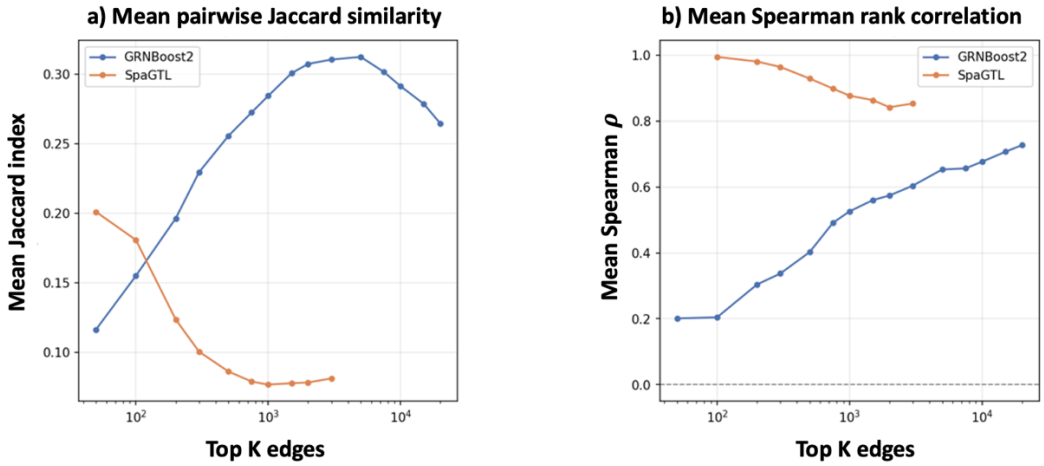

Figure 4.4: Pairwise top K edge membership and conditional rank stability across ten runs. *The x axis in both panels gives K, the number of highest ranked edges compared, on a logarithmic scale. (A) The y axis gives the mean Jaccard index between the top K edge sets of two runs, averaged over all 45 run pairs. Jaccard runs from 0 to 1, where 1 means two runs returned exactly the same edge set at that K and 0 means they shared no edges, so a rising curve indicates that agreement in edge identity improves as more of the ranking is included. (B) The y axis gives the mean Spearman rank correlation calculated only among edges present in both top K sets, again averaged over run pairs. This ranges from −1 to +1, where values near +1 mean the shared edges carried almost the same importance ordering in both runs. One line is drawn per method in each panel.*

Frequency of TF and exact edge recurrence across runs

The previous analysis compared edge selection and ranking within matched top K sets. This analysis asked whether repeated runs recovered the same regulators and whether those regulators retained the same target connections.

TFs recurred more consistently than exact TF target edges in both methods (fig. 4.5). At a threshold of five runs, 52% of GRNBoost2 TFs remained, compared with only 15% of its exact edges. The corresponding proportions for SpaGTL were 29% and 3%. When detection in all ten runs was required, GRNBoost2 retained 22% of its TFs and 2% of its edges. SpaGTL retained 4% of its TFs, while fewer than 0.5% of its edges remained. Thus, both methods repeatedly identified some of the same regulators, but the exact target connections assigned to those regulators were much less consistent.

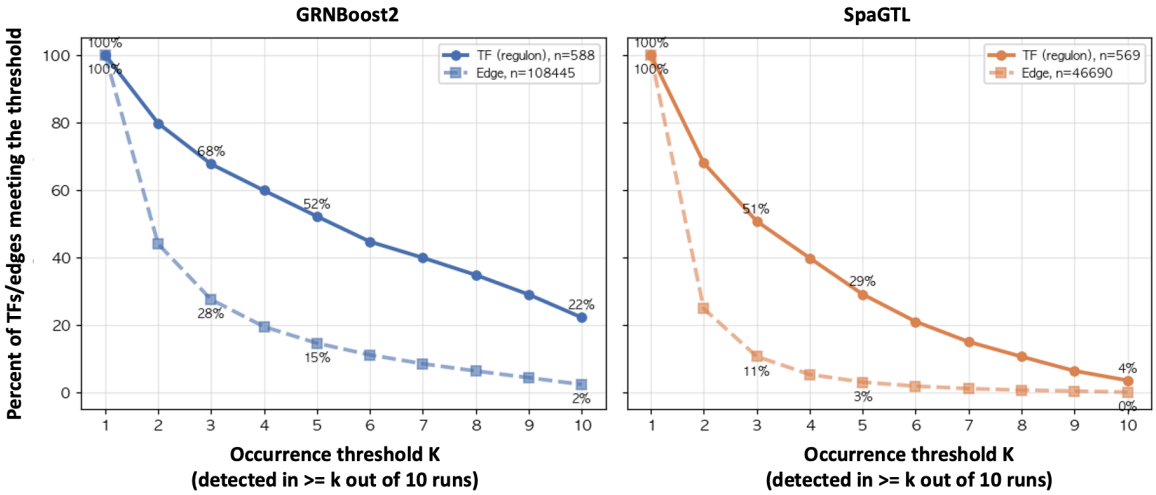

Figure 4.5: Cumulative survival of TFs and exact TF–target edges across occurrence thresholds. *The curves show the percentage of all unique TFs and exact TF target edges detected in at least K of the ten runs. Edge rank was not considered in this analysis. Higher values at stricter occurrence thresholds indicate greater recurrence across runs.*

The preceding analysis showed that 22% of GRNBoost2 TFs and 4% of SpaGTL TFs were detected in all ten runs. In contrast, only 2% and fewer than 0.5% of their exact edges, respectively, appeared in every run. Target set overlap was therefore examined to determine whether repeatedly detected TFs shared some of the same target genes across runs. TFs were grouped by the exact number of runs in which they were detected, and the mean target set Jaccard index was calculated for each group (fig. 4.6).

Target set overlap fluctuated without a clear trend among TFs detected in exactly two to five runs. From five runs onward, however, the mean Jaccard index increased consistently in both methods. GRNBoost2 rose from 0.084 at five runs to 0.231 at ten runs, while SpaGTL increased from 0.065 to 0.202. GRNBoost2 had higher mean Jaccard values at most occurrence counts. The exceptions were eight runs, where the methods were nearly identical, and nine runs, where SpaGTL was higher. Even among TFs detected in all ten runs, their target sets overlapped only partially. Recurrent detection of a TF therefore did not imply that the same targets were assigned in every run.

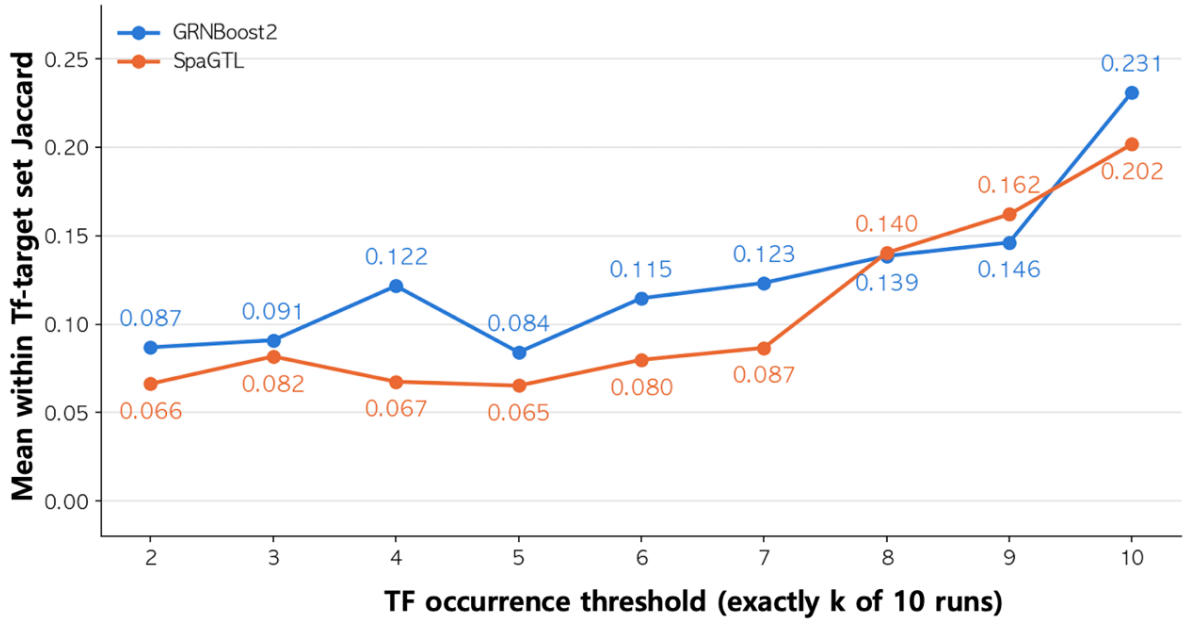

Figure 4.6: Target set consistency by exact TF occurrence count. *TFs were grouped by the exact number of runs in which they were detected, ranging from 2 to 10. For each TF, pairwise Jaccard indices were calculated between its target gene sets across the runs in which it appeared. These values were averaged across run pairs and then across TFs within each occurrence group. A value of 1 indicates identical target sets, whereas 0 indicates no shared targets. TFs detected in only one run were excluded because no pairwise comparison was possible. Lines represent GRNBoost2 and SpaGTL.*

Summary of reproducibility

Neither stochastic method produced exactly the same motif pruned network across the ten runs. GRNBoost2 maintained a more consistent overall network size. Its edge agreement increased up to an intermediate ranking depth before declining slightly, while rank consistency improved as more edges were included. SpaGTL showed greater edge overlap at the top of its ranking, but this overlap declined rapidly as K increased. Its rank consistency decreased gradually but remained relatively high.

Both methods recovered TF identity more consistently than exact TF target edges. Target set overlap varied among less recurrent TFs and improved among TFs detected more consistently across runs. However, even highly recurrent TFs retained only partly overlapping target sets. Repeated detection of the same TF therefore did not imply stable regulon composition.

A single run represents one possible inferred network rather than a fixed result. Subsequent interpretation was therefore based on information summarized across repeated runs. Database validation scores were averaged across runs, while consensus regulons were constructed for regulon activity analysis.

Reproducibility describes internal consistency and says nothing about whether the recovered edges are biologically accurate. The next section therefore examines agreement with independent regulatory evidence as a complementary measure of biological accuracy.

4.2.2 Validation against external regulatory evidence

Database based validation compared the inferred networks with CollecTRI, KnockTF, and ChIP Atlas (Müller-Dott et al., 2023; Feng et al., 2024; Oki et al., 2018). An evaluation framework based on BEELINE, a benchmark for GRN inference methods, was implemented to assess ranked edge recovery (Pratapa et al., 2020). Fixed top K cutoffs were used to trace early precision and recall as lower ranked edges were added. Separately, K_{ref} was defined as the number of edges in each reference network and was used to calculate summary EP and EPR values. AUPRC measured how effectively reference-supported edges were prioritized across different edge-score thresholds within full edges. GO enrichment assessed the functional coherence of the top consensus regulons.

Coverage of the reference networks

The three reference networks were filtered to retain only edges whose TF and target gene were both present in the gene universe used for GRN inference. After filtering, CollecTRI, KnockTF, and ChIP Atlas contained broadly similar numbers of edges, ranging from approximately 15,727 to 20,920. However, their TF and target gene coverage differed markedly (fig. 4.7). CollecTRI included approximately 4,000 genes and more than 550 TFs, providing the broadest TF coverage. KnockTF covered approximately 8,600 genes but only around 55 perturbed TFs. ChIP Atlas included approximately 9,000 genes and around 340 TFs.

Similar edge counts across the reference networks therefore did not indicate equivalent evaluation coverage. Each reference evaluated a different portion of the inferred networks according to the TFs and target genes it contained. Method performance was therefore compared within each reference rather than by directly comparing scores across different references.

The networks inferred by the three methods also differed substantially in size and composition. SpaGTL contained the fewest TFs and target genes and consequently had the fewest edges. SpaGRN included more TFs but fewer target genes than GRNBoost2. Nevertheless, SpaGRN contained substantially more edges than either of the other methods. This indicates that SpaGRN generated a denser network with more connections between TFs and target genes.

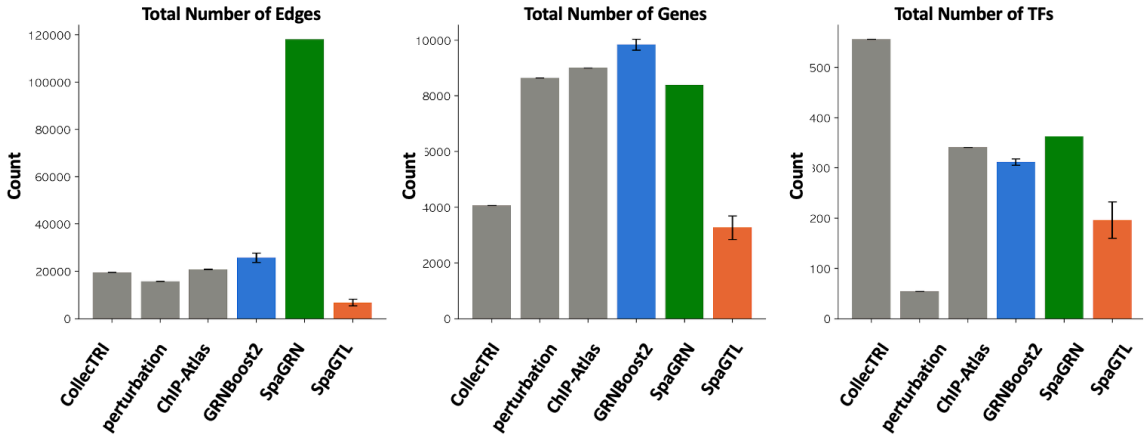

Figure 4.7: Predicted network size and filtered reference network coverage. *Bars show the numbers of edges, TFs, and unique target genes in each inferred and reference network. The y axis shows the corresponding raw counts. For GRNBoost2 and SpaGTL, bars represent the mean across ten runs and error bars show one standard deviation. SpaGRN represents a single deterministic run and therefore has no error bars. The reference networks were restricted to edges whose TF and target gene were present in the gene universe used for GRN inference.*

Precision and recall across ranking depth

Against CollecTRI, SpaGRN showed the highest early precision at $K = 500$ (fig. 4.8). Its lead over GRNBoost2 gradually narrowed as K increased. GRNBoost2 reached a higher precision at approximately $K = 5000$, suggesting that its reference support was maintained across a broader part of the ranking. In contrast, support for SpaGRN was more concentrated among its highest ranked edges. SpaGTL had the lowest early precision at every cutoff and showed relatively little change as K increased. CollecTRI supported edges were therefore not strongly concentrated at the top of the SpaGTL ranking.

A common pattern was also observed across all three methods. Early precision decreased as K increased, whereas early recall rose. Since the top K analysis accumulated predictions from the highest rank downward, this pattern indicates that CollecTRI supported edges were more concentrated among the highest ranked predictions. The increase in recall showed that some supported edges were also present at lower ranks. However, these lower ranked edges matched CollecTRI less frequently than the highest ranked edges. CollecTRI does not contain every true TF target relationship, so a lack of reference support does not necessarily indicate a biologically incorrect prediction.

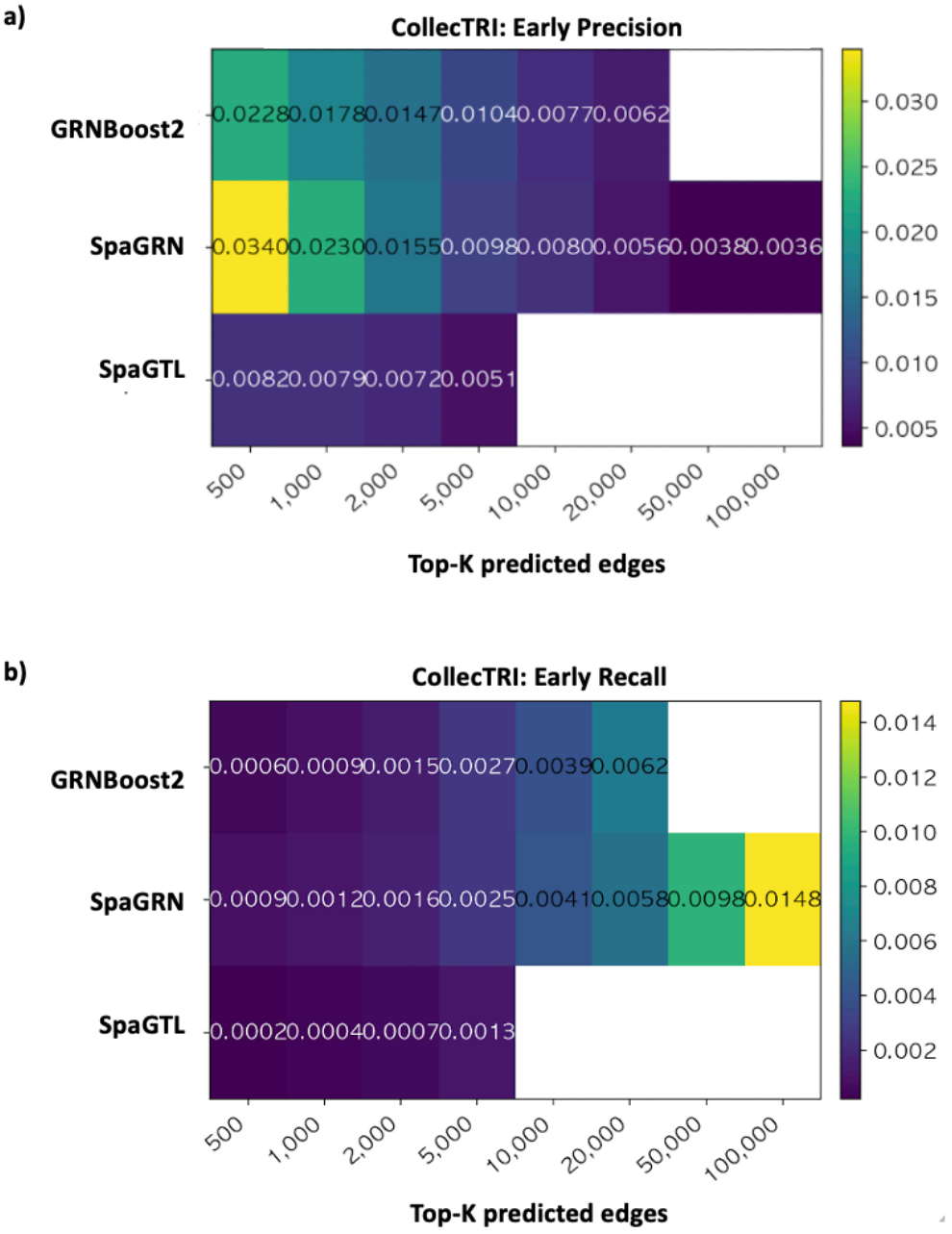

Figure 4.8: CollecTRI early precision and early recall across top-K predictions. *In both panels, columns show K and rows show the three methods; cell color and the printed value report the metric at each available cutoff. (A) The upper panel shows early precision, the proportion of selected predictions supported by CollecTRI. (B) The lower panel shows early recall, the proportion of CollecTRI edges recovered. White cells indicate cutoffs larger than the eligible predicted network.*

The KnockTF perturbation benchmark produced a different ordering of the methods (fig. 4.9). GRNBoost2 had the highest early precision at $K = 500$, reaching 0.0504. Its precision then declined and remained between 0.0395 and 0.0396 from $K = 2000$ onward. SpaGRN started below GRNBoost2 but rose to approximately 0.0452 at $K = 5000$, where it took the lead. Its precision remained close to 0.0455 at larger cutoffs. SpaGTL had the lowest early precision at every shared cutoff.

The methods also differed in early recall. GRNBoost2 initially performed better than SpaGRN, but SpaGRN moved ahead at $K = 2000$. GRNBoost2 then reached a plateau near 0.0046, whereas SpaGRN increased sharply to approximately 0.029. SpaGTL had the lowest recall throughout the evaluation and showed only a small increase.

Changes across K followed different patterns. For GRNBoost2, early precision declined while recall increased and then reached a plateau. This suggests that perturbation supported edges were relatively concentrated near the top of its ranking, with few additional reference edges recovered beyond a certain depth. In contrast, both early precision and recall increased for SpaGRN after $K = 1000$. Its perturbation supported edges were therefore not limited to the highest ranks. Expanding the ranking increased both the number and the proportion of supported predictions. SpaGTL showed small increases in both measures, but its overall support remained low.

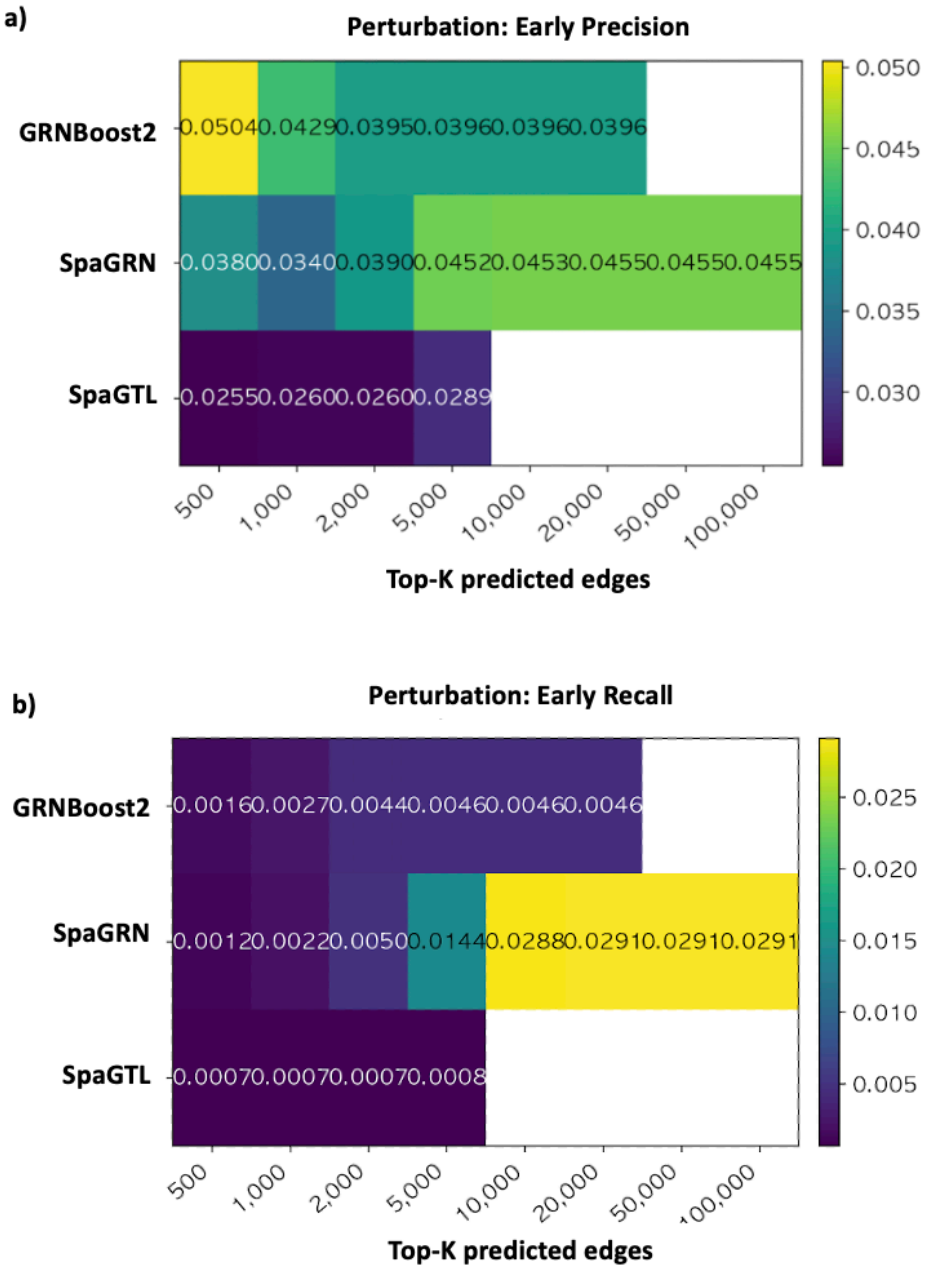

Figure 4.9: KnockTF perturbation early precision and early recall across top-K predictions. *Columns, rows, color scale, printed values, and white cells are interpreted as in Figure 4.8. (A) The upper panel shows early precision. (B) The lower panel shows early recall. KnockTF covers only around 55 perturbed TFs, so the evaluable fraction and recall remain low for all methods.*

Against ChIP Atlas, GRNBoost2 had the highest early precision at every cutoff shared by all three methods (fig. 4.10). Its precision increased from 0.0102 at $K = 500$ to a peak of 0.0119 at $K = 2000$, then gradually declined. SpaGTL had higher precision than SpaGRN from $K = 500$ to $K = 2000$ but SpaGRN moved ahead at $K = 5000$. GRNBoost2 also showed the highest early recall across the shared cutoffs. At smaller values of K, SpaGTL recall was similar to or slightly higher than that of SpaGRN. This pattern reversed at $K = 5000$.

Changes across K differed among the methods. For GRNBoost2, early precision increased until $K = 2000$ and then declined, while recall continued to rise. ChIP Atlas supported edges were therefore relatively concentrated within its top 2,000 predictions. Adding lower ranked edges recovered further reference interactions, but at a lower rate. SpaGRN showed low precision within its top 2,000 edges, followed by an increase at $K = 5000$. Precision then remained between approximately 0.005 and 0.006, while recall continued to rise and reached 0.0189 at the largest cutoff. Binding support for SpaGRN was therefore spread across a broader part of its ranking rather than concentrated at the very top. SpaGTL reached its highest precision at $K = 1000$ after which precision declined and recall increased slightly. This indicates that some of its ChIP Atlas supported edges occurred near the top of the ranking, although its overall recovery remained limited.

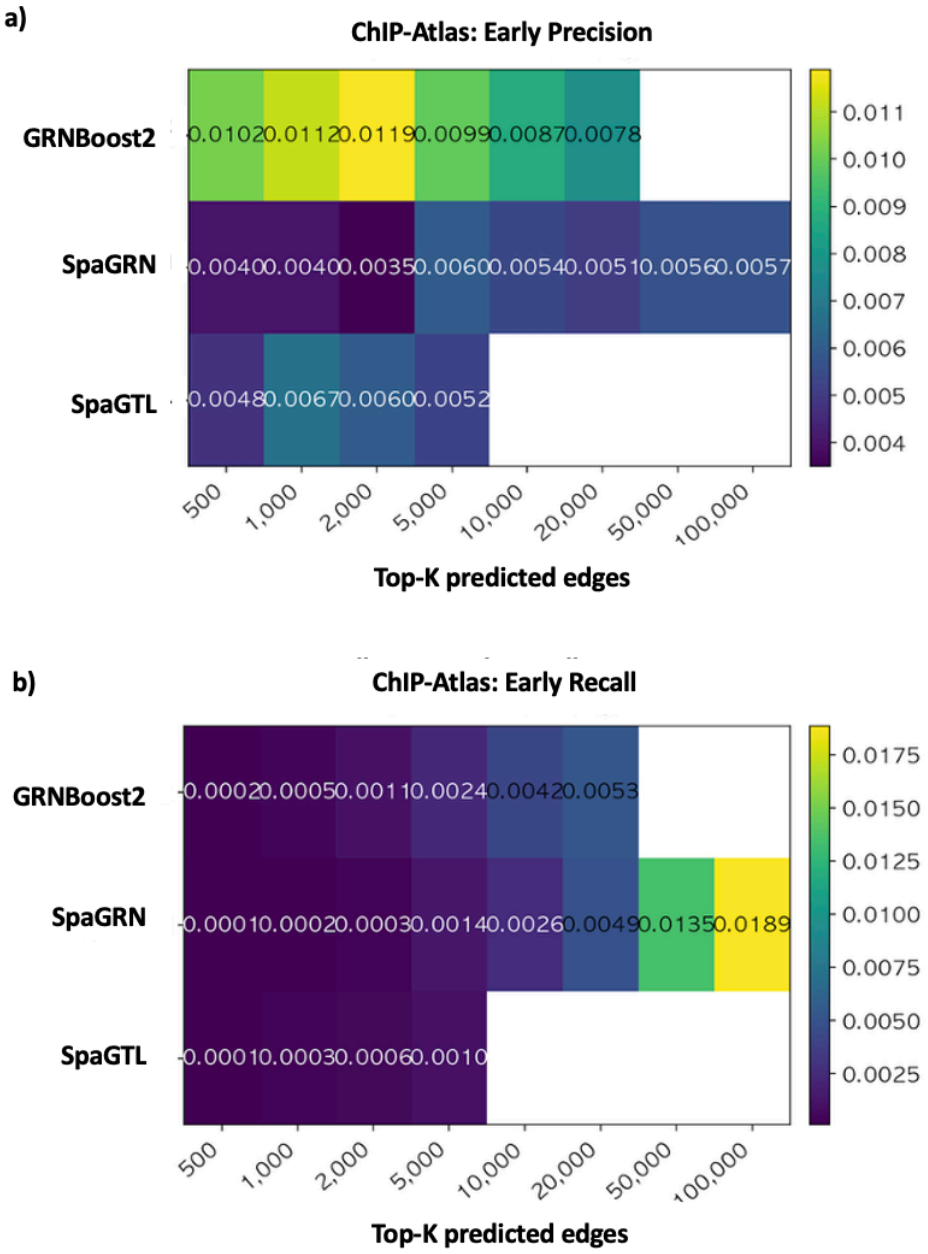

Figure 4.10: ChIP-Atlas early precision and early recall across top-K predictions. *Columns, rows, color scale, printed values, and white cells are interpreted as in Figure 4.9. (A) The upper panel shows early precision. (B) The lower panel shows early recall.*

Summary metrics at the reference-size cutoff

The preceding top K analysis ranked the TF target edges inferred by each method according to absolute edge importance. The value of K was then increased step by step to examine how reference support changed as lower ranked edges were added. This showed whether supported edges were concentrated near the top of the ranking or distributed across a broader range. fig. 4.11 provides a separate summary of overall validation performance at a fixed cutoff matched to the size of each reference database and across the full ranking.

Early precision and EPR were calculated at K_{ref}, where the cutoff matched the number of edges in each reference network. Early precision gives the proportion of the top K_{ref} predictions that matched the reference. EPR compares this value with the precision expected from random edge selection within the same candidate edge space. An EPR above 1 therefore indicates that the predictions contained a higher proportion of reference edges than expected at random.

AUPRC summarized the relationship between precision and recall across the full TF target edge ranking rather than at a single cutoff. The observed AUPRC was divided by its random baseline within the same candidate edge space to obtain the normalized AUPRC, reported here as the AUPRC ratio.

The ordering of the methods by early precision and EPR differed across references. For CollecTRI, the order was GRNBoost2, SpaGRN, and SpaGTL. For KnockTF, SpaGRN ranked first, followed by GRNBoost2 and SpaGTL. ChIP Atlas produced another ordering, with GRNBoost2 first, SpaGTL second, and SpaGRN third. The method with the strongest support among its upper ranked edges therefore depended on the type of regulatory evidence. Absolute EP values remained very low. Nevertheless, all three methods achieved EPRs above 1 across all reference datasets, although the degree of enrichment was modest, with values only slightly exceeding 1.

The leading method based on AUPRC and the AUPRC ratio also depended on the reference. Within each reference, however, the method with the highest AUPRC was the same method that ranked first by EPR. GRNBoost2 led against CollecTRI and ChIP Atlas, with AUPRC ratios of 10.1 and 15.4, respectively. SpaGRN led against KnockTF with a ratio of 28.4. Across all combinations of methods and references, AUPRC ratios ranged from 6.7 to 28.4 and remained above the random baseline of 1. The fixed cutoff and full ranking evaluations therefore identified the same leading method within each reference. All three methods also recovered reference interactions above random expectation across their full edge rankings.

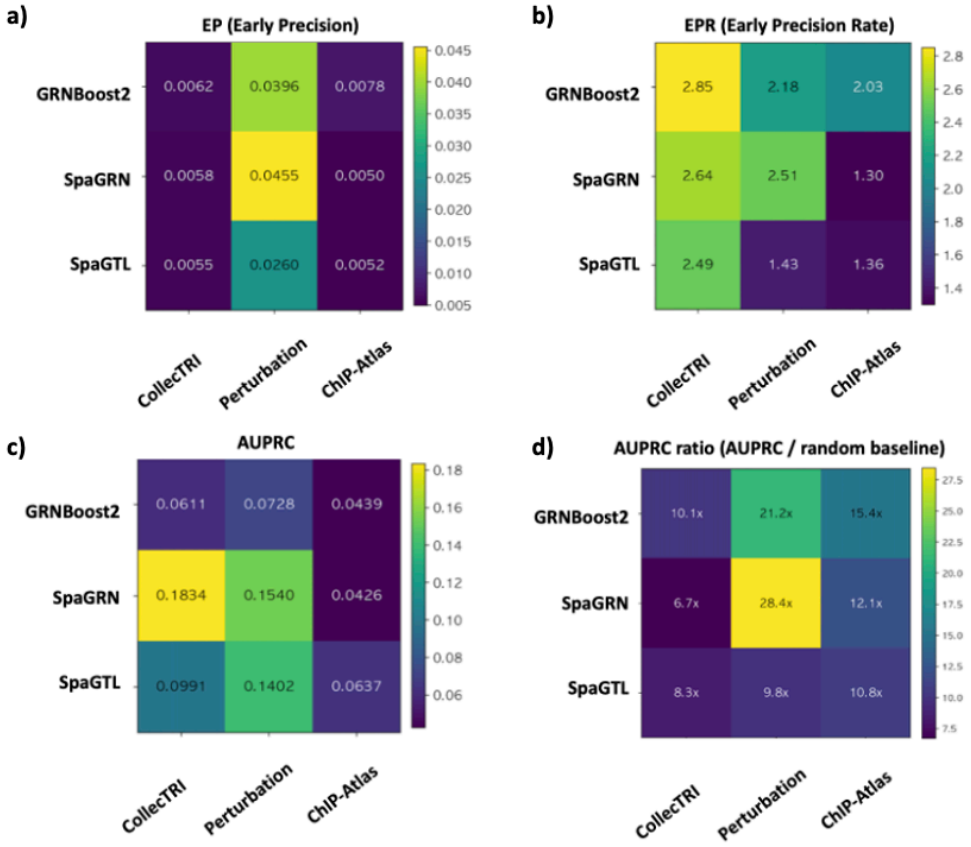

Figure 4.11: Summary metrics for the three external evidence resources. *Rows denote methods and columns denote CollecTRI, perturbation, and ChIP-Atlas; cell color and printed values show metric magnitude. (A) Early precision at K equal to the filtered reference size. (B) Early precision ratio, where 1 indicates random expectation. (C) Macro AUPRC across TF-level precision-recall curves. (D) AUPRC divided by the random baseline; values above 1 indicate better-than-random recovery.*

Functional coherence of top regulons

All three methods had brain-relevant biological processes represented among their 20 highest-ranked regulons, although the annotated functions varied across ranks (fig. 4.12). In GRNBoost2, top-ranked regulons were enriched for terms related to axon myelination and neural development, whereas broader developmental and cellular processes appeared at lower ranks. SpaGRN regulons in the upper and middle ranks were enriched for synaptic transmission and synaptic vesicle-related processes, while ion transport and glutamatergic signaling were represented at lower ranks. The highest-ranked SpaGTL regulons were enriched for neuronal ensheathment and synaptic signaling, followed by membrane potential regulation and glutamate receptor signaling. Some regulators shared across methods showed broadly concordant functional annotations. Sox10 regulons were associated with myelination-related terms in both GRNBoost2 and SpaGTL, consistent with its established role in oligodendrocyte differentiation and myelin gene regulation (Hornig et al., 2013). Although the Nkx6-2 regulons received different representative terms in GRNBoost2 and SpaGRN, both were related to myelinated axons or paranodal organization, consistent with its established role in CNS myelin paranode organization (Southwood et al., 2004). Other shared transcription factors were associated with different representative functions across methods, suggesting possible differences in their inferred target sets. However, functional annotation alone was insufficient to establish such differences.

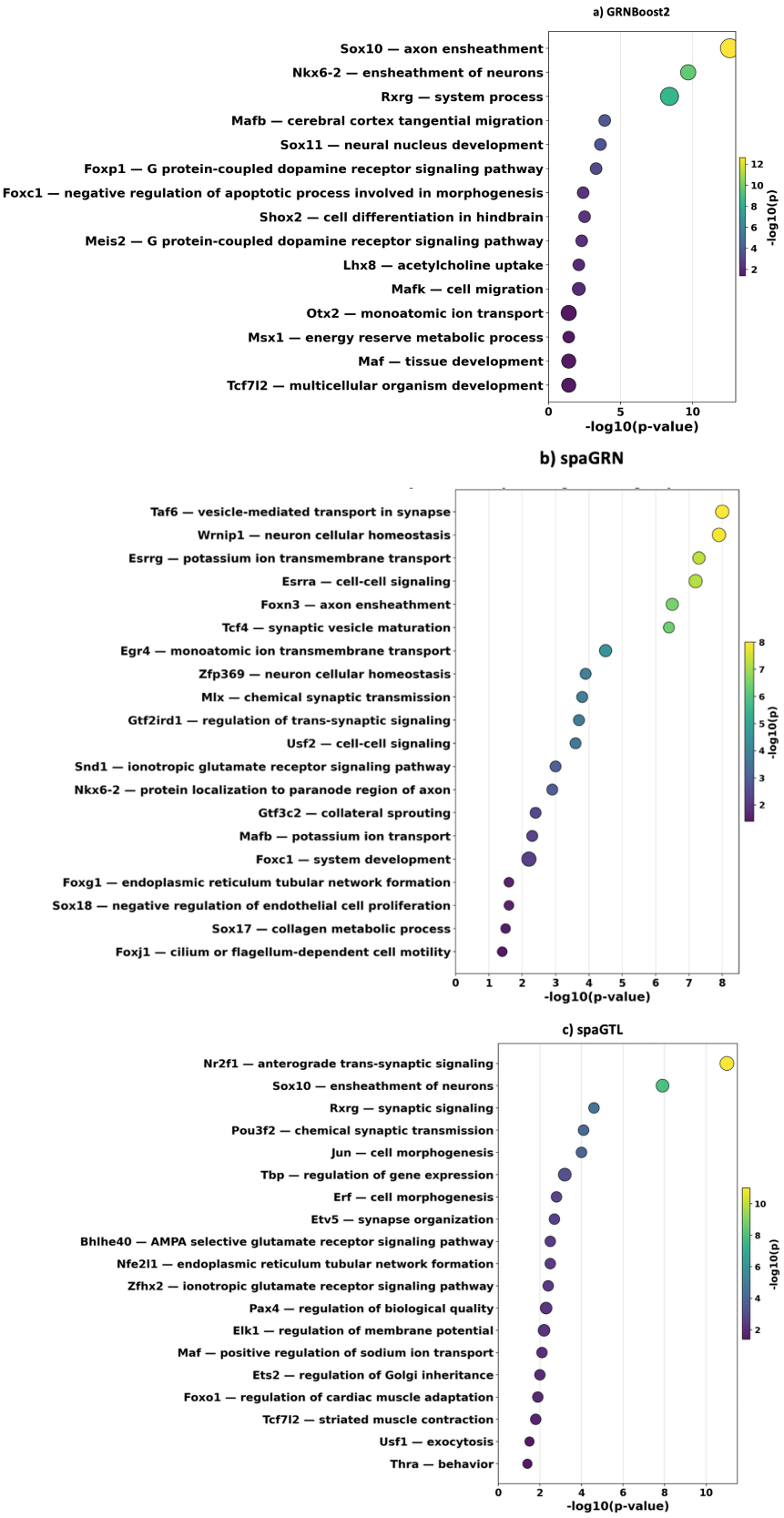

Figure 4.12: Gene Ontology biological process enrichment of the top regulons ranked by mean target importance. *Each point represents one regulon and its most significant enriched Gene Ontology term, with regulons ordered by mean target importance. Bubble size indicates the number of overlapping target genes, while color represents the adjusted p value, with stronger color indicating greater significance.*

Summary of database-based validation

The three types of regulatory evidence did not identify a single overall winner. When each method was evaluated using a number of top ranked predictions equal to the number of edges in the corresponding reference, GRNBoost2 had the highest EPR for CollecTRI and ChIP Atlas. SpaGRN led against KnockTF. The AUPRC ratio across the full edge ranking identified the same leading methods for each reference. EPR and the AUPRC ratio exceeded the random baseline of 1 for every combination of method and reference. SpaGTL did not lead against any reference and often produced the lowest values. Nevertheless, both normalized metrics remained above 1 in every case, showing that SpaGTL also performed better than random prediction.

Adding lower ranked edges generally recovered more reference interactions and therefore increased recall. Precision, however, usually declined or remained at a similar level. Reference supported edges were therefore present at lower ranks but occurred at a higher proportion near the top of the ranking.

GO enrichment identified biological processes relevant to the brain among the top regulons of all three methods. This indicates that their target genes formed functionally interpretable groups.

CollecTRI, KnockTF, and ChIP Atlas are not complete ground truths containing every true TF target relationship. Each reference represents a different type of evidence and covers a limited set of TFs and target genes. An edge that matches a reference can therefore be considered externally supported, but an unmatched edge is not necessarily incorrect. These evaluations indicate the level of support available from current reference resources rather than the absolute accuracy of each method. The reference networks and GO enrichment also contain no spatial information. The following analyses therefore examine whether regulon activity is coherent within tissue space.

4.2.3 Spatial validation of regulon activity

Whereas database-based validation assessed whether inferred TF–target relationships were supported by existing regulatory resources, it did not evaluate how the resulting regulon activity was distributed across the tissue. Spatial validation therefore examined how strongly similar activity values clustered across neighboring bins, using Moran's I and Geary's C, and whether different regulons produced distinct spatial patterns. Regulon size was evaluated as a potential influence on this spatial clustering, while individual maps were inspected for compatibility with known anatomical and functional contexts. Finally, broad regional activity patterns were compared with Allen ISH TF-expression profiles to provide an external spatial reference.

Internal spatial organization of regulon activity

The first spatial analysis examined whether the final regulon activity maps produced by the three methods differed in the degree of spatial clustering. Gene expression in the brain is spatially structured by anatomical regions, cell-type composition, and local tissue context. Therefore, if an inferred regulon represents a program associated with these biological processes, bins with similar AUCell activity would be expected to occur near one another rather than being scattered across the tissue. Moran's I and Geary's C were used to quantify this spatial clustering.

Spatial neighborhoods were defined using the K nearest bins, with $K = 5$, 10, and 15 tested. Moran's I and Geary's C were then used to quantify the strength of spatial clustering at each neighborhood size. At the main setting of $K = 10$, 97.7% of GRNBoost2 regulons and all SpaGRN and SpaGTL regulons exceeded an absolute z-score of 1.96. Because nearly all regulons exceeded this threshold, the proportion above the threshold did not distinguish the methods; differences were instead evident in score magnitude. Median Moran's I z-scores at $K = 10$ were 49.7 for GRNBoost2, 180.7 for SpaGRN, and 162.7 for SpaGTL. The same method order was observed at $K = 5$ and $K = 15$ as well as for the sign-reversed Geary's C z-score (table 4.2 and fig. 4.13).

Table 4.2: Spatial organization across neighborhood sizes. *TFs give the number of eligible regulons in each method. Significant percentages use an absolute z-score above 1.96. Median zI is the Moran's I z-score and median zC is the sign reversed Geary's C z-score. Larger values indicate stronger spatial organization for both measures.*

Method	K	TFs	% sig. Moran	Median zI	% sig. Geary	Median zC
GRNBoost2	5	133	97.0	43.4	96.2	26.7
GRNBoost2	10	133	97.7	49.7	97.0	31.7
GRNBoost2	15	133	99.2	55.2	97.7	36.4
SpaGRN	5	318	100.0	141.3	100.0	124.9
SpaGRN	10	318	100.0	180.7	100.0	159.8
SpaGRN	15	318	100.0	206.5	100.0	179.0
SpaGTL	5	36	100.0	123.9	100.0	115.8
SpaGTL	10	36	100.0	162.7	100.0	149.2
SpaGTL	15	36	100.0	189.5	100.0	175.0

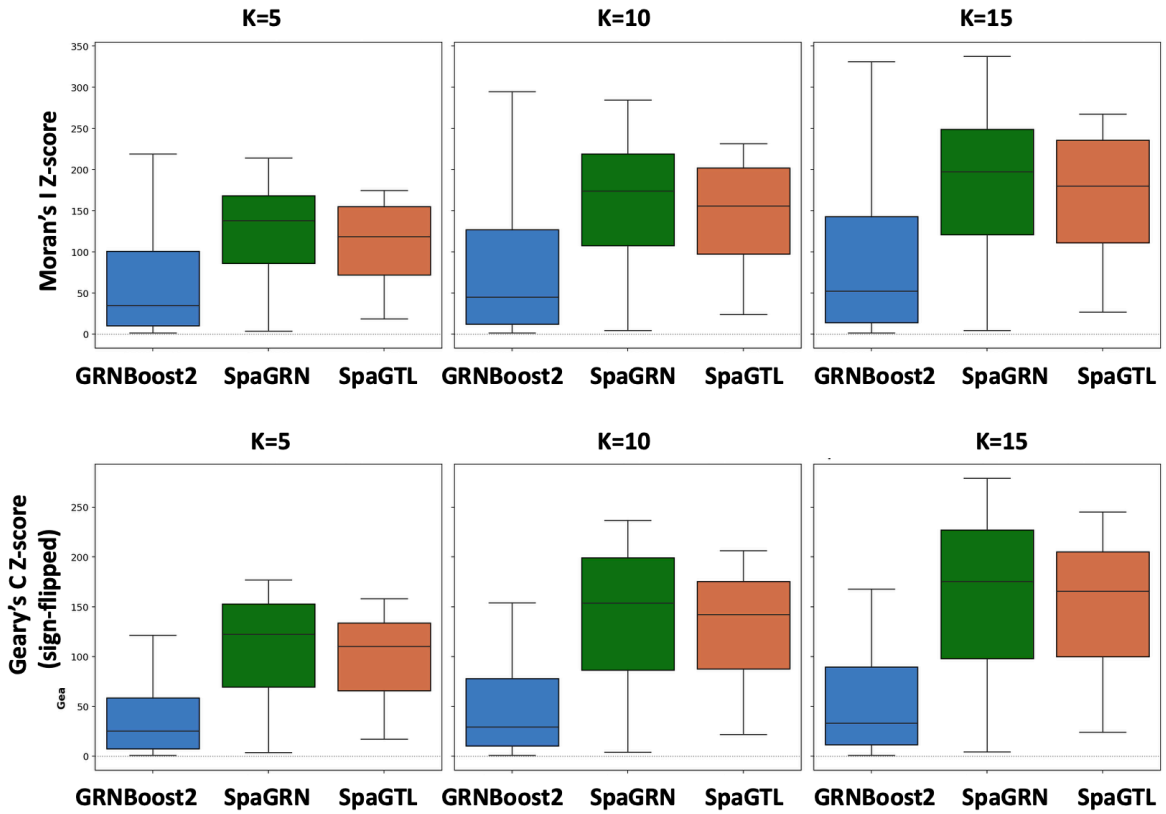

Figure 4.13: Neighborhood size sweep for the full regulon set of each method. *Each column corresponds to the number of nearest neighbors used to construct the spatial weight graph (K=5, 10, or 15). The top row shows Moran's I z-scores, and the bottom row shows sign-reversed Geary's C z-scores. Within each panel, the boxplots show the score distributions across the full eligible regulon sets of GRNBoost2, SpaGRN, and SpaGTL. Larger values indicate stronger spatial clustering for both measures.*

Because the methods recovered different sets of TFs, the full-set comparison was influenced by both TF-set composition and method-specific differences in the activity maps. Holding TF identity constant narrowed the differences but did not change the method order. Across the 22 shared TFs, median Moran's I z-scores were 78.7 for GRNBoost2, 227.7 for SpaGRN, and 176.2 for SpaGTL. SpaGRN showed the strongest spatial clustering for the shared TFs, followed by SpaGTL and GRNBoost2.

Influence of regulon size on spatial coherence

Regulon size was examined because larger regulons may produce higher spatial scores. AUCell activity is calculated from the target genes within a regulon, so larger target sets may produce more continuous regional activity patterns.

SpaGRN had the largest regulons, with a median of 66 targets, compared with 30 for GRNBoost2 and 28 for SpaGTL (table 4.3). Within each method, regulon size was positively associated with the Moran's I z-score. The Spearman correlations between log10 regulon size and the Moran's I z-score were +0.61 for GRNBoost2, +0.75 for SpaGRN, and +0.45 for SpaGTL. Regulon size was therefore associated with spatial-score magnitude, with the strongest association observed for SpaGRN.

However, regulon size did not fully account for the differences between methods. After target counts were matched across the 22 TFs shared by all three methods, the median Moran's I z-scores were 28.7 for GRNBoost2, 182.2 for SpaGRN, and 166.1 for SpaGTL. Thus, under this spatial-validation criterion, SpaGRN retained the strongest spatial clustering after regulon-size matching, followed by SpaGTL and GRNBoost2.

Table 4.3: Influence of regulon size on Moran's I z-scores at K = 10. *Median regulon size and ρ between log10 target count and zI use each method's full eligible regulon set. Matched-size medians use the 22 shared TFs after each regulon was truncated to the smallest target count available for that TF. Larger values indicate stronger spatial organization, but the matched-size analysis is a sensitivity check rather than a correction that establishes biological accuracy.*

Factor	GRNBoost2	SpaGRN	SpaGTL	Interpretation
Median regulon size (targets)	30	66	28	SpaGRN produced larger regulons
ρ (log10 size, zI)	+0.61	+0.75	+0.45	Size was positively associated with zI
Matched-size median zI	28.7	182.2	166.1	The method order remained unchanged

Map similarity and TF specific distinction

Before interpreting the individual regulon maps, the complete set of spatial activity maps for the 22 TFs shared by all three methods was visually inspected (see Appendix A). Several SpaGRN maps showed similar broad spatial patterns. The same tendency appeared in SpaGTL, although it was less pronounced. To quantify this observation, pairwise Spearman correlations were calculated between the spatial AUCell profiles across all bins within each method. For each regulon, the median correlation with the other 21 regulons from the same method was used as a descriptive measure of map similarity. Higher values indicate greater similarity to other maps produced by the same method. Values near zero indicate a more distinct spatial pattern.

Correlations within each method were generally highest for SpaGRN, intermediate for SpaGTL, and close to zero for GRNBoost2 (fig. 4.14). Based on map distinctness alone, GRNBoost2 produced the most distinct regulon activity maps. SpaGTL showed an intermediate level of distinction, while SpaGRN showed the strongest repetition across regulons.

Two explanations are possible. First, target gene programs assigned to different TFs may genuinely show high activity in the same cell populations or anatomical regions. Second, SpaGRN and SpaGTL use spatial structure during network inference. As a result, regulons associated with different TFs may capture the same dominant regional signal rather than distinct TF specific patterns.

Spatial coherence alone should therefore not be interpreted as evidence of a TF-specific spatial pattern. When multiple regulons show similar maps, the signal may instead reflect shared cell populations or broad anatomical organization. Spatial autocorrelation should thus be interpreted together with map distinctness across regulons.

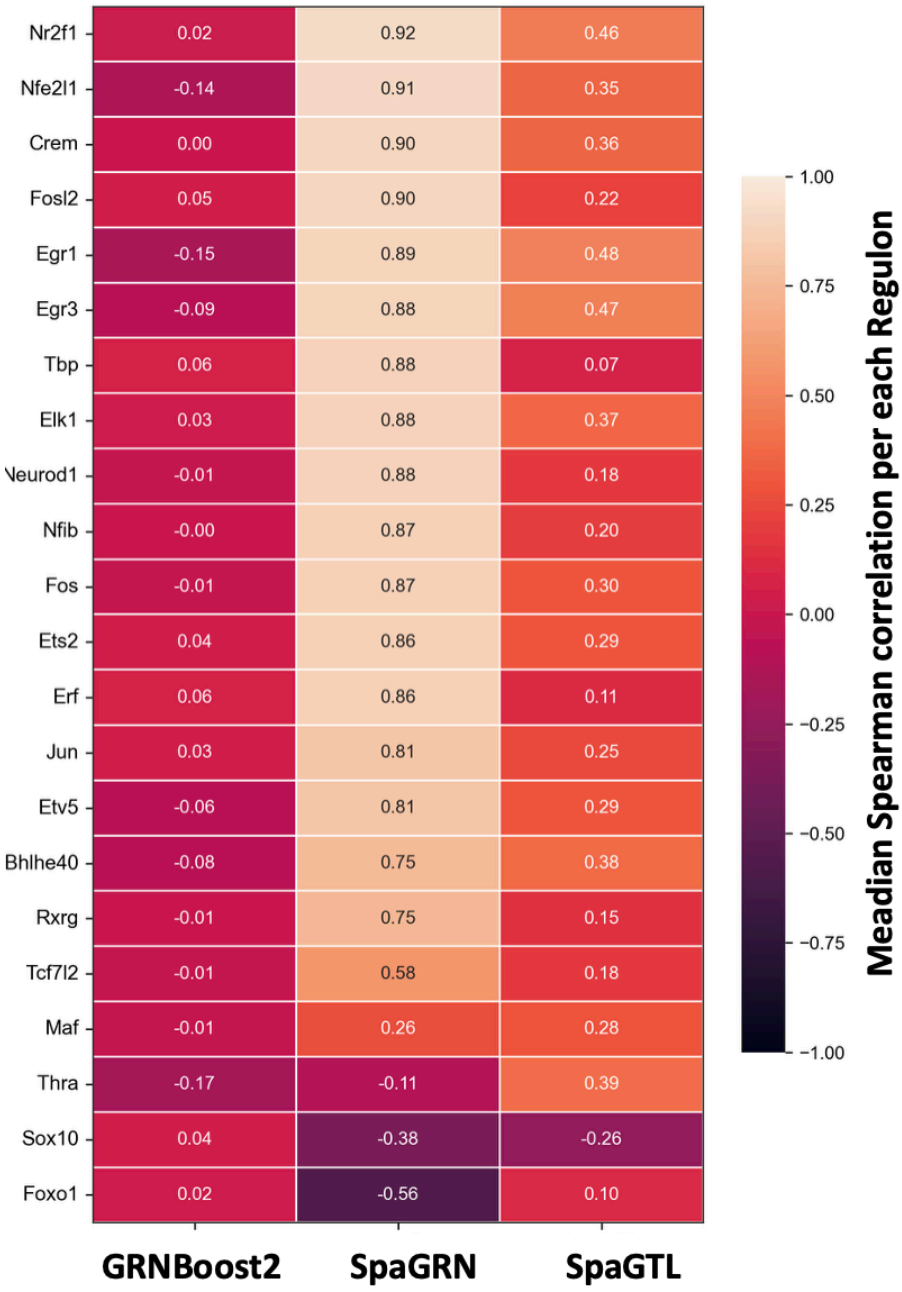

Figure 4.14: TF level similarity of regulon activity maps within each method. *For each TF, the displayed value is the median Spearman correlation between its AUCell profile across bins and the profiles of the other 21 shared regulons inferred by the same method. Higher positive values indicate greater similarity to other regulon maps. Values near zero indicate little shared spatial ordering, while negative values indicate opposing spatial patterns.*

Anatomical inspection of individual regulons

Earlier analyses compared the three methods using the overall distributions of Moran's I z scores, regulon size, and similarity among activity maps. The focus then shifted to differences among individual TFs. Spatial maps for all 22 shared TFs were visually reviewed and are provided in the Appendix A. However, a detailed comparison with anatomical literature was not carried out for every TF.

At K = 10, the Moran's I z score for each TF was compared directly across the three methods (fig. 4.15). SpaGRN produced the highest value for 11 of the 22 TFs. GRNBoost2 produced the highest value for 6 TFs, while SpaGTL produced the highest value for 5. No method showed the strongest spatial clustering for every TF. Instead, the results varied across individual regulons.

These differences in Moran's I z score provided a starting point for selecting maps for closer inspection. A high value only shows that bins with similar regulon activity tend to occur near one another. It does not establish that the pattern is specific to the TF or biologically meaningful. The earlier analysis of map distinctness was therefore also considered. When a map closely resembled many other regulon maps produced by the same method, agreement with a broad anatomical region was treated as limited support for a TF specific pattern.

Nine examples were selected post hoc and interpreted using relevant anatomical or functional literature. They were grouped into three descriptive types. The first showed increased spatial clustering together with clearer anatomical compatibility. The second showed increased clustering without clear anatomical support specific to the TF. The third included regulons that already showed strong spatial structure in GRNBoost2. Agreement with published evidence was treated as biological compatibility, not as proof of regulation between the TF and its inferred targets.

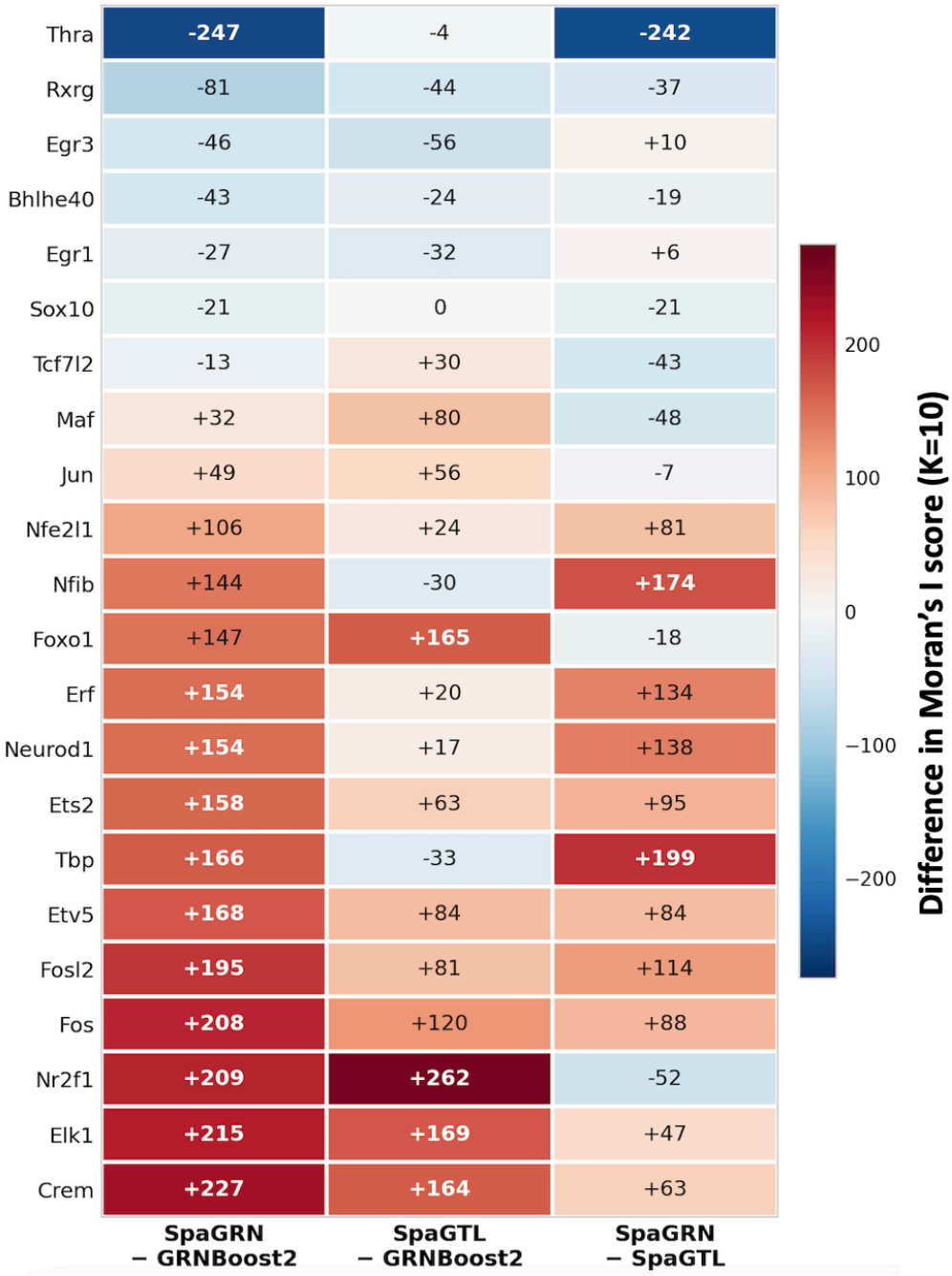

Figure 4.15: Change in Moran's I z-score for every shared TF at K equal to 10. *The three columns show SpaGRN minus GRNBoost2, SpaGTL minus GRNBoost2, and SpaGRN minus SpaGTL. Each value compares the two methods for the same TF. Positive values shown in red mean that the method named first had the higher Moran's I z score. Negative values shown in blue mean that the method named second had the higher value. The full comparison provides context for the selected regulon maps discussed below.*

The first type included regulons whose spatial scores increased relative to GRNBoost2 in at least one of two spatially informed methods (SpaGRN and SpaGTL) and whose maps became more consistent with broad anatomical expectations (fig. 4.16).

Foxo1 provided the clearest case. Its central high activity region became more continuous, while its Moran's I z-score rose from 106.6 in GRNBoost2 to 253.4 in SpaGRN and 271.6 in SpaGTL. FoxO1 expression has been reported in the adult mouse striatum and in subsets of hippocampal neurons (Hoekman et al., 2006) and the observed central domain overlapped the broad cerebral nuclei annotation. The Foxo1 maps produced by both SpaGRN and SpaGTL remained relatively distinct from the other regulon maps produced by their respective methods. These observations were therefore read as compatibility with a more clearly organized cerebral nuclei associated Foxo1 target program, rather than as validation of individual TF to target edges.

Nr2f1 showed broader cortical compatibility but less TF specific support. Its map changed from diffuse in GRNBoost2 to stronger outer cortex like patterns in SpaGRN and SpaGTL, and its Moran's I z-score increased from 25.0 to 234.4 and 286.7. NR2F1 has established roles in neocortical development and cortical arealization (Bertacchi et al., 2020). The observed cortical location is therefore broadly compatible with known NR2F1 biology. The SpaGRN Nr2f1 map, however, closely resembled many other SpaGRN regulon maps, which limits its interpretation as an Nr2f1 specific spatial domain, and the anterior to posterior gradient described in the literature cannot be tested in a single coronal section.

Tcf7l2 provided a mixed rather than a clearly positive example. SpaGTL increased the score from 162.0 of GRNboost2 to 192.5 and made an internal interbrain like contour somewhat clearer, while SpaGRN lowered the score to 149.3. TCF7L2 regulates thalamic differentiation programs (Lipiec et al., 2020), so the SpaGTL pattern is broadly compatible with this context. The interbrain label is nevertheless wider than the thalamus, and the two spatially informed methods moved the score in opposite directions, so Tcf7l2 was treated as partial support rather than a clear positive case.

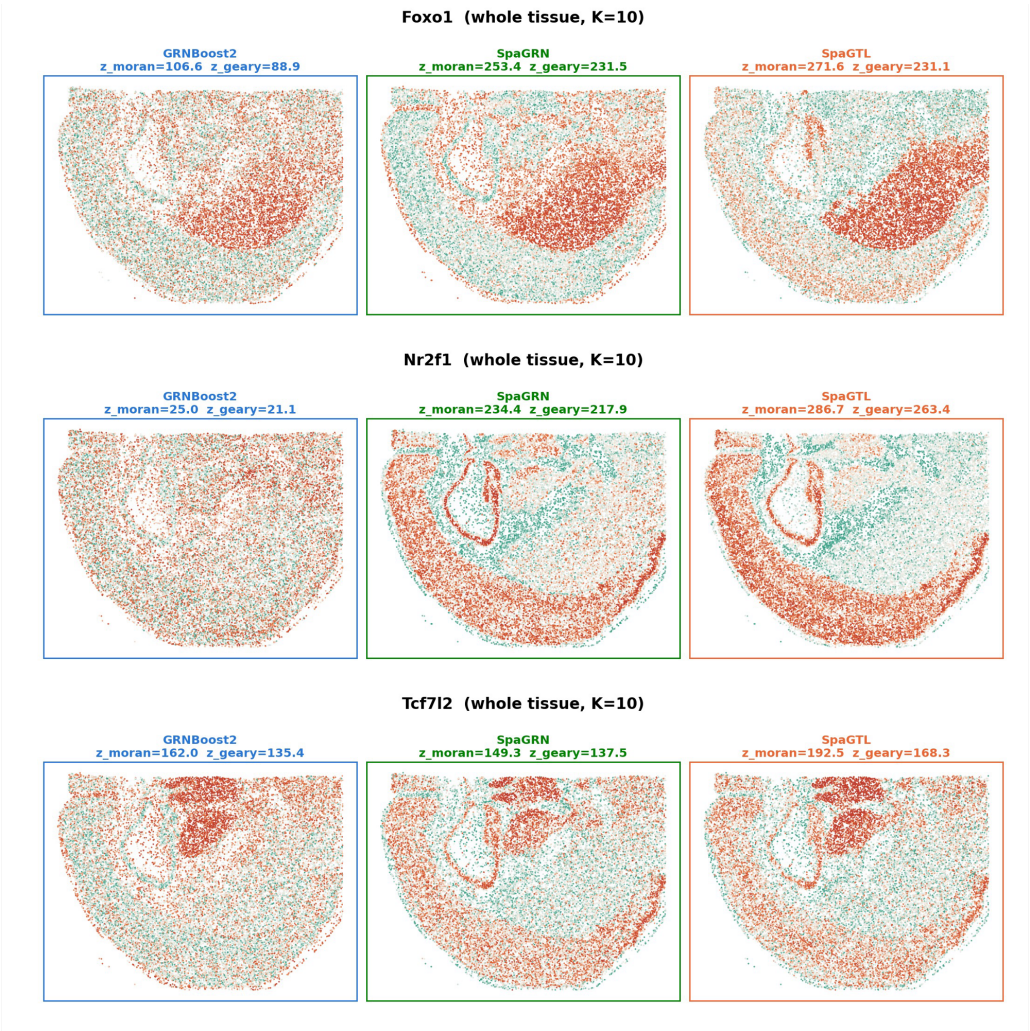

Figure 4.16: Regulon activity maps showing spatial gain with different levels of anatomical support. *Rows show Foxo1, Nr2f1, and Tcf7l2. Columns show GRNBoost2, SpaGRN, and SpaGTL. Color encodes the within regulon percentile rank of AUCell activity across bins. Standardized Moran and sign reversed Geary scores are printed above each panel. Broad anatomical labels were used for description only and were not supplied during GRN inference.*

The second response type included regulons for which SpaGRN, SpaGTL, or both produced a higher Moran's I z-score than GRNBoost2, but whose maps did not give clear TF-specific anatomical support (fig. 4.17).

Tbp rose from 44.4 in GRNBoost2 to 209.9 in SpaGRN, but fell to 11.2 in SpaGTL. TBP has a general role in transcription initiation by RNA polymerases I, II, and III (T. Kim and Roeder, 1994). This broad transcriptional role does not provide a clear expectation for activity in a specific brain region, although the particular target set selected by a method could still be enriched in specific cell types. A concentrated TBP map therefore does not by itself show a TBP-specific regional pattern. The SpaGRN Tbp map also closely resembled several other SpaGRN regulon maps, further limiting a TF-specific interpretation.

Fos showed a higher Moran's I z-score in SpaGRN (213.5) and SpaGTL (125.4) than in GRNBoost2 (5.7). Fos is an immediate early factor whose expression can increase in highly active neurons (Yassin et al., 2010). It also acts with Jun family proteins as part of AP-1, a broadly acting complex that can have a pioneer-like role in opening chromatin in some contexts (Vierbuchen et al., 2017). If neuronal activation was spatially concentrated in this tissue section, coordinated expression of Fos target genes could have contributed to the observed regulon pattern. However, neuronal activity was not measured directly, and the SpaGRN Fos map resembled several other SpaGRN regulon maps. Its similarity to Jun and other regulon maps may therefore reflect a shared AP-1-related program rather than an anatomical pattern specific to Fos. The high spatial score therefore does not establish a Fos-specific regional pattern.

Nfib had a higher Moran's I z-score in SpaGRN (228.5) than in GRNBoost2 (84.5) or SpaGTL (54.6). NFIB is expressed in adult cortical neurons and astrocytes, and NFI proteins have distinct laminar distributions in the adult mouse neocortex (K.-S. Chen et al., 2017). The cortical organization in the SpaGRN map is therefore broadly compatible with the reported expression of NFIB. However, the study measured NFIB protein expression, whereas the present map represents activity inferred from NFIB target genes. The laminar patterns reported by Chen et al. were not assessed directly in the present analysis. The SpaGRN Nfib map also closely resembled the SpaGRN Tbp and Fos maps. The high score therefore indicates strong spatial organization, but the available evidence is insufficient to identify it as an NFIB-specific pattern.

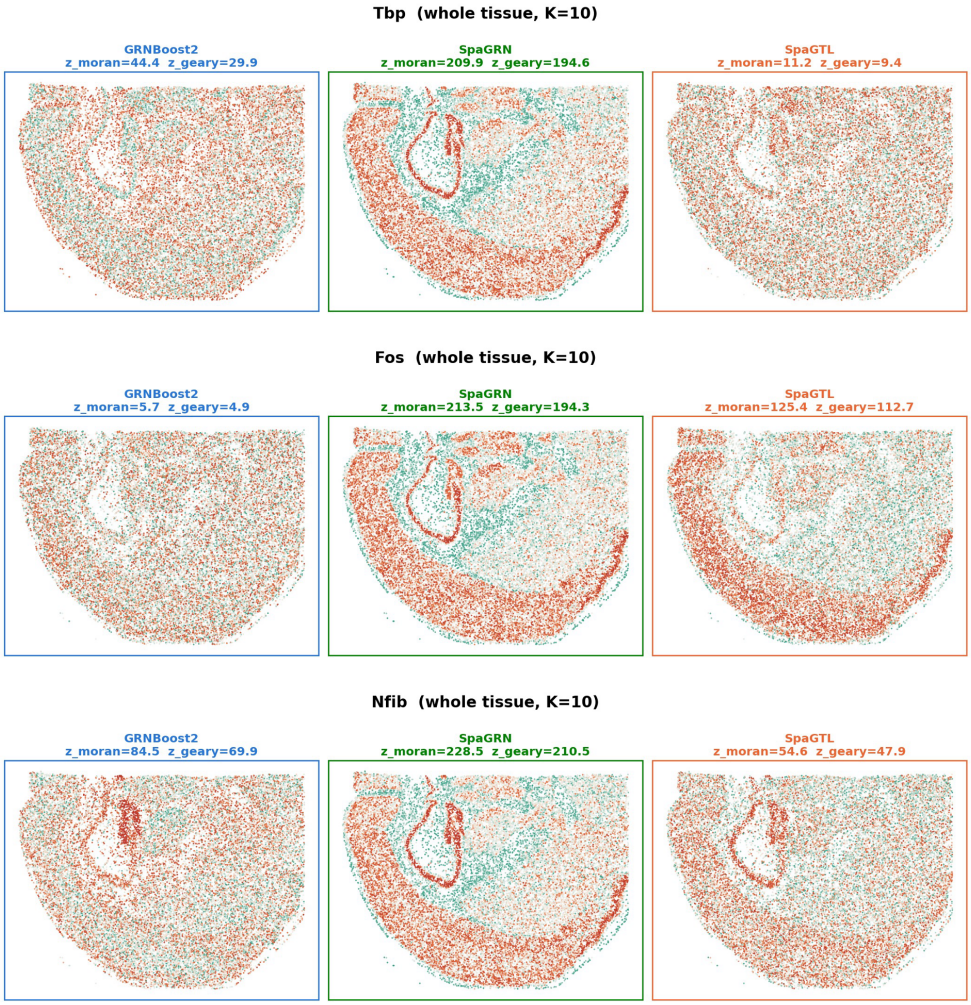

Figure 4.17: Regulon activity maps in which a higher spatial score lacks clear TF specific anatomical support. *Rows show Tbp, Fos, and Nfib. Columns and color encoding follow Figure 4.16. These examples illustrate that a strong spatial score need not be accompanied by evidence of TF specific anatomical support.*

The third response type included regulons that already showed strong spatial structure in GRNBoost2, even though GRNBoost2 did not use bin coordinates. This can occur when the inferred target genes are expressed together within the same spatially localized cell population or tissue region. Their combined AUCell activity can then form a spatial map even when the inference method uses expression alone.

Rxrg provides the clearest example. GRNBoost2 produced a compact CNU-like domain and gave the highest Moran's I z-score of the three methods at 335.4, compared with 254.2 for SpaGRN and 291.6 for SpaGTL. Rxrg is reported in discrete central nervous system regions, including the striatum, during fetal development, and is involved in dopaminergic signaling in striatal circuits (Dollé et al., 1994; Krzyżosiak et al., 2010). The observed central domain is therefore consistent with the known regional context of Rxrg.

Sox10 showed similarly high spatial scores in all three methods, at 259.4, 238.1, and 259.1. SOX10 is an established regulator of oligodendrocyte development (Stolt et al., 2002). Its target genes may therefore be co-expressed in oligodendroglial cells and associated white matter regions, and shared expression of this kind can produce a spatially organized regulon map without coordinate input.

Egr3 represents a different case. It is linked to neuronal activity programs (Lin Li et al., 2007). If neuronal activity is concentrated in one part of the section, its target genes can also increase together in that area. GRNBoost2 can capture this shared expression pattern without using spatial coordinates. However, neuronal activity was not measured directly, so this remains a possible explanation rather than direct evidence for an Egr3-specific spatial program.

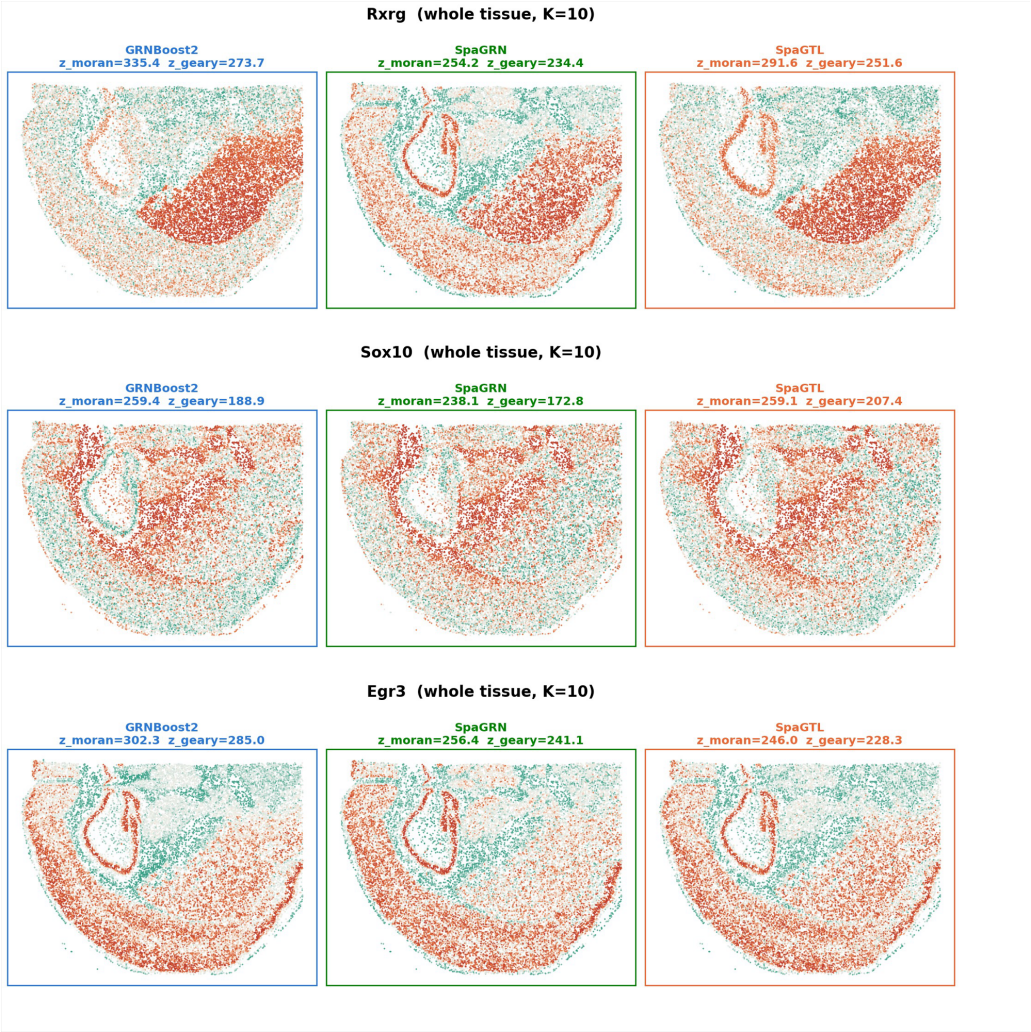

Figure 4.18: Regulon activity maps with strong spatial structure recovered without coordinate input. *Rows show Rxrg, Sox10, and Egr3. Columns and color encoding follow Figure 4.16. GRNBoost2 used the same expression matrix as the spatial methods but did not use bin coordinates or anatomical labels.*

Agreement with Allen ISH regional ordering

The Allen in situ hybridization (ISH) reference provided an independent regional profile of TF transcript expression. For each TF, the ordering of regulon activity across the three broad regions defined during annotation was compared with the ordering of TF expression across the corresponding Allen regions. Exact agreement was observed for 9 of 66 TFs (13.6%) in GRNBoost2, 42 of 109 (38.5%) in SpaGRN, and 15 of 26 (57.7%) in SpaGTL (fig. 4.19). Thus, regional concordance was highest for SpaGTL, followed by SpaGRN and GRNBoost2.

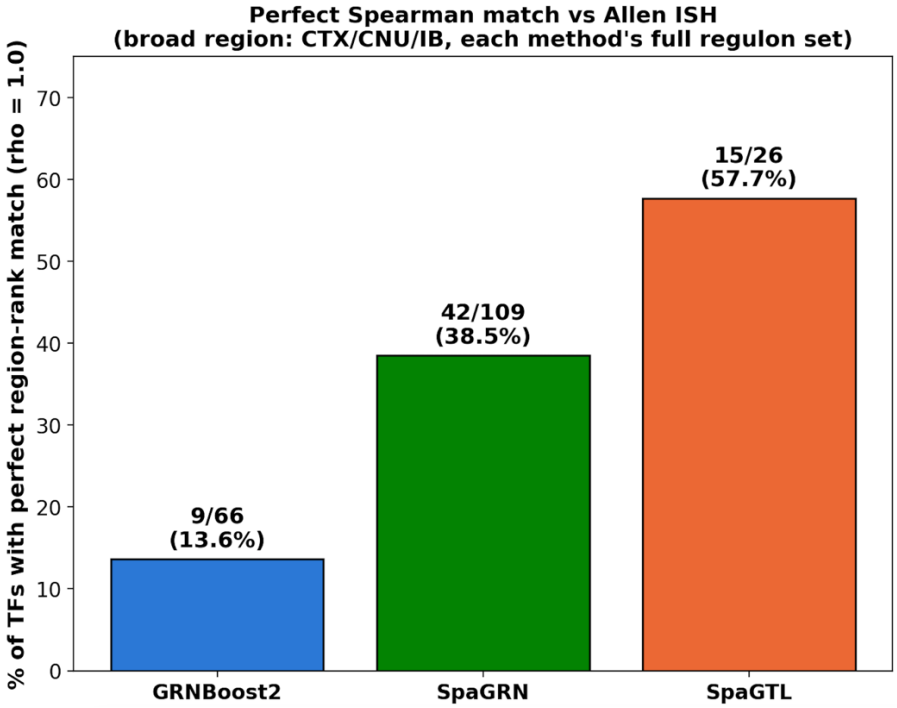

Figure 4.19: Proportion of TFs whose broad region ranking matched the Allen ISH ranking exactly. *For each method, the bar gives the percentage of that method's evaluable regulons whose ranking of the three broad regions (CTX, CNU, IB) reproduced the Allen ISH ordering with a Spearman correlation of 1.0. Counts above each bar give the matching TFs out of the TFs evaluable for that method.*

Among the 22 TFs shared by all three methods, 17 had corresponding Allen ISH reference data. For each of these TFs, the three methods were ranked according to their agreement with the Allen reference (fig. 4.20). SpaGTL ranked first or tied for first for 8 TFs and ranked second for a further 8 TFs. It ranked last or tied for last for only 1 TF. SpaGRN ranked first or tied for first for 7 TFs, but ranked last or tied for last for 6 TFs. GRNBoost2 ranked first or tied for first for 5 TFs and ranked last or tied for last for 10 TFs.

Thus, SpaGTL showed the most consistent agreement with the Allen regional reference across the shared TFs. It was not always the highest-ranking method, but it was almost always ranked first or second. SpaGRN achieved the second highest agreement for a similar number of TFs, although its performance varied more across TFs. GRNBoost2 showed the least consistent agreement in this matched comparison. The per-TF regional orderings and correlation values behind this comparison are given in table 4.4.

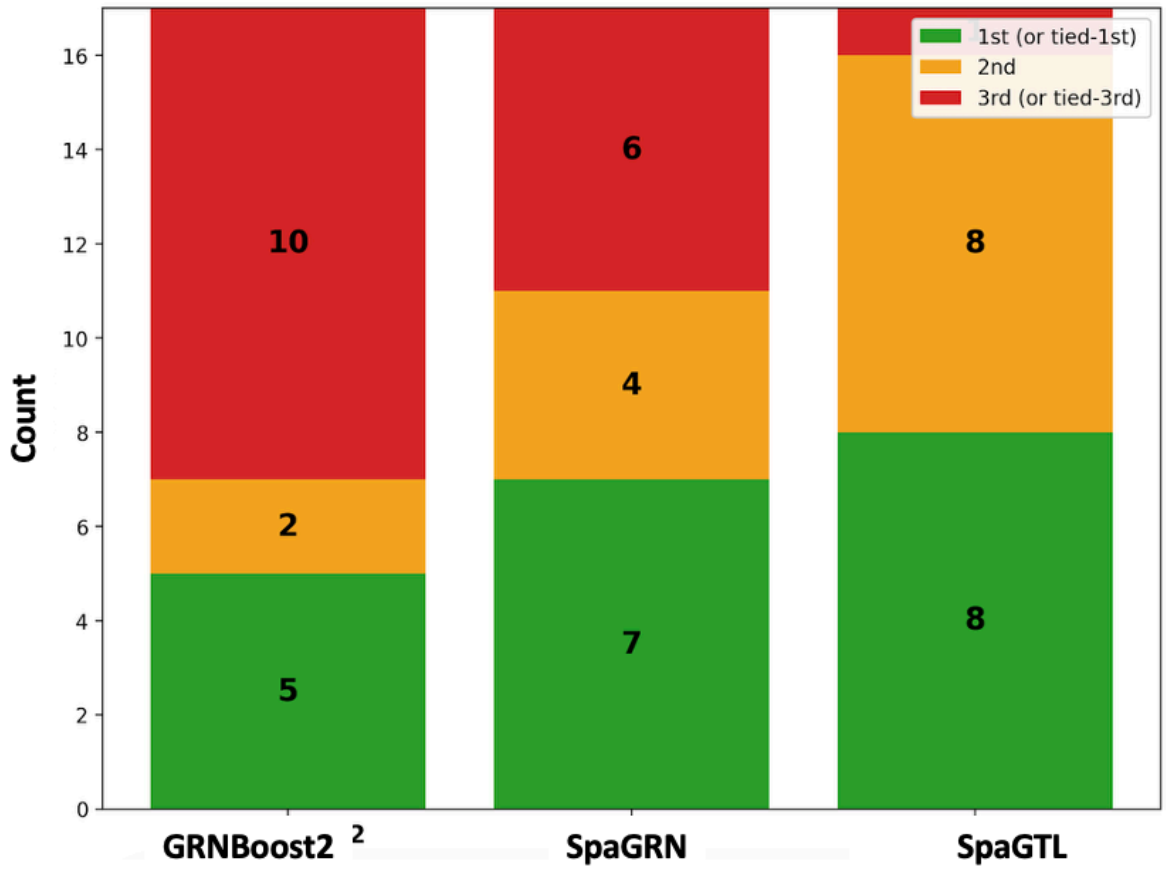

Figure 4.20: Method ranks for broad region agreement with Allen ISH across 17 shared TFs. *Each stacked bar counts the TFs for which a method ranked first, second, or third. Ranks were assigned within each TF using Spearman correlation between predicted and Allen three region profiles, with Pearson correlation after min max scaling used to break ties. Ties were retained, so a method can share a rank.*

Table 4.4: Broad region ordering of regulon activity compared with Allen ISH for the shared TFs. *For each TF, the Allen ISH column gives the ordering of the three broad regions in the Allen in situ hybridization reference, and each method column gives the ordering obtained from mean AUCell regulon activity, followed in brackets by the Spearman rank correlation and the Pearson correlation between the predicted and Allen three-region profiles. Pearson values were calculated after min-max scaling. Regions are the cerebral cortex (CTX), cerebral nuclei (CNU), and interbrain (IB).*

TF	Allen ISH	GRNBoost2 spearman ρ / pearson r	SpaGRN spearman ρ/ pearson r	SpaGTL spearman ρ / pearson r
Bhlhe40	CTX>IB>CNU	CTX>CNU>IB (+0.5 / +0.67)	**CTX>IB>CNU (+1.0 / +0.99)**	CTX>IB>CNU (+1.0 / +0.88)
Egr1	CTX>CNU>IB	**CTX>CNU>IB (+1.0 / +0.99)**	CTX>CNU>IB (+1.0 / +0.89)	**CTX>CNU>IB (+1.0 / +0.99)**
Egr3	CTX>CNU>IB	**CTX>CNU>IB (+1.0 / +0.99)**	CTX>CNU>IB (+1.0 / +0.83)	**CTX>CNU>IB (+1.0 / +0.99)**

(continued from previous page)

TF	Allen ISH	GRNBoost2 spearman ρ / pearson r	SpaGRN spearman ρ/ pearson r	SpaGTL spearman ρ / pearson r
Elk1	CTX>IB>CNU	CNU>IB>CTX (-1.0 / -0.95)	**CTX>IB>CNU (+1.0 / +1.00)**	CTX>CNU>IB (+0.5 / +0.64)
Erf	CTX>CNU>IB	CNU>CTX>IB (+0.5 / -0.10)	**CTX>CNU>IB (+1.0 / +1.00)**	CNU>CTX>IB (+0.5 / +0.34)
Etv5	CTX>CNU>IB	CTX>IB>CNU (+0.5 / +0.88)	CTX>IB>CNU (+0.5 / +0.80)	**CTX>CNU>IB (+1.0 / +1.00)**
Fos	CTX>CNU>IB	IB>CTX>CNU (-0.5 / -0.37)	CTX>IB>CNU (+0.5 / +0.51)	**CTX>IB>CNU (+0.5 / +0.71)**
Foxo1	CNU>CTX>IB	CNU>IB>CTX (+0.5 / +0.96)	CNU>CTX>IB (+1.0 / +0.99)	**CNU>CTX>IB (+1.0 / +1.00)**
Jun	CTX>CNU>IB	CTX>IB>CNU (+0.5 / +0.59)	CTX>IB>CNU (+0.5 / +0.70)	**CTX>IB>CNU (+0.5 / +0.78)**
Neurod1	CTX>IB>CNU	IB>CTX>CNU (+0.5 / -0.13)	**CTX>IB>CNU (+1.0 / +0.96)**	CTX>CNU>IB (+0.5 / +0.98)
Nfib	CTX>CNU>IB	**CTX>CNU>IB (+1.0 / +0.97)**	CTX>IB>CNU (+0.5 / +0.96)	CTX>IB>CNU (+0.5 / +1.00)
Nr2f1	CTX>IB>CNU	IB>CNU>CTX (-0.5 / -0.14)	**CTX>IB>CNU (+1.0 / +0.95)**	CTX>IB>CNU (+1.0 / +0.91)
Rxrg	CTX>CNU>IB	CNU>CTX>IB (+0.5 / -0.32)	**CNU>CTX>IB (+0.5 / +0.09)**	CNU>CTX>IB (+0.5 / -0.18)
Sox10	IB>CNU>CTX	CNU>IB>CTX (+0.5 / +0.95)	IB>CTX>CNU (+0.5 / +0.40)	**IB>CNU>CTX (+1.0 / +1.00)**
Tbp	CTX>IB>CNU	CNU>CTX>IB (-0.5 / -0.70)	**CTX>IB>CNU (+1.0 / +1.00)**	IB>CTX>CNU (+0.5 / +0.35)
Tcf7l2	IB>CNU>CTX	**IB>CNU>CTX (+1.0 / +1.00)**	IB>CTX>CNU (+0.5 / +0.86)	IB>CTX>CNU (+0.5 / +0.95)
Thra	CTX>CNU>IB	CTX>CNU>IB (+1.0 / +0.99)	IB>CNU>CTX (-1.0 / -0.97)	**CTX>CNU>IB (+1.0 / +1.00)**

Summary of spatial validation

SpaGRN produced the strongest clustering, followed by SpaGTL and GRNBoost2. Regulon size was positively associated with Moran's I z scores, but the method order remained after target counts were matched.

Individual maps showed that high spatial scores could arise in different ways. Some maps became more continuous in regions compatible with published evidence. Others followed broad patterns shared by several regulons. Strong spatial patterns also appeared in selected GRNBoost2 regulons without coordinate input.

Map similarity was used to place these results in context, not to define which method was better. Shared patterns may reflect regulons involved in the same biological programs. They may also arise when a method repeatedly captures the same broad tissue structure. These possibilities could not be separated by the present analysis. The greater repetition observed in SpaGRN therefore calls for caution when interpreting its high spatial scores as TF specific patterns.

Allen ISH served as an external reference rather than a ground truth. It compared TF transcript expression with inferred target activity and did not validate individual TF to target links. SpaGTL showed the strongest regional agreement with this reference, followed by SpaGRN and GRNBoost2.

Overall, the methods emphasized different spatial properties. SpaGRN produced the strongest clustering but also the greatest similarity among its maps. SpaGTL showed slightly weaker clustering and the strongest Allen agreement. GRNBoost2 had lower overall clustering and less similarity among its maps, while still recovering several anatomically compatible patterns without coordinate input.

Chapter 5

Discussion

This study compared GRNs inferred by GRNBoost2, SpaGRN, and SpaGTL, which differ in their use of spatial information, using the same quality-controlled Visium HD mouse brain section. The comparison examined both how the inferred GRNs differed and how complementary evaluations could be used to interpret those differences. The following sections discuss the evidence provided by each evaluation perspectives and the limits of their interpretation.

5.1 Reproducibility of stochastic network inference

Across the full edge ranking, GRNBoost2 was more stable than SpaGTL in both network size and edge recovery. SpaGTL retained its highest ranked edges and their order relatively well. However, agreement between runs decreased when more edges from lower positions in the ranked list were included.

TFs detected in several runs did not consistently recover the same TF to target edges. The same TF could therefore be detected again while being connected to different targets. This instability was observed for both GRNBoost2 and SpaGTL. A single output from a stochastic method was therefore better understood as one possible network rather than the only network supported by the data.

Previous studies have addressed the stochasticity of GRNBoost2 by running pySCENIC repeatedly and building a consensus from recurrent regulons and target genes (Garner et al., 2023). The five run consensus used in this study was also chosen to reduce the influence of variation between individual runs.

5.2 External regulatory evidence and reference dependence

All three methods showed more support than expected at random in CollecTRI, KnockTF, and ChIP-Atlas. GRNBoost2 showed the strongest support in CollecTRI and ChIP-Atlas. SpaGRN showed the strongest support in the KnockTF perturbation reference. SpaGTL also exceeded

random expectation in all three references, but it produced the smallest number of edges. This limited the number of reference edges that it could recover.

In the Top K analysis, the leading method changed depending on the reference and cutoff. The conclusion obtained from a small set of the highest ranked edges was therefore not always the same as the conclusion obtained from a wider network. These differences did not reflect just the quality of the inferred networks, but also the properties of the reference used for evaluation. Previous GRN benchmarks have similarly shown that method performance depends on the reference and dataset, while accurate recovery of individual edges remains difficult (Pratapa et al., 2020).

CollecTRI combines several curated resources, but it is not a context specific mouse brain network. The human interactions were also converted to mouse orthologues in this study, which introduced further uncertainty (Müller-Dott et al., 2023). KnockTF provides gene expression changes following TF perturbation. However, only a limited number of TFs could be evaluated, and the affected genes may include indirect downstream effects as well as direct targets (Feng et al., 2024). ChIP-Atlas offers evidence of TF binding, but binding alone does not establish functional regulation (Oki et al., 2018). An edge absent from these references was therefore treated as unsupported by the available reference rather than incorrect.

A matched comparison restricted to TFs that could be evaluated in all three references was not performed. Such an analysis could test whether differences in method ranking were partly caused by differences in TF coverage. It could also show whether predicted edges for the same TF received consistent support from curated knowledge, perturbation responses, and physical binding evidence.

5.3 Spatial organization and repeated regulon maps

The Moran's I z-scores and sign reversed Geary's C z-scores calculated from bin level AUCell activity were highest for SpaGRN, followed by SpaGTL and GRNBoost2. Regulon size was positively related to the Moran's I z-score. However, the order of the methods remained unchanged in the sensitivity analysis that matched each shared TF to its smallest target count. Regulon size was therefore related to the spatial score, but it did not change the overall method ranking.

SpaGRN uses spatial autocorrelation to prioritize spatially patterned genes and TF–target relationships. Although this does not guarantee that the resulting regulon activity maps will be spatially clustered, it makes such clustering more likely. SpaGRN's high spatial scores may therefore partly reflect its spatial selection procedure. (Y. Li et al., 2025).

Map similarity within each method was highest in SpaGRN. SpaGTL showed a weaker repeated pattern, while the median similarity for GRNBoost2 was close to zero. There were two possible explanations for this repetition. Target gene programs assigned to different TFs may genuinely have been active in the same cell populations or anatomical regions. Alternatively, different regulons may have repeatedly captured the same broad regional signal rather than patterns

that were distinct for each TF. The high spatial scores in SpaGRN and SpaGTL therefore had to be interpreted with care. It was important to consider whether clustering reflected a biological pattern related to the TF or a broad tissue pattern occurring across many regulons.

Several patterns were observed in the selected maps of the 22 TFs shared by all three methods. In some cases, the spatially informed methods produced more continuous patterns that were compatible with broad anatomy. In other cases, clustering increased but clear anatomical support specific to the TF was limited by map repetition or the broad function of the TF. Strong spatial structure was also present in some GRNBoost2 regulons, even though GRNBoost2 did not use coordinate information. High spatial grouping therefore did not always produce a clearer anatomical pattern or a pattern specific to the TF.

However, the 22 shared TFs were not selected as anatomical marker genes. They were simply the regulons recovered by all three methods. Repeating this analysis with clear regional or cell type markers would allow more direct testing of agreement with expected spatial patterns.

5.4 Allen regional agreement and anatomical resolution

Among the regulons that could be evaluated within each method, SpaGTL reproduced the complete Allen ISH ordering across the three regions most often. SpaGRN ranked second, followed by GRNBoost2. SpaGTL also showed the most consistent agreement among the 17 TFs shared by all three methods, as it was almost always ranked first or second. SpaGRN showed high agreement for many TFs, but its performance varied more between TFs. GRNBoost2 showed the lowest agreement overall.

Allen ISH measures TF transcript expression, whereas AUCell summarizes the activity of target genes assigned to that TF. These values are not the same. However, if a target program is also high in a region where the TF is strongly expressed, the regulon map can be considered indirectly compatible with the known location of the TF.

TF transcript abundance does not always represent TF activity. Actual activity can be affected by protein abundance, localization, post translational regulation, and cofactors. Activity estimated from target genes can therefore provide regulatory information that differs from TF expression (Garcia-Alonso et al., 2019). Allen agreement was interpreted as evidence of anatomical compatibility. It did not prove that a regulon was correct, and disagreement did not mean that the regulon was incorrect.

Spearman and Pearson correlations calculated from only three regions provided limited statistical information. These values were therefore treated as descriptive evidence supporting the regional ordering rather than as strict quantitative validation.

The anatomical comparison could be improved in two ways. First, STalign could register the coordinates and tissue shape of the Visium HD section directly to the Allen Common Coordinate Framework. Atlas region labels could then be transferred to each bin, allowing the regional mean AUCell activity and Allen ISH ordering to be recalculated (Clifton et al., 2023). Second,

spatial grouping methods such as BANKSY could identify tissue domains independently. These domains could then be manually annotated using marker genes and histology (Singhal et al., 2024). Neither approach would replace Allen ISH. Instead, they would improve the reliability of assigning each bin to an anatomical region.

Table 5.1: Summary of method performance across the validation categories used in this study.

Criterion	GRNBoost2	SpaGRN	SpaGTL
Reproducibility	More stable than SpaGTL across broader edge sets	Deterministic in this workflow	Top ranked edges were stable, but agreement decreased when more edges were included
CollecTRI	Highest in the baseline normalized comparison	Lower than GRNBoost2	Lowest but still above random expectation
KnockTF perturbation	Strong at some small Top K cutoffs	Highest in the overall normalized comparison	Lowest but still above random expectation
ChIP-Atlas	Highest in the baseline normalized comparison	Lower than GRNBoost2	Lowest but still above random expectation
Moran's I and Geary's C	Lowest	Highest	Second highest
Map similarity within each method	Median close to zero	Strongest repetition, with a median of about 0.78	Weaker repetition, with a median of about 0.25
Complete Allen ordering across three regions	Lowest agreement	Second highest agreement	Highest agreement
Allen top region agreement	Lowest	Second highest	Highest

5.5 Limitations and future directions

Limitations directly related to reproducibility, external references, and the interpretation of spatial maps were discussed in the relevant sections above. This section considers limitations that affected the overall study design rather than one validation measure. These included the use of a single tissue section, the difficulty of identifying regulation within individual cells from 8 µm bins, the inability to isolate the effect of spatial information in a comparison of complete pipelines, and the possible influence of feature selection on method ranking.

5.5.1 Dataset scope and independent validation

The main limit to generalization was the use of one mouse brain section from a single animal. The observed method ranking may have been affected by the orientation of the section, the anatomical structures present, technical quality, or the expression pattern of that sample. The full benchmark should therefore be repeated using biological replicate animals, adjacent sections, and different brain levels. Samples from different sexes and ages could also be included where possible.

Independent single-cell RNA-sequencing data could provide complementary evidence about the cell-type associations of the inferred regulons. For each regulon, its inferred target-gene list could be used to calculate AUCell scores for individual brain cells in Tabula Muris. Comparing these scores across annotated cell types could show whether the target-gene program was enriched in biologically plausible cell populations and whether the corresponding TF was expressed in the same populations. However, as Tabula Muris does not contain spatial coordinates, this analysis could not validate anatomical location or spatial coherence. It would also not establish direct or causal regulation between the TF and its predicted targets (Tabula Muris Consortium, 2018).

The whole-brain MERFISH atlas could be used to assess whether regulon activity patterns were reproducible across matched brain regions and cell types. The regulon target sets inferred from the Visium HD data could be used to calculate cell-level AUCell scores in the whole-brain MERFISH atlas. The resulting activity patterns could then be compared across matched brain regions and cell types to assess whether the patterns observed in the Visium HD data were reproduced. Imputed transcriptome-wide expression profiles could additionally be used to increase regulon target coverage although results based on imputed genes would provide less independent evidence than those based on genes measured directly by MERFISH(M. Zhang et al., 2023).

5.5.2 Cellular resolution and mixed bin interpretation

The 8 µm bins used in this analysis were close to single cell scale, but they were not the same as segmented cells. One cell could extend across several bins. A bin at a cell boundary could also contain transcripts from more than one cell.

An inferred TF to target relationship could therefore not be assumed to represent regulation within the same cell. Some relationships may instead have reflected cell type composition, the presence of neighboring cells, or differences between tissue domains. The current networks were more appropriately interpreted as spatial regulatory hypotheses inferred from bin level expression than as confirmed GRNs within individual cells.

This limitation could be addressed by using 2 µm bins and histological images to segment cells. Bins assigned to the same cell could then be combined into a cell by gene matrix. Bin2cell and ENACT support this type of reconstruction for Visium HD data (Polański et al., 2024; Kamel

et al., 2025). Within areas where segmentation remained uncertain, a deconvolution method such as RCTD could estimate the mixture of cell types within each bin (Cable et al., 2022).

The three GRN pipelines could then be run again using reconstructed cells or observations with high confidence of representing individual cells. Comparing the bin level and reconstructed cell level networks would show which edges remained at cellular resolution and which depended on the mixed structure of the bins.

5.5.3 Isolating the contribution of spatial information

This study compared the complete outputs of three pipelines. It could therefore not separate differences caused by spatial information from those caused by different inference algorithms or other parts of the pipelines.

A spatial permutation analysis could examine this question while keeping the expression matrix and other analysis settings unchanged. SpaGRN could be run once with the real coordinates and again after randomly assigning expression profiles to spatial locations. SpaGTL could similarly be run with the real spatial graph and with a graph rebuilt from permuted coordinates. GRNBoost2 does not use coordinates, so it could remain an expression only reference with the same expression input and random seed.

The same reproducibility, external evidence, and spatial validation analyses could then be applied to every condition. This would allow direct comparison of how much the results changed when the real spatial information was used.

The original SpaGTL study also used ablation experiments during pretraining. Individual model components were removed to test their contribution to performance (W. Huang et al., 2024). This provides an example of an analysis design that compares a complete model with an altered version to assess the effect of a specific component.

5.5.4 Feature selection and input alignment

A common HVG or SVG filter was not applied to the three methods. This choice kept the same set of genes after quality control and prevented spatial information from entering GRNBoost2 before inference. However, genes with low variation or little useful information may have remained in the candidate set. Biologically informative genes were also not given explicit priority.

A future sensitivity analysis could compare two other input settings. The first would use the same HVG set. The second would apply the feature selection procedure recommended for each method. This comparison could show whether method ranking depended on feature selection or remained stable when the input genes were aligned.

Providing SVGs to GRNBoost2 would introduce spatial information during preprocessing. A common SVG analysis should therefore be presented as a sensitivity analysis of feature selection rather than as the main comparison with an expression only method. SpatialDE, SPARK-X,

or nnSVG could be used to select SVGs (Svensson, Teichmann, and Stegle, 2018; Zhu, Sun, and X. Zhou, 2021; Weber et al., 2023).

5.5.5 Indirect nature of spatial evaluation

The spatial analyses provided indirect evidence rather than direct validation of regulatory correctness. Moran's I and Geary's C quantified the spatial organization of target-based AUCell activity but did not establish that the activity occurred in biologically expected locations or was specific to the assigned TF. Literature-guided inspection and the Allen ISH comparison were also indirect because published evidence and ISH describe TF expression, localization, or function, whereas regulon maps summarize the expression of predicted target genes.

Recent spatial perturbation data can partially resolve this. Spatial Perturb-Seq links CRISPR knockouts to spatially resolved transcriptome-wide responses in the adult mouse hippocampus (Shen et al., 2026), while Perturb-FISH links perturbation identities to targeted transcript measurements through imaging in other experimental systems (Binan et al., 2025). Although current datasets do not provide broad coverage of TF perturbations across mouse brain regions and cell types, future expansion of such data could allow the predicted targets from this study to be tested in matched spatial contexts. Consistent target-gene responses following TF perturbation would provide functional support for an inferred regulon, although they would not establish direct TF binding.

5.5.6 Contributions and future perspectives for spatial GRN inference

The main contribution of this study was the direct comparison of GRNBoost2, SpaGRN, and SpaGTL in the same Visium HD mouse brain section. It showed whether inferred edges by three different models showed different networks, regulon composition, and spatial activity patterns. The use of different types of validation evidence also clarified the range and limits of the biological interpretations supported by each method. Several regulons showed spatial activity that was compatible with known anatomical or functional contexts. These regulons represent candidate regulatory programs for future experiments, not confirmed TF to target relationships.

Recent spatial GRN research is expanding from the use of location alone towards models that also capture changes over time, multiple molecular layers, and information learned from large datasets. One direction is spatiotemporal GRN inference using time series spatial transcriptomics. Connecting spatial datasets from different stages of development or disease could allow cell states and tissue structures to be followed together. Recent methods based on optimal transport have connected spatial structures spanning time to study how regulatory programs change across both time and location (Bryan, Farhi, and Cleary, 2026).

A second direction is the combination of spatial multiomics. Technologies can now measure spatial RNA together with ATAC sequencing or CUT&Tag in the same tissue section. This al-

lows TF and target expression to be linked with the accessibility of regulatory elements and histone states at the same location. Such data could extend GRNs based mainly on expression correlation into more mechanistic networks that include TF to regulatory element to target relationships (Loers and Vermeirssen, 2024; Y. Li et al., 2025).

A third direction is the development of base models and transfer learning models pretrained on large spatial atlases. Methods such as SpaGTL show the potential to transfer representations learned from large spatial datasets to new tissues. Recent studies also describe pretrained models as an important direction for the field. However, generalization between tissues and platforms, model explainability, and bias in the training data remain unresolved challenges (W. Huang et al., 2024; T. Liu et al., 2025). Overall, spatial GRNs are developing towards models that combine changes over time, multiple molecular layers, and spatial information that can be transferred across datasets.

Chapter 6

Conclusion

This thesis compared GRNBoost2, SpaGRN, and SpaGTL using the same Visium HD mouse brain dataset and evaluated their inferred networks in terms of reproducibility, external regulatory support, and the spatial properties of regulon activity. The purpose was to determine how GRN inference strategies that use spatial information differently performed under a common experimental setting and what could be learned from complementary evaluation criteria.

The comparison did not produce a single ranking of the methods. GRNBoost2 showed greater reproducibility across a broader part of the inferred network and received the strongest support from curated and binding resources. SpaGTL showed greater stability among its highest ranked edges and the most consistent agreement with broad regional TF expression in the Allen Mouse Brain Atlas. SpaGRN received the strongest support from perturbation data and produced the highest spatial coherence.

The spatial analyses further showed that strong spatial coherence did not by itself establish that a pattern was specific to the assigned TF or biologically plausible. SpaGRN produced highly organized regulon activity maps, but maps from different regulons frequently showed similar broad patterns. The available literature also did not always support a clear biological interpretation specific to the assigned TF. Some GRNBoost2 regulons were also spatially organized despite being inferred without coordinate information.

These findings should be interpreted within the limits of the study design. The analysis used one mouse brain section from a single sample, and the 8 µm bins did not represent segmented cells. The comparison also assessed complete pipelines rather than isolating the contribution of spatial information. Finally, the spatial evaluations relied mainly on indirect evidence rather than direct measurements of regulatory activity. The conclusions are therefore comparative rather than confirmatory.

Overall, the results show that no single criterion can provide a complete assessment of method performance. The evaluation criteria offered complementary information, but each had its own interpretive limits. The most relevant evidence therefore depends on the analytical aim. The inferred regulons should be regarded as candidate regulatory programs for further validation rather than as confirmed relationships between TFs and target genes.

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Appendix A

A.1 Additional spatial validation figures

The figures below give the whole-tissue spatial autocorrelation results (K = 10) for GRNBoost2, SpaGRN, and SpaGTL for transcription factors not shown in the main text. Standardized Moran's I(z_{moran}) and sign-reversed Geary's C ($-z_{\text{geary}}$) are printed above each panel.

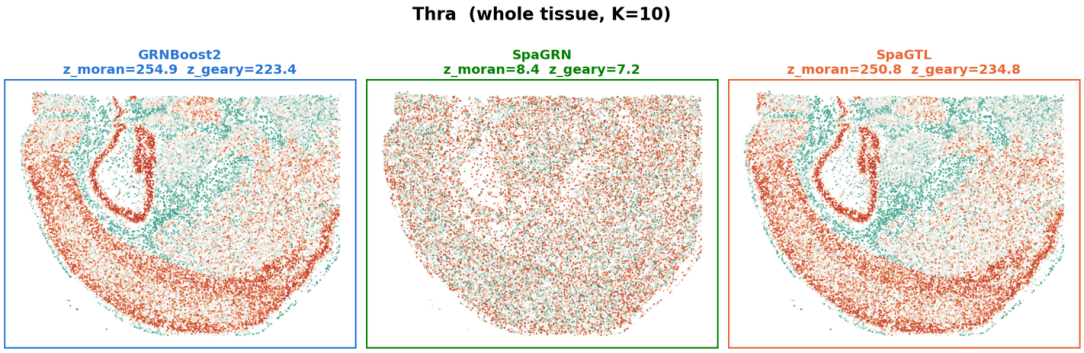

Figure A.1: Spatial autocorrelation of the Thra regulon.

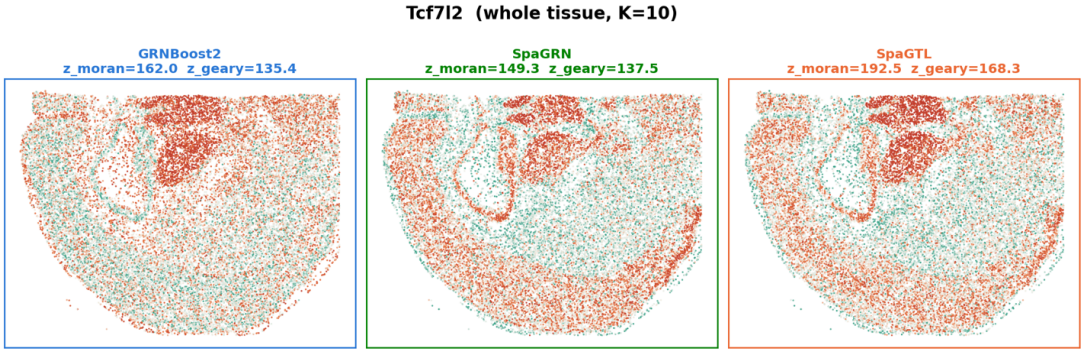

Figure A.2: Spatial autocorrelation of the Tcf7l2 regulon.

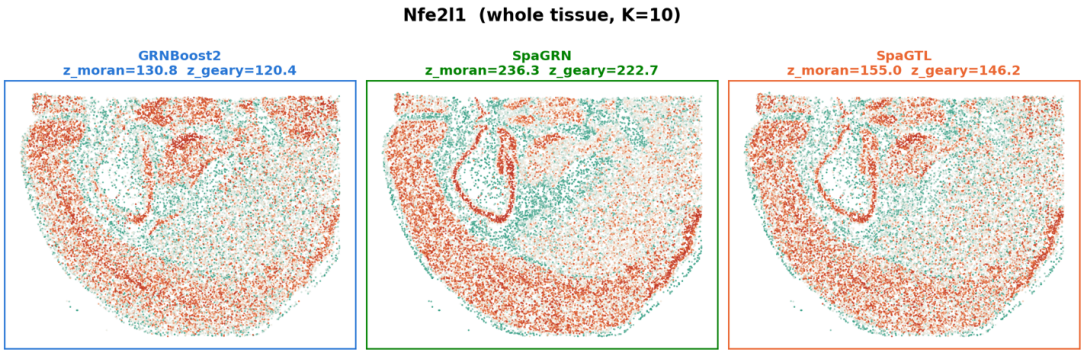

Figure A.3: Spatial autocorrelation of the Nfe2l1 regulon.

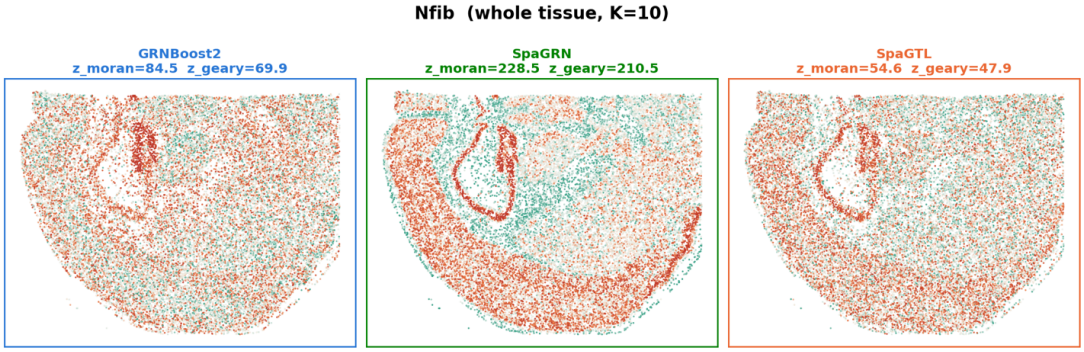

Figure A.4: Spatial autocorrelation of the Nfib regulon.

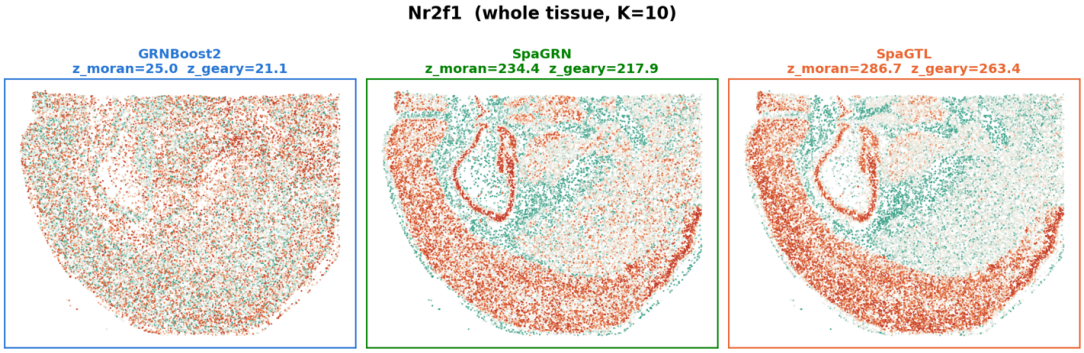

Figure A.5: Spatial autocorrelation of the Nr2f1 regulon.

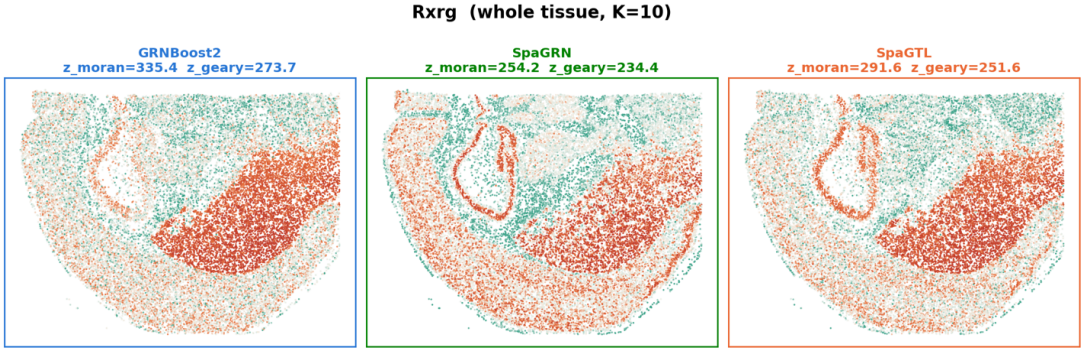

Figure A.6: Spatial autocorrelation of the Rxrg regulon.

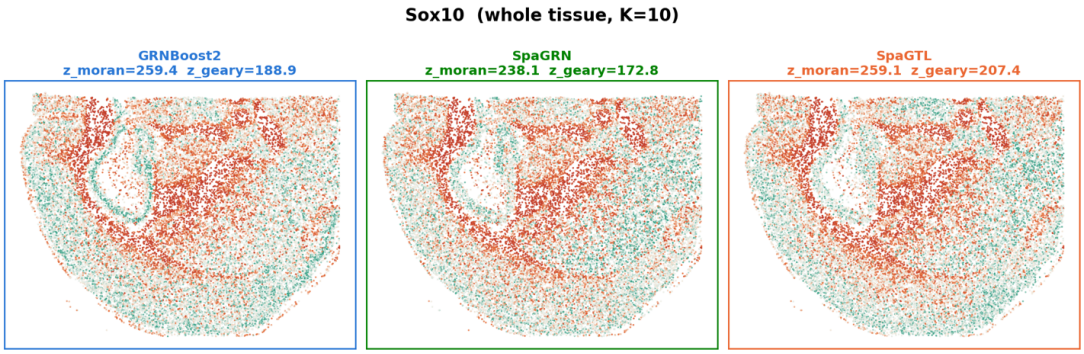

Figure A.7: Spatial autocorrelation of the Sox10 regulon.

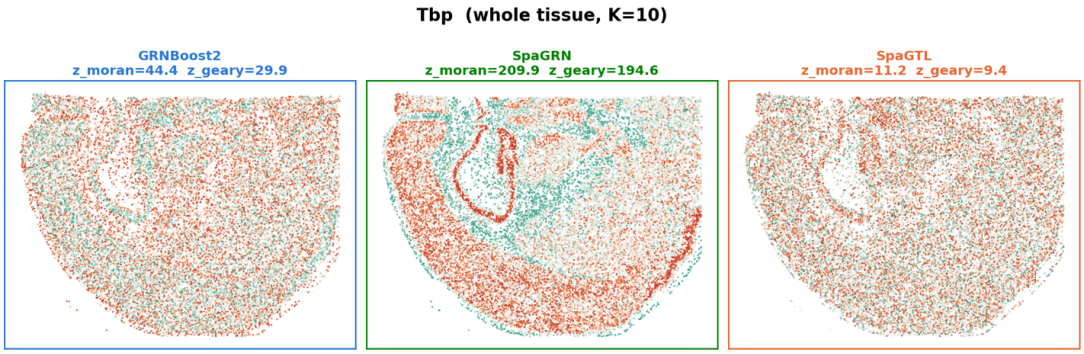

Figure A.8: Spatial autocorrelation of the Tbp regulon.

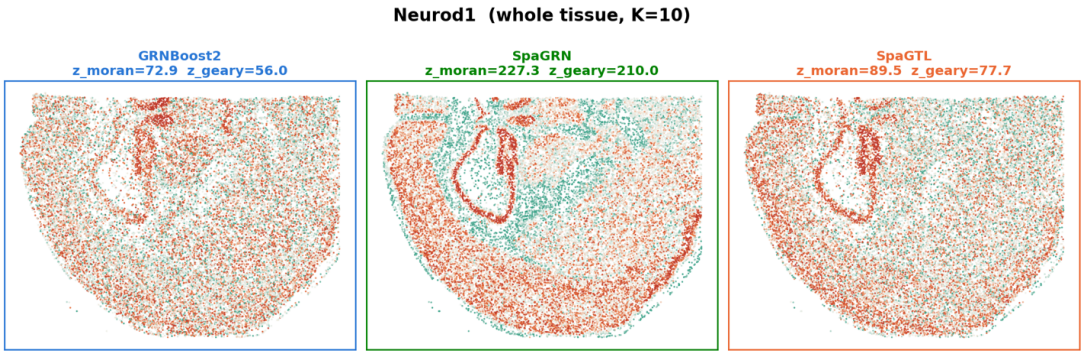

Figure A.9: Spatial autocorrelation of the Neurod1 regulon.

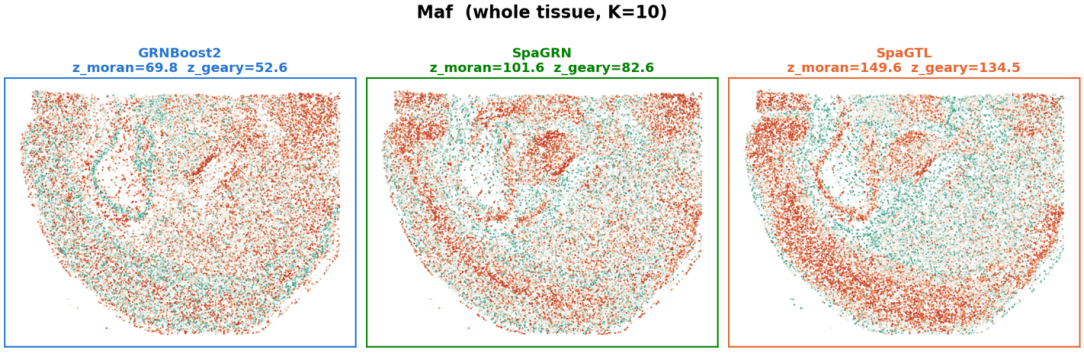

Figure A.10: Spatial autocorrelation of the Maf regulon.

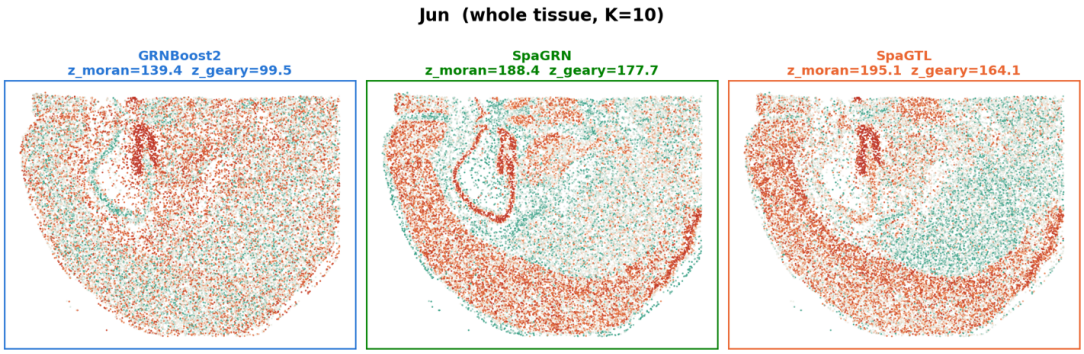

Figure A.11: Spatial autocorrelation of the Jun regulon.

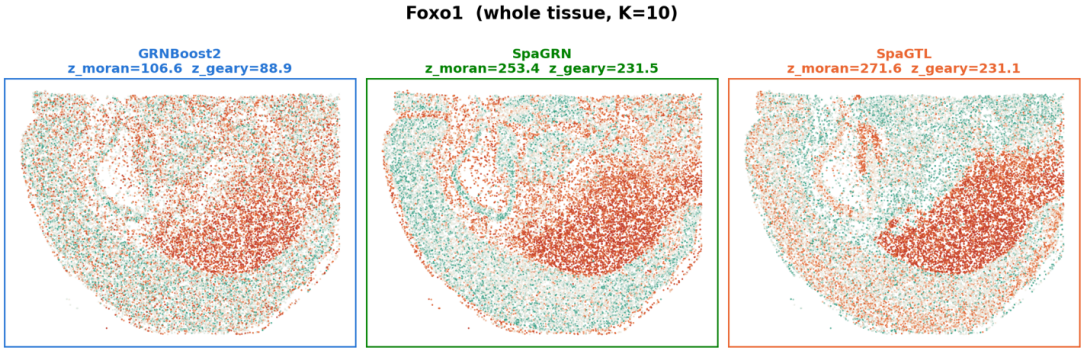

Figure A.12: Spatial autocorrelation of the Foxo1 regulon.

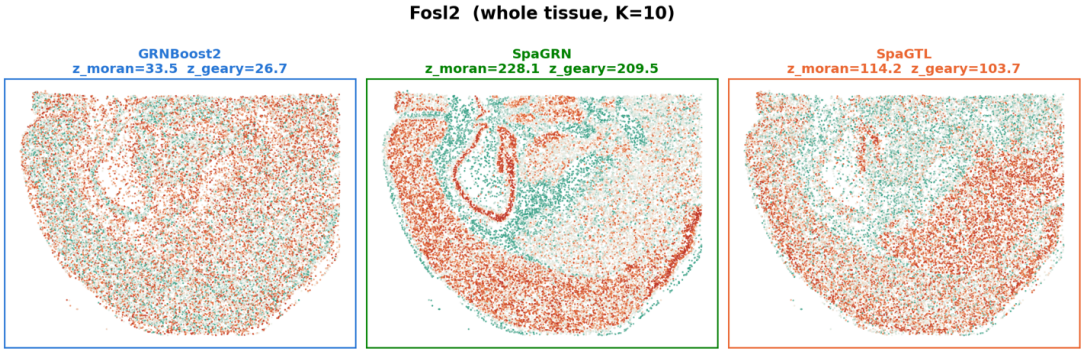

Figure A.13: Spatial autocorrelation of the Fosl2 regulon.

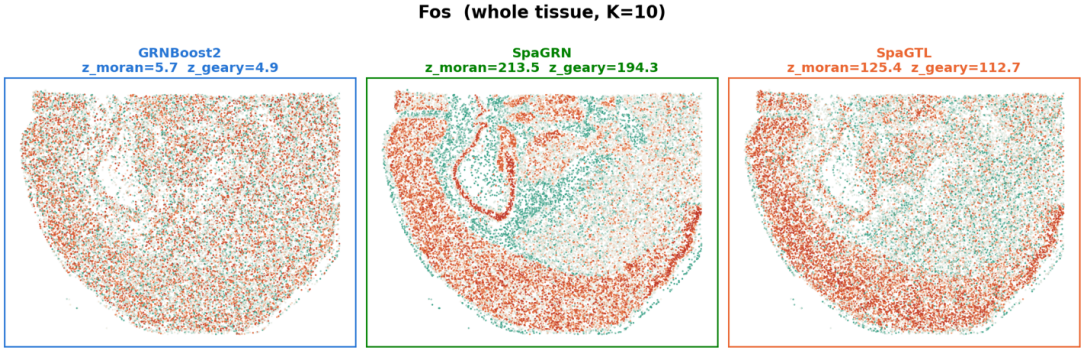

Figure A.14: Spatial autocorrelation of the Fos regulon.

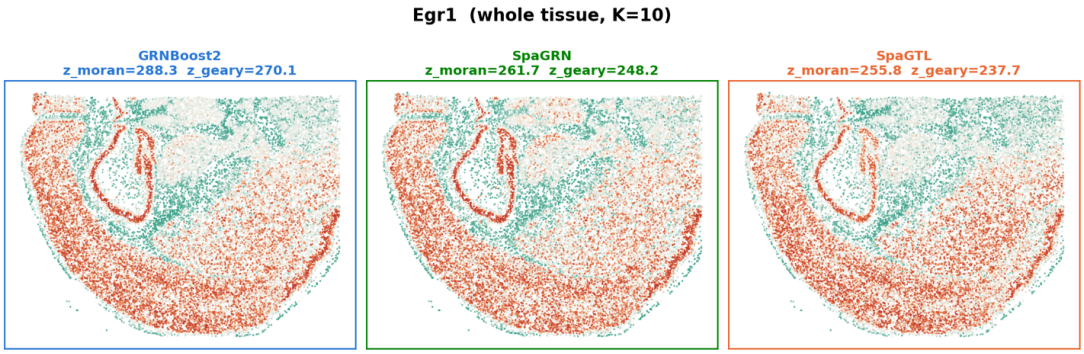

Figure A.15: Spatial autocorrelation of the Egr1 regulon.

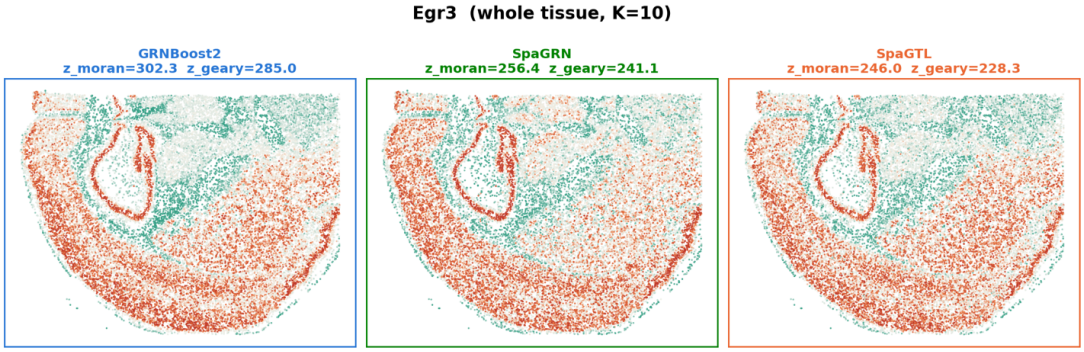

Figure A.16: Spatial autocorrelation of the Egr3 regulon.

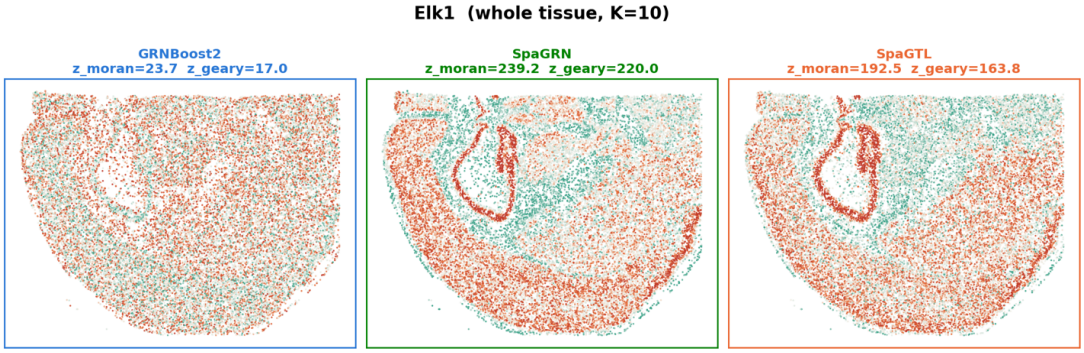

Figure A.17: Spatial autocorrelation of the Elk1 regulon.

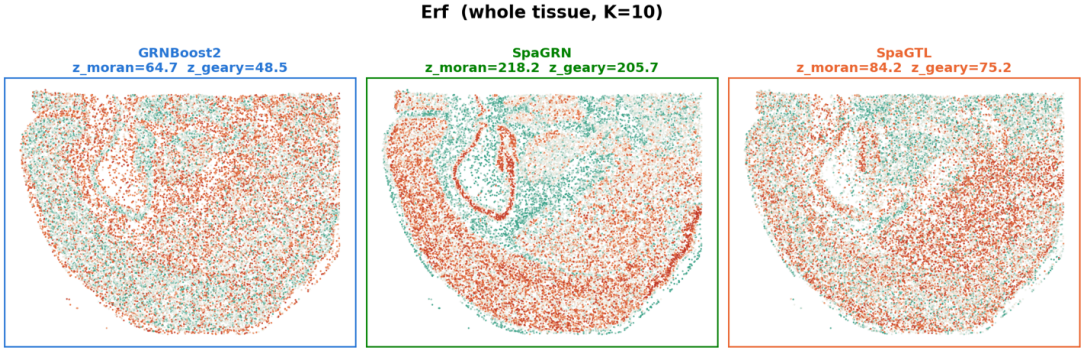

Figure A.18: Spatial autocorrelation of the Erf regulon.

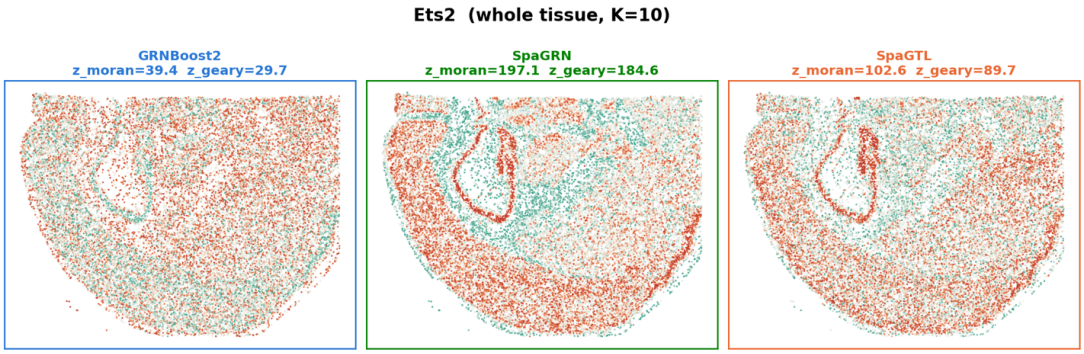

Figure A.19: Spatial autocorrelation of the Etv2 regulon.

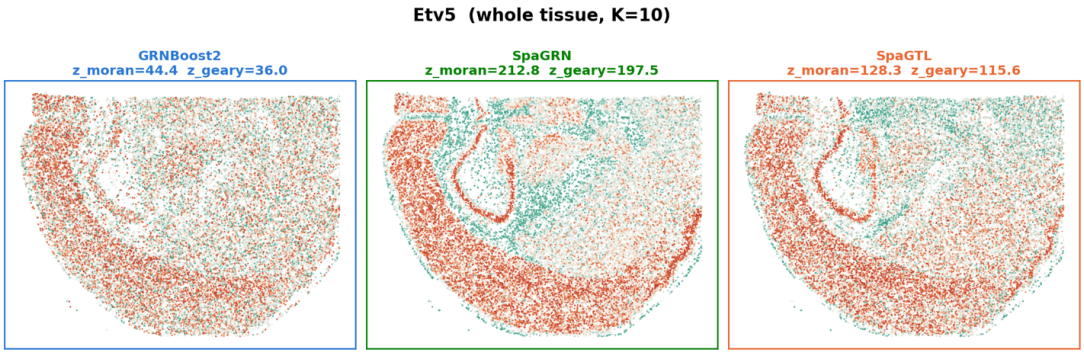

Figure A.20: Spatial autocorrelation of the Etv5 regulon.

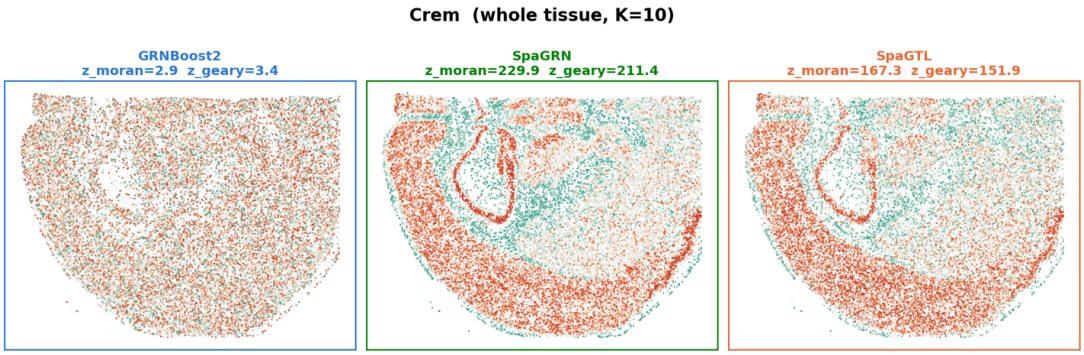

Figure A.21: Spatial autocorrelation of the Crem regulon.

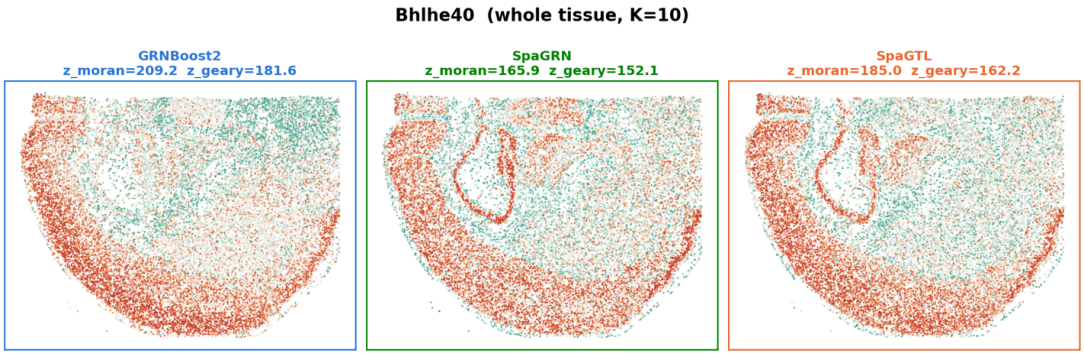

Figure A.22: Spatial autocorrelation of the Bhlhe40 regulon.

Appendix B

All source code, scripts, and data processing workflows developed and used in this project are publicly available for reference and reproducibility. The full repository can be accessed at the following link: `https://github.ugent.be/vermeirssenlab/MasterThesis_Chaeyoung`